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Progress in Materials Science 48 (2003) 171–273

Progress in
Materials
Science

www.elsevier.com/locate/pmatsci

Physics and phenomenology of strain hardening: the FCC case

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Accepted 1 January 2002

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1. Introduction

The theory of work hardening has had a turbulent history. In 1953, Cottrell [1] remarked that “it was the first problem to be attempted by dislocation theory and may be the last to be solved”. The first attempt [2] was certainly inadequate, both in terms of the predicted work-hardening rate and in terms of the stability of the dislocation structure. In the late 50s and early 60s, work hardening was one of the main themes to be discussed at international meetings. While there were many researchers contributing to the debate, it was dominated by three towering personalities: Alfred Seeger [3], Peter Hirsch [4], and (to a more subdued extent) Jacques Friedel [5]. Despite all the controversy, the two cardinal facts of work hardening had already been established and were generally accepted: the existence of an athermal hardening

rate (of a magnitude much smaller than a first back-of-the-envelope calculation would give); and the existence of a phenomenon called, even then, ‘dynamic recovery’, which characterizes the strongly temperature- and rate-dependent decrease of the hardening rate at larger strains, is not dependent on diffusion, and is strongly coupled to the stacking-fault energy (SFE) [6]. These two phenomena were associated, respectively, with the “stages” II and III of single-crystal hardening in fcc metals.

Inasmuch as this period certainly had its ‘baroque’ aspects, one may call the following era the ‘classical’. It began with the superb review article by Nabarro, Basinski and Holt [7], continued with Doris Kuhlmann-Wilsdorf [8,9], and also included the present authors. While Kuhlmann-Wilsdorf’s theory was quite detailed in many respects, it had, for the longest time, the major draw-back that the temperature dependence of large-strain hardening was not included. Mecking and Kocks concentrated on the parallels between single-crystal and polycrystal hardening, and aimed at an integration of physical and phenomenological approaches.

The one new discovery, in the last three decades was the identification of a ‘stage IV’ at large strains (usually seen in torsion tests) [10]. It appears to have a number of variants, depending on material and temperature.

The primary goal of a theory of work hardening is a prediction (or explanation) of the stress–strain curve. This is linked, of course, to the development of dislocation substructure. A fundamental fact, however, is that even a rudimentary description of the substructure requires many parameters, whereas a description of work hardening requires few. Thus, any model of work hardening that starts with observed substructures must involve a substantial contraction of information as a basic part of the model; conversely, any theory that begins with the stress–strain curve can only hope to be in accord with a very coarse approximation to substructural features. In the 1980s, a number of models appeared that emphasized the well-known heterogeneity of dislocation substructures (after large strains, in fcc metals) and abstracted this into a ‘composite’ substructure [11–13]: do we dare call this the ‘romantic’ period? They all have in common that additional parameters are involved, and sometimes diffusion [12–13].

Finally, the 1990s have seen some (impressionistic) efforts that appear to us a reversion to the pre-1950s period. One of these is based on computer simulations of dislocation kinetics at very low strains [14], and sees as one of its goals to explain the transition from “stage I” (a single-crystal phenomenon) to stage II. A similar goal uses statistical mechanics and new dislocation observations to establish work hardening behavior ‘from scratch’ [15]. The other recent (classicistic) revival is a reversion to a Bailey–Orowan type theory of dynamic recovery as simultaneous static, diffusion dependent, recovery [16].

In our own developing view, one new insight seems worth emphasizing. While the introduction of the concept of different ‘stages’ of work hardening was certainly a milestone in the developing understanding of work hardening mechanisms [17]—at this point, it seems more appropriate to refer to the different *mechanisms* themselves (e.g., athermal hardening, dynamic recovery, and perhaps ‘stage IV *processes*’): while one of these mechanisms may occasionally dominate in a particular regime, they no doubt all occur simultaneously.

An interesting change in terminology—and indeed in the underlying physics—has occurred between the 1970s and now: one used to talk about work hardening ‘theories’; now one talks about work hardening ‘models’. Cottrell’s pessimistic remark with which we started this Introduction, did properly refer to what even today would be called a ‘theory’: a quantitative prediction of a particular macroscopic behavior based, in a complete sequence, on well established microscopic mechanisms. A ‘model’ is more modest—and more ambitious at the same time: it aims to rationalize a whole set of macroscopic observations on the basis of well-established physical principles—but without the quantitative prediction of a number of parameters (which can be measured experimentally). A *theory* of work hardening is today as hopeless as ever; but a *model* of work hardening of pure fcc material we would consider now virtually available.

A parallel development (primarily in the 1970s) to the evolution of physical theories and models has been the interaction of the Materials Science and the Solid Mechanics communities. Classical plasticity treated materials as continua—a valid approximation for polycrystals—and, per force, in a phenomenological way. Only one of these phenomenological approaches could survive: the one based on a state-variable description of material behavior. Earlier, the most common descriptions postulated the flow stress to be a function of strain (‘strain hardening’) or of plastic work (‘work hardening’) [18]; but neither the plastic strain nor the plastic work is a state variable. In physical treatments, on the other hand, from the very first broad treatment of single-crystal strain hardening [3] onward, one starts with the strain-hardening *rate*

$$\theta \equiv d\tau/d\gamma \quad (1)$$

with τ the critical resolved shear stress (CRSS) on the active slip system(s) and γ the shear on that system. Hart [19] pointed out that a differential form such as (1) does not imply that θ is a derivative of a state function $\tau(\gamma)$: an *integration* of the instantaneous strain-hardening rate over some path of strain and time yields the stress–strain curve for that path. A particularly interesting part of the path is the change in the orientations of the crystal axes (the texture of the polycrystal) through plastic deformation. For this reason alone, the integration over an actual path is important for a comparison with experiments—which measure the integrated quantities. For a specification of the constitutive behavior of the material, on the other hand, only a differential form of the hardening law is admissible. Accordingly, plots used to describe it are typically θ versus τ . These quantities can be readily obtained in single-crystal deformation but in polycrystals texture effects have to be filtered out. Two tests done on the same material but different texture will exhibit different macroscopic stress–strain curves, although the hardening, i.e. θ versus τ , on a grain level might be virtually the same.

The plan of the article is as follows. Section 2 will introduce experimental and theoretical fundamentals; Section 3 will present a detailed analysis and abstraction of the experimental facts concerning strain hardening; followed by a detailed review of (our) ‘basic model’, comprising the mechanisms of athermal hardening (Section 4)

and dynamic recovery (Section 5). All this is concerned with pure face-centered cubic (fcc) metals, since most is known about these. But different fcc metals behave differently—in a sequence dictated by the stacking-fault energy (SFE); this material dependence will be treated in detail. It will be augmented by a discussion, to show that the developed concept provides the framework also for other material classes such as solution hardened alloys as well as hexagonal and bcc metals. Section 6 will give a summary, with an emphasis on fundamental aspects of future plasticity research.

We wish to emphasize that this article contains very little in the way of new proposals: it is meant to summarize the comprehensive model we have published, in various aspects, in the years 1966–1992 [20–50], marked by bold numbers in the list of references. Inasmuch as there have been some variations in our views over time, the present exposition gives a ‘mature’ version of the model. Whereas, during the development, there was an intimate alternation of experimentation and modeling, we shall here present the analysis of experiments first (Section 3), the summary of the models later (Sections 4 and 5), and preceding it all, a brief summary of salient background (Section 2).

The model was developed for, and most of the experiments were done on, pure face-centered-cubic (FCC) metals with an emphasis on polycrystals. Inasmuch as the mechanisms used, however, relate to general dislocation behavior (in presence of many slip systems of similar strength), we believe—and have found in many cases—that our approach is also relevant for other materials, so long as they are single phase. A brief discussion of this is given in Section 3.4.2.

2. Fundamental facts

In this section, we summarize the salient fundamental features of strain hardening behavior (in pure fcc metals), of the associated microstructural changes, and of the dislocation theoretical basis for modeling this behavior.

2.1. Stress–strain curves of single crystals and polycrystals

2.1.1. The conventional ‘stages’ of strain hardening

A major milestone in the evolution of strain-hardening theory was the proposal by J. Diehl [17,51]: that his experiments on copper single crystals and all other experiments on fcc single crystals up to that time (e.g. Lücke and Lange [52], Blewitt et al. [53]) could be abstracted into the existence of a number of different *stages* of strain hardening. Stage I or ‘easy glide’ depends very strongly on the orientation of the crystal and does not occur when the deformation takes place by multiple slip from the beginning (Fig. 1). Stage II characterizes the steepest rate of strain hardening, of order $\mu/200$ (with μ an appropriate shear modulus), which depends on orientation but is otherwise rather insensitive to all kinds of variables.

Finally, stage III exhibits a decreasing slope and is very sensitive to the temperature and rate of deformation (Fig. 2 [53]). A further characteristic of these two stages

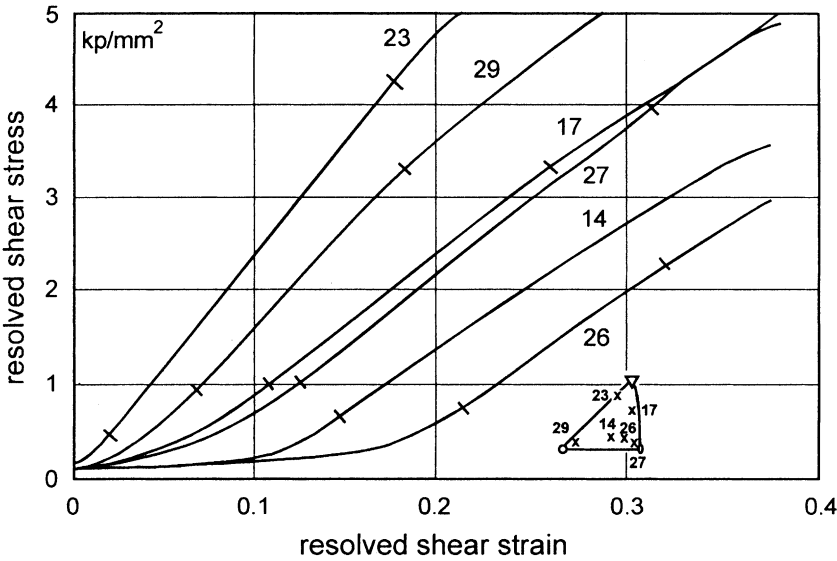


Fig. 1. Shear stress versus shear strain curves for single crystals of various orientations in 99.98% copper at room temperature and an initial shear strain rate of $2 \cdot 10^{-3} \text{ s}^{-1}$ [17]. The beginning and end of stage II are marked. (Note that the beginning of double slip occurs in the middle of stage II for orientations 23 and 29, way into stage III for all others.)

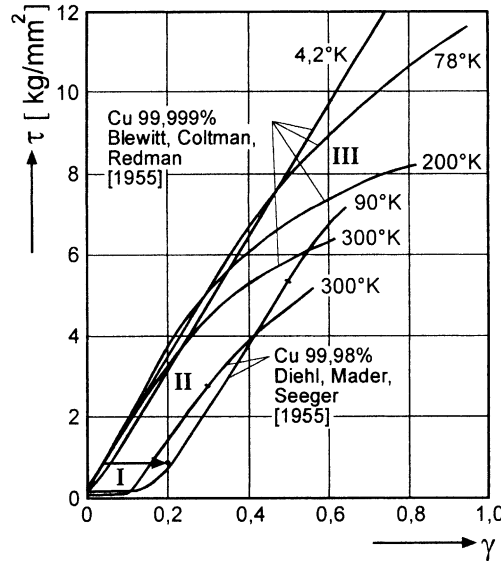


Fig. 2. Shear stress versus shear strain curves of single crystals [orientation near (100)] of 99.999% copper at various low temperatures [53] and for a single-slip orientation [51].

is that stage II is material insensitive, whereas stage III depends strongly on material, even within the fcc metals—in a sequence that suggests a dependence on the (normalized) stacking-fault energy (SFE) [54–56]. The presence of stage III limits the extent of stage II and may in fact eliminate it as a separate stage, especially at high temperatures and for materials of high SFE.

These characteristics were later also identified in polycrystals. Fig. 3 shows a set of copper polycrystal stress–strain curves [47] for five different temperatures and two strain rates. It is seen that they all start with the same slope at yield and then disperse; moreover, the initial slope is similar to that of the single crystals, once appropriate average orientation factors have been incorporated.

This kind of observation is quite typical for fcc materials [27], and it has led to an interpretation of the stages of strain hardening that covers both single crystals and polycrystals [35]. In this view, the characteristic of stage II is not that it exhibits a substantial range of linear hardening, but that it is athermal (of a shear hardening rate of about $\mu/200$, equivalent to $d\sigma/d\varepsilon \approx E/50$ in polycrystals, where E is Young's modulus—all within a factor of 2 or so). In fact, it may not be a 'stage' at all, but more like an asymptote for low-strain multiple-slip behavior. Similarly, stage III is characterized by a specific, strong dependence on temperature and strain rate, which is observed in both single crystals and polycrystals. It has always been difficult to define its 'beginning' (called τ_{III} in the literature). We will find below (Sections 3.2 and 3.3) that a general scaling of the stress for all of 'stage III' can provide a parameter that can be used to characterize the dependence on temperature, strain rate, and material. Detailed analyses of high strain behavior later revealed a stage IV [10,57] of low hardening rate. It can be used as an asymptote for high strains, just as

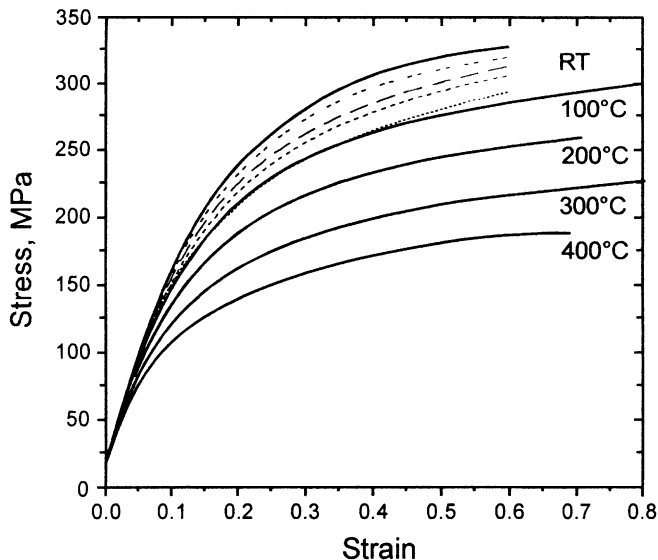


Fig. 3. Compressive stress strain curves for OFE copper polycrystals (grain size $\cong 25 \mu\text{m}$). The strain rate is 1 s^{-1} for all solid curves, at room temperature also lower rates down to 10^{-4} s^{-1} [47] in decades.

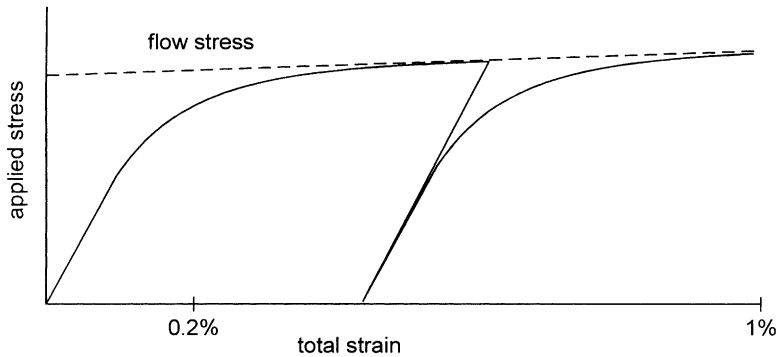


Fig. 4. Applied stress versus total strain, at a scale that illustrates the definition of the *flow stress* as relating to plastic properties only. All strain hardening is in the slope of the dashed line.

‘stage II’, in the new view, has degenerated into an asymptote for low strains. The *mechanisms* that have been formerly identified with stages II, III and IV then operate *simultaneously* throughout the entire range of the stress–strain curve.

2.1.2. The elasto-plastic transition. Definition of flow stress and strain hardening

The curves in Figs. 1–3 showed *plastic* strain only, in the abscissa. The initial elastic region is not included—but if were, it would not show in these plots: it is so steep in comparison to all measured strain-hardening rates

Fig. 4 shows a schematic curve of applied stress versus total strain at a very different scale; here, it is the rate of strain hardening that cannot be seen—because it is so *low*. What is conventionally called the *flow stress* relates to plastic properties only; for example, we define it from the back-extrapolation of the plastic curve; often, it is defined at a particular plastic strain (0.2%). Similarly, under *strain hardening* we understand only the change in *flow stress* with pre-strain (which could be measured by the back-extrapolation after repeated un- and re-loading) [58,59].

2.2. Microstructure fundamentals

2.2.1. Slip line observations

The first micrographic information that was obtained on plastically deformed samples concerns the appearance of slip lines on the surface. It served to identify the crystallographic nature of slip and also gave significant clues on the heterogeneity and the dynamics of slip. A crucial technique in the early investigations (i.e., before the invention of the SEM) was the shadowing of surface replicas for electron-microscopic studies by Heinz Wilsdorf [60].

Fig. 5 shows two pictures from [3,61]. In the first, the slip steps appear to be quite homogeneously distributed; in the second, one sees bundles or ‘slip bands’, which may often be resolved into a number of parallel, slightly offset steps, perhaps connected by ‘cross slip’. Look for the few slip line terminations: from these, the average slip line length can be inferred. In polycrystals (Fig. 6a), slip lines look qualitatively

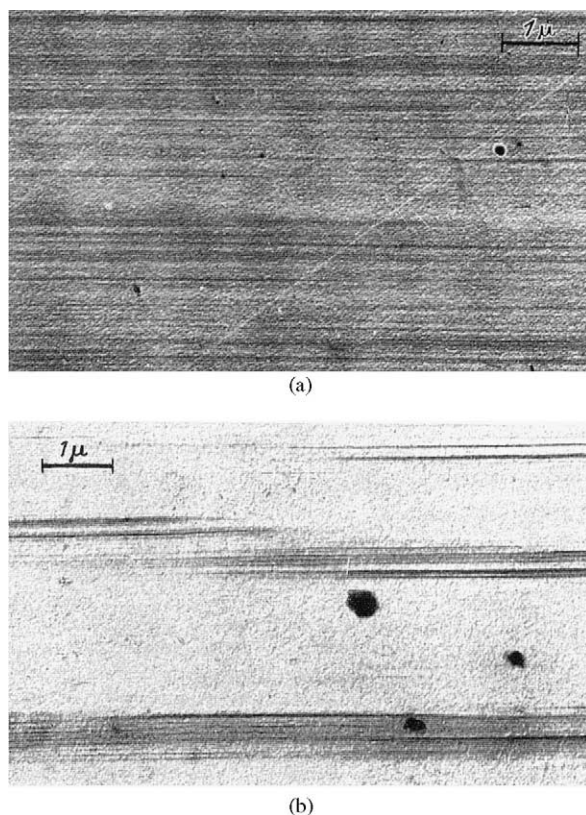


Fig. 5. (a) Fine slip on a copper single crystal during a small strain interval in stage II. Note some slip line endings. (b) Slip bands formed in the same crystal in stage III [3,61].

similar, except that typically more than one set is observed (even though the constraints on the surface are presumably not as severe as in the grain interiors). For this reason, when polycrystal deformation is compared with single-crystal deformation, the latter should be in multiple-slip orientations [62]. It is also worth noting that the effect of the grain boundaries is primarily to enforce macroscopic compatibility (more easily attained with many slip systems), rather than the penetration of individual slip lines (or dislocations) through the boundary. This is demonstrated in Fig. 6 for aluminum and brass [60]. Only when slip is very coarse, restricted to a few very strong slip lines (as in α -brass [63]) is there a strong correlation between the two grains at the boundary.

Another crucial property of slip line formation is that it happens quite rapidly—in the order of 10^{-4} – 10^{-3} s [63], compared to the average time between slip events of about 1 s. In some cases (brass single crystals [63] or neutron-irradiated crystals [64], for example) the formation of one slip line is followed by a Lüders-like lateral propagation; in pure metals, the sequence is random. In either case, there is a tendency of slip lines to form at a spacing of not less than the average dislocation spacing at that stress. Such behavior would be expected as a compromise between the two

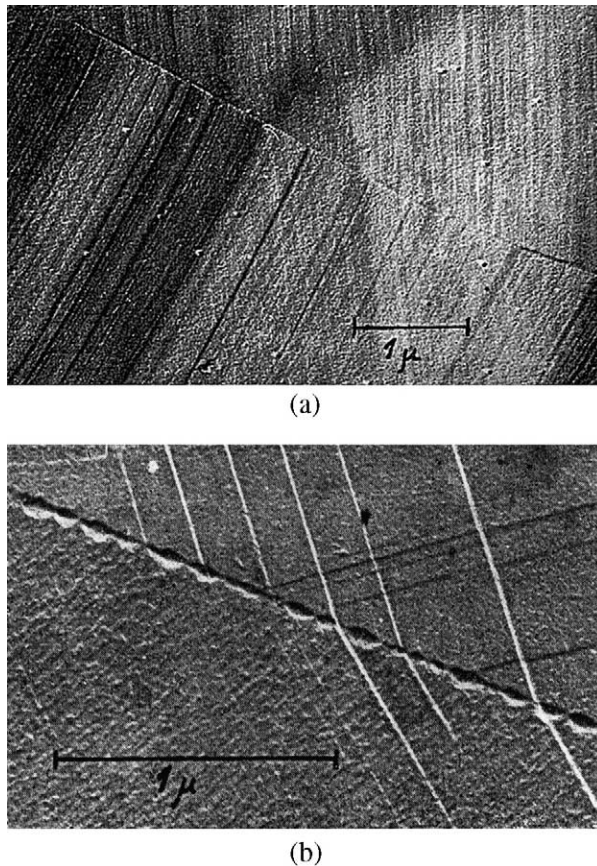


Fig. 6. (a) Multiple slip line sets near a grain boundary in aluminum and (b) in brass [60].

tendencies to maximize the macroscopic dissipation rate by having slip as homogeneous as possible and, on the other hand, to avoid regions that have been preferentially hardened [65]: then “intraplane hardening equals interplane hardening”.

From this point of view, the heterogeneity of slip is about as low as conceivable (in the absence of more macroscopic heterogeneity such as ‘deformation bands’ of various sorts [66,67]. However, there is a separate observation that does not follow from this argument; namely, the slip step height is typically about 20–30 Burgers vectors. In Fig. 7, one can appreciate this height qualitatively, and one can see that the glide bands formed fully at different times [68].

2.2.2. TEM observations

While slip lines are indicators of where dislocations have moved, transmission electron micrographs show where dislocations have been stored (which is the cause of strain hardening). Fig. 8 shows typical arrangements of dislocations observed by TEM. Fig. 8a, taken after deformation into the end of stage I [69] in a single-slip

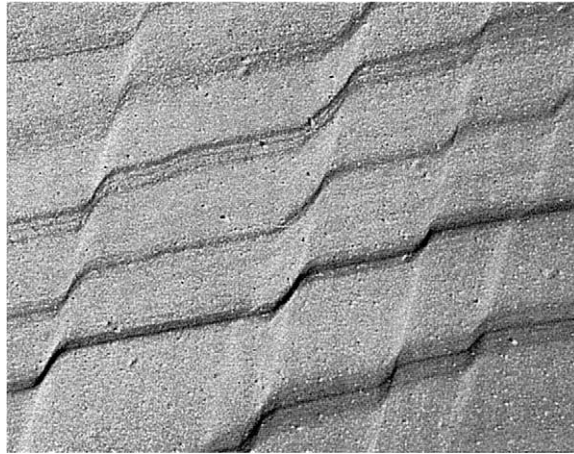


Fig. 7. Shadowgraph of slip bands of two intersecting slip systems on aluminum deformed at room temperature [68].

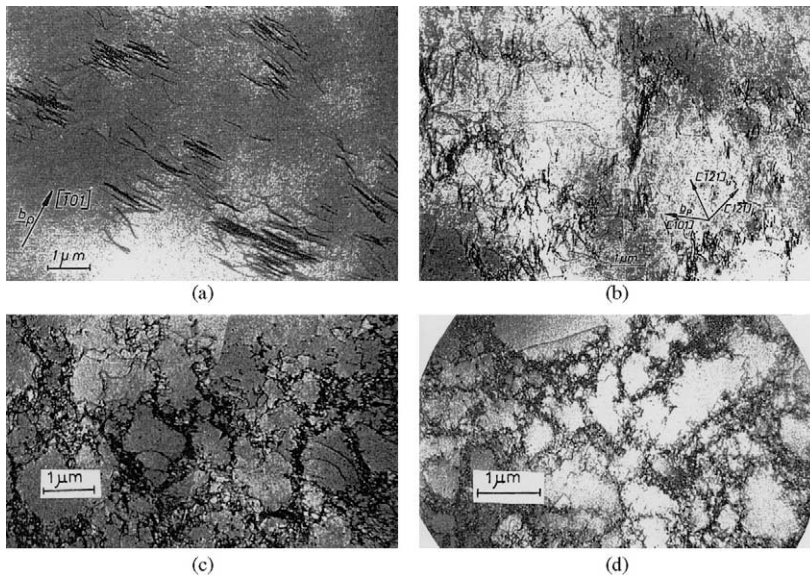


Fig. 8. Typical electron transmission micrographs: (a) in late stage I and (b) in stage II of a Cu single crystal deformed in single slip [69], (c) and (d) are from a $\langle 100 \rangle$ crystal [70]. All tests were at room temperature and the flow stress levels were in MPa: 1.1 (a); 14 (b); 28 (c); and 69 (d).

single crystal, shows essentially straight dislocations (which are often in dipole or multipole arrangements). Fig. 8b displays a low density of dislocations, perhaps randomly arranged, it was taken in early stage II also in a single-slip orientation. Fig. 8c and d shows dislocation patterns obtained at somewhat higher stress levels in a multiple-slip orientation (100) [70]. There are regions denser with dislocations

(‘tangles’) and regions less dense; the tangles are interconnected into (diffuse) ‘cell walls’ and the less dense regions may be called ‘cells’; this is typical for deformation in multiple slip. This situation is also typical for single slip orientations in the regime where the strain-hardening rate is very low, or of samples that have been statically recovered after deformation. Needless to say, the variety of detail in observed structures is virtually infinite; the above is our (and the above authors’) assessment of what is essential.

Similar patterns could have been obtained in a polycrystal. Inasmuch as the dislocation substructure depends very much on the stage of strain hardening that dominates the behavior, a comparison of different curves *at constant strain* is problematical. For example, a comparison at 50% strain in Fig. 2 would have corresponded to different parts of the hardening curve at different temperatures. A comparison *at constant hardening rate* might come closer to a study of the behavior at a given stage of deformation.

We have undertaken one small study in this direction [71]. A number of samples from the extensive investigation of copper polycrystals over many temperatures and strain rates, introduced in Fig. 3 above [47], were prepared for TEM observation. They were picked such as to have the same strain-hardening rate. The resulting pictures were magnified in proportion to the flow stress (which was of course different for different temperatures); Fig. 9. It is seen that the *scale* of the images is about the same; that is, there are cells of similar size (although different character in detail), and the dislocation densities are about the same (as will be detailed in connection with the flow-stress/dislocation-density correlation in the next section).

The fact that both the average dislocation spacing and the cell diameters maintain a fixed relationship is one aspect of a multi-faceted phenomenon called *similitude* [8]. For the assessment of the dislocation patterns one must bear in mind that only the depicted area but not the thickness of the TEM foils could be adjusted to the average dislocation distance; also if similitude were strictly observed, the projected total dislocation length would be somewhat larger at the higher dislocation densities.

Quantitative measurement of dislocation densities and cell sizes is difficult. Dislocation densities are hardest to determine in very dense areas; on the other hand, cell sizes are most definitive when the cell walls are well-developed and not too much spread out. These two tendencies complement each other: if a proportionality between dislocation spacing and cell size can be established, in a region where both can be reasonably well measured, then their joint relation with the flow stress can be established over a wide range. This proportionality has been widely studied; Fig. 10 shows an example from Staker and Holt [72]. If the symbol l is used for the mean dislocation spacing in the slip plane (which is the inverse square-root of *half* the *volume* density of dislocation length used by these authors), and the symbol λ for the square-root of the average area per cell (=0.89 times the “average cell diameter d ” used by them), their results yield, to amazing accuracy

$$\lambda = 10l \quad (2)$$

Note that the data in Fig. 10 refer to different temperatures. To the extent that the ‘stages’ are similar at all temperatures, it is thus possible to get indirect microstructural

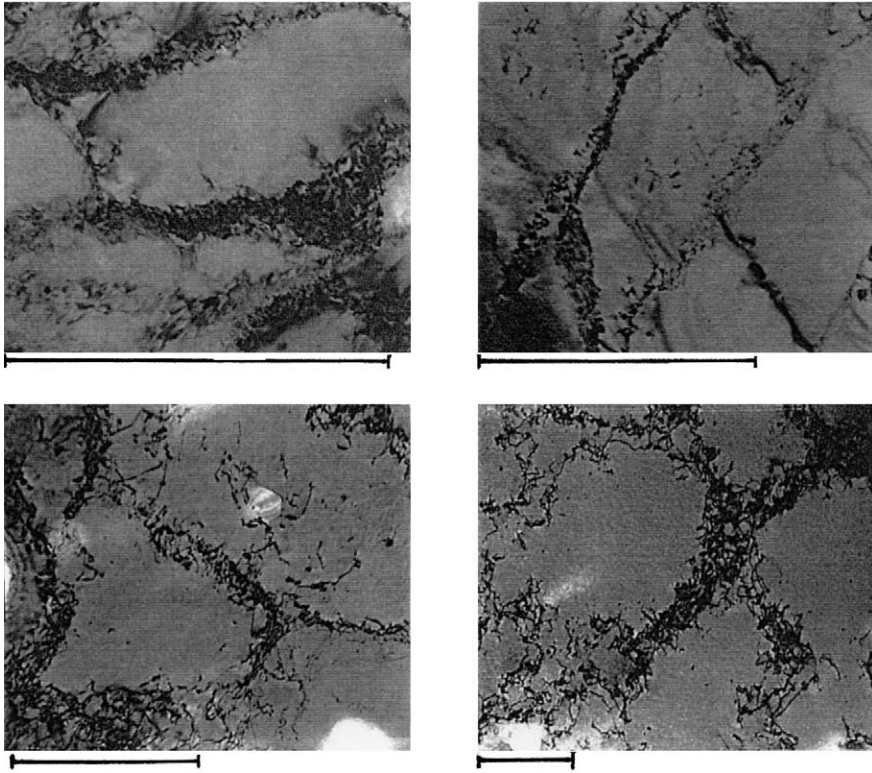


Fig. 9. TEM foils of copper polycrystals deformed to the same strain-hardening rate at four temperatures: 200, 300, 400, and 500 °C, from left to right and top to bottom—magnified in proportion to the flow stress attained, a measure of the average dislocation distance [71]; each bar represents 1 μm .

information on ‘high-strain behavior’ (at room temperature: i.e. low-hardening behavior) by measuring lower dislocation densities at higher temperature.

Finally, it is interesting to compare the cell sizes to the slip line lengths (inferred from the termination in figures like Fig. 5): a careful study by Ambrosi and Schwink [73] has shown that the (fine) slip-line length is typically 3 (or more) times the cell size.

2.3. Dislocation theory fundamentals

2.3.1. Flow stress

One of the most surprising observations in the dislocation theory of plasticity is that the relation between flow stress τ and dislocation density ρ [2]

$$\tau = \alpha \mu b \sqrt{\rho} \quad (3)$$

(with μ the shear modulus and b the magnitude of the Burgers vector) holds, with little regard to the *arrangement* of dislocations, with such generality for all those

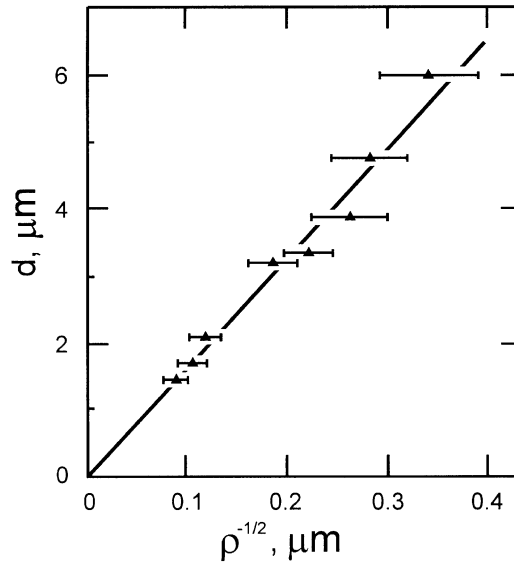


Fig. 10. Dislocation cell size versus average dislocation spacing in copper polycrystals after straining at various temperatures to 10% strain [72].

cases where the flow stress is solely controlled by dislocation–dislocation interaction. This is true for pure fcc material, where the lattice resistance to dislocation motion is always negligibly small, for hexagonal metals with a close packed lattice and also for less densely packed lattices, such as bcc at high enough temperature where the Peierls–Nabarro force vanishes.

On account of thermal activation the constant α depends on temperature and therefore it is useful to split it into two parts

$$\alpha = s(\dot{\gamma}, T)\alpha_0 \quad (4)$$

α_0 is the value at $T=0$ K where $s=1$; s decreases with increasing temperature and decreasing strain rate for each material in a characteristic way (Fig. 12). The actual value of α_0 is determined by the geometrical arrangement of the dislocation and, assumedly, it decreases continuously with deformation when the dislocation structure is gradually converted from a random into a cellular arrangement. But the variations are in a 10% range (see Section 5.4) and thus much below the accuracy with which α can be determined experimentally by investigations such as reported in Fig. 11. The value of the constant α does, of course, depend on the precise specification of how ρ was measured. If one uses the intersection density (rather than the length per unit volume), and the value of μ that enters into the screw dislocation energy¹ [25], one

¹ Using the average of the screw and edge energies for isotropic materials is not as appropriate; it would lower α . Similarly, using the volume density of dislocations would lower α , and its relation to the areal density (which actually matters) depends on the arrangement.

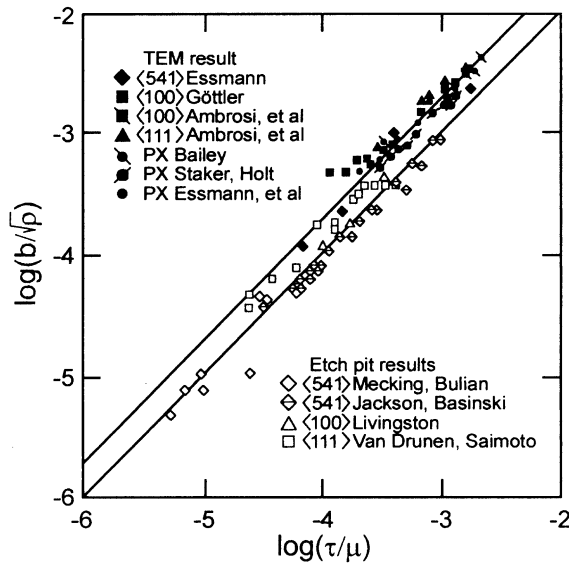


Fig. 11. Square root of the dislocation density versus the flow stress. Data collected in [35].

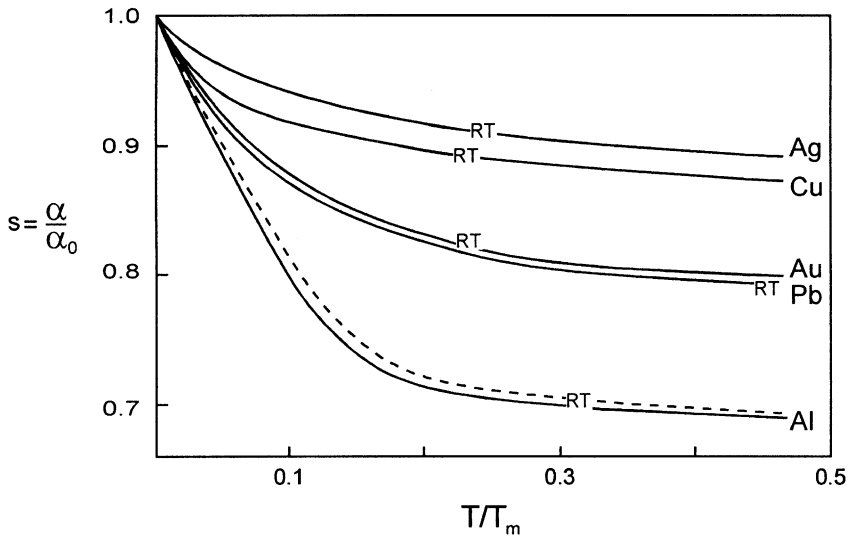


Fig. 12. The effect of thermal activation on the flow stress of some pure fcc metals at low temperatures (for medium strain rates around 10^{-4} s^{-1}) from data compiled in [80] (own measurements and literature data [199–201]). The dashed line shows for Al the behavior for a 10 times higher strain rate. The Cottrell–Stokes ratio, $\tau(T)\mu_0/\tau(0) = s = \alpha/\alpha_0$, obtained by temperature changes, is plotted versus the homologous temperature. Thereby the values of $\tau(0)/\mu_0$ were determined by back extrapolation to zero temperature, due to lack of low-temperature data, a somewhat ambiguous procedure for Au and, in particular, for Pb where merely two data points were lying below RT. At higher temperature—above about $0.4T_m$ —in most of the experiments an increase of the curves has been observed which can be attributed to stain aging effects by impurities and which has been ignored in the present evaluation.

obtains for copper at room temperature [35] $\alpha \approx 1/2$ to 1. This value holds for single crystals and polycrystals, for stage II and all of stage III that can be measured; in stage I, the relation holds [74–76] with same value of α , if the *forest* density ρ_f is entered (and a small stress is subtracted, assumed to be due to solutes). The interaction with forest dislocations does in fact dominate the flow stress in all stages, in all current models (see Section 4).

It was pointed out in Section 2.2.2 that cell sizes are proportional to dislocation spacing; thus, the flow stress can also be expressed in terms of cell size. In fact, this proportionality makes it possible to determine τ from whichever quantity is easier to measure. An extensive compilation of data for subgrain sizes, mostly at high temperature [77] showed that, even though regression analysis may yield slightly different results, a linear relation between flow stress and inverse cell wall spacing fits about as well as any other. These authors find (converted to shear stress and with the definitions of λ and of the shear modulus μ used here [25])

$$\tau/\mu = 7.5 \cdot b/\lambda \quad (5)$$

The flow stress discussed here relates to dislocation interactions only. In general, there are other contributions to the yield strength, and the superposition law is not trivial [44,78]. Also, in solid solutions, the concentration may affect (3) itself, and thus the strain hardening rate [79]. To simplify the present discussion of strain hardening, we identify τ with *the dislocation contribution to the flow stress in pure materials only*.

2.3.2. Thermal activation

In such cases, dislocation *motion* is ‘areal’: a picture of N dislocation segments *spreading* over an area a_0 is more appropriate than one of (straightish) mobile dislocations moving at an average velocity [39,79]. The kinetics is determined by the waiting time t_w for thermal activation, so that the plastic strain rate becomes [25,80,81]

$$\dot{\gamma} = bNa_0/t_w = \dot{\gamma}_0^* \exp\{-\Delta G(\tau/\hat{\tau})/kT\} \quad (6)$$

(This equation holds for the individual slip systems as well as for the strain rate produced by many; we have used the symbol $\dot{\gamma}$, the crystallographic shear rate, in a general sense—the precise geometrical meaning to be determined in each case [50].) The quantity $\hat{\tau}$ is the ‘mechanical threshold’: the flow stress when the activation energy $\Delta G=0$, i.e. ‘at 0 K’, $\hat{\tau}/\mu \equiv \tau(0)/\mu_0$. It is determined by the state of the material, and is really what was meant by τ in (3).²

The flow stress τ at a finite temperature and a given strain rate can be obtained by inverting (6):

$$\tau = \hat{\tau} \cdot s \{kT \ln(\dot{\gamma}_0^*/\dot{\gamma})\} \quad (7)$$

² In the other limit, when $\tau=0$, there could, in principle be back-jumps over the activation hump, which has led many investigators to write (6) in terms of a sinh-function rather than an exponential. However, the area a_0 swept out after each activation event is so large that this is quite impractical.

where s is the inverse of the function ΔG in (6). The coupling between the dependencies on temperature and on strain rate is the hallmark of thermally activated processes. It must be slightly modified by the fact that the shear modulus—to which τ is proportional—depends on temperature (but not strain rate) and, for this reason [25], also ΔG must be divided by μ ; to attain non-dimensionality, we write

$$\frac{\tau}{\mu} = \frac{\hat{\tau}}{\mu} \cdot s \left(\frac{kT}{\mu b^3} \ln \frac{\dot{\gamma}_0^*}{\dot{\gamma}} \right) \quad (8)$$

The effect of thermal activation on the flow stress due to dislocation interactions is not very large. Fig. 12 shows experimental data for single crystals of a number of pure fcc metals; they are normalized to the flow stress at zero temperature (obtained by back extrapolation). It turns out that the dependence of *strain hardening* is also thermally activated—and to a much more sensitive extent. For this reason, we neglect the temperature dependence of the flow stress in most of the following discussion and treat strain hardening as the evolution of τ , not of $\hat{\tau}$ [50,191]. (A remark in advance: $\hat{\tau}$ is the proper quantity to be used in any analysis of the flow stress in which the dislocation density is the exclusive state variable. We will see in Section 5, that the effect of thermal activation on the work-hardening coefficient requires the introduction of a second state variable. This is linked to the dislocation arrangement and to θ , with the implication that τ , not $\hat{\tau}$, is the appropriate variable to be used in the analysis of work hardening behavior.) In the past, it was often assumed [3] that, even if the flow stress were dominated by forest interactions, there would be an athermal ‘plateau’ level (τ_μ) to which (7) would be added. Such an assumption is not warranted within a framework of phenomenological discipline, when the ‘Cottrell–Stokes law’ holds; it does in pure metals, as is best demonstrated by a ‘Haasen plot’. Fig. 13 shows it for a number of fcc metals [25].

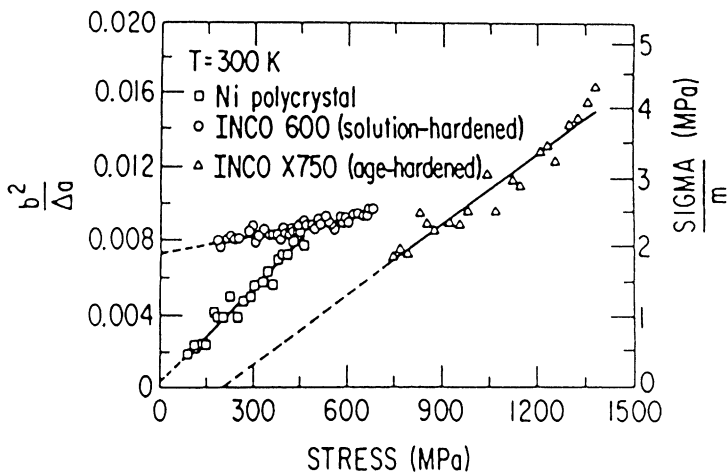


Fig. 13. “Haasen plot” for pure nickel and two of its alloys: solution hardened (Inconel 600) and precipitation hardened (Inconel X 750) [81].

When the line of data extrapolates to the origin, there is no need for defining an additional ‘internal stress’. (When the intercept is positive on the τ axis, there is; e.g. in dispersion strengthened alloys. When the intercept is negative on the τ -axis, this is evidence for an additional, more strongly thermally activated mechanisms, such as lattice or solute resistance [81].)

2.3.3. Strain hardening

As the (dislocation-caused) flow stress is determined by (3), its increase with strain is due to dislocation storage. The rate of accumulation of dislocations can formally be written as

$$\frac{d\rho}{d\gamma} = \frac{dL}{bda} = \frac{1}{b\Lambda} \quad (9)$$

where dL means the length of dislocation stored per area swept da , and Λ may be called a ‘mean free path’ [20,39].

This dislocation accumulation rate is a fundamental quantity in the dislocation theory of strain hardening [22,29,82]. It is accessible from experiment: differentiating (3) gives

$$\tau \frac{d\tau}{d\gamma} = \frac{(\alpha\mu b)^2}{2} \frac{d\rho}{d\gamma} \quad (10)$$

With the definition $\theta \equiv d\tau/d\gamma$ (1;3), and inserting (9), one may also write this as

$$\tau\theta = \frac{(\alpha\mu)^2}{2} \frac{b}{\Lambda} \quad (11)$$

Finally, inserting (3) again, the strain-hardening rate itself becomes

$$\frac{\theta}{\mu} = \frac{\alpha}{2} \frac{l}{\Lambda} \quad (12)$$

where

$$l \equiv 1/\sqrt{\rho} \quad (13)$$

Fig. 14 shows a schematic plot according (11) as a function of the flow stress, for a number of special cases. First, assume the mean free path, Λ , to be a constant; for example, as if all dislocations went through the entire grain in a polycrystal and get stored at a grain boundary, without any storage in the interior; this makes Λ the grain size, and $\tau\theta$ should be constant (dotted, horizontal line). Second, assume ‘similitude’—i.e., that the mean free path Λ is proportional to the dislocation spacing; then, according to (12), θ would be a constant and $\tau\theta$ should be proportional to τ (dashed line, extrapolating through the origin). In reality, there may be contributions to

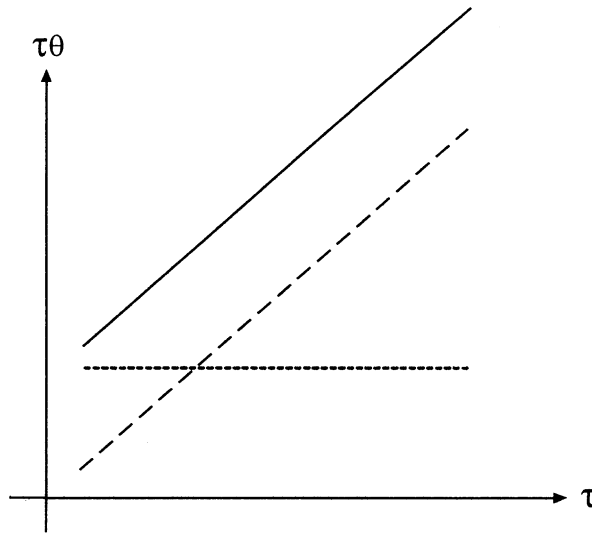


Fig. 14. Dislocation storage rate when the mean free path is constant (dots); when it is proportional to the dislocation spacing (dashes); and when both superpose (solid).

dislocation accumulation from *both* a constant mean free path and one proportional to l ; then, you get the solid line in Fig. 14: straight, but with a positive intercept on the $\tau\theta$ -axis.³ Fig. 15 shows an experimental observation of this effect, for two different grain sizes [83].

Fig. 15 also demonstrates a general departure from the schematic in Fig. 14: while the lines tend to start out straight, they then curve downward; this corresponds to the declining slope in stress–strain curves. Phenomenologically, one may describe this as an increase in Λ/l with strain; under some circumstances, this may make physical sense. More commonly, one introduces a new mechanism, ‘dynamic recovery’, which emphasizes the strong dependence on temperature and strain rate in this regime. In that case, Λ remains associated with the (athermal) *storage* mechanisms, while $d\rho/d\gamma$ remains the *net* accumulation rate; then one can write

$$d\rho = \frac{d\gamma}{b\Lambda} - d\rho_r \quad (14)$$

We have seen that the first term in (14) leads to a constant hardening term, θ_0 . One can therefore recast (14) in terms of the strain hardening rate:

$$\theta = \theta_0 - \theta_r(T, \dot{\gamma}) \quad (15)$$

where θ_r is the dynamic recovery term.

³ The intercept may also be positive on the τ -axis: when there is an additional contribution to the flow stress, such as from solution hardening; then, the stress–strain curves exhibit an initial concave-up region [29].

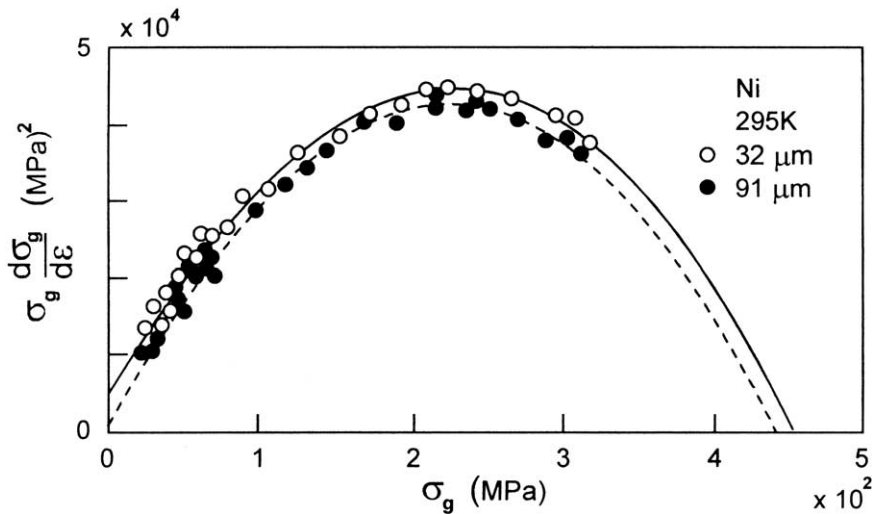


Fig. 15. A measured grain size dependence [83].

Eqs. (14) and (15) embody two mechanisms, not two stages: both mechanisms are going on from the beginning to the end—although the first term dominates at low strains and is compensated by the second at large strains. In Section 3, we shall present a detailed analysis of experimental facts. They show that the first term indeed corresponds to an athermal, constant contribution to the hardening, and the second term represents a contribution that depends strongly on temperature and mildly on strain rate—and does *not* depend directly on *time*.

In Sections 4 and 5, we address the modeling. Section 4 treats the athermal hardening rate. Its value, $\approx \mu/200$ (Section 2.1.1), has been the subject of many theoretical attempts [3,4,8,20,84]. We note here only that the mean free path, that would follow from (12) or (14), would have to be quite large with respect to the average dislocation spacing (100–300 times, depending on α); thus, it is also much larger than the typical ‘cell size’ (10–30 times)—and actually, it is even larger than the typical slip line length (3–10 times larger). To explain this order of magnitude, without losing its insensitivity to so many effects, has been the major challenge.

2.3.4. Similitude

The concept of ‘similitude’ has been used in a number of different ways [7,9,85]. From a *structural* point of view, the term may be used to characterize the observed proportionality between dislocation spacing and cell size (2), which appears to hold along the entire stress–strain curve. We have noted (Fig. 9), however, that the cell wall *thickness* does not scale in the same way. In general, any number of microstructural parameters may evolve in different ways—but they may perhaps not be relevant to any particular macroscopic property being investigated.

The *flow stress* relation (3) or (5) also holds, as far as one can determine, over the entire range; this means that there is but a single independent parameter that deter-

mines it. ‘Similitude’ is sometimes used to characterize single-parameter relations. The ‘principle of similitude’ introduced by Kuhlmann-Wilsdorf [8] went further than the above two points: it claimed a proportionality between the *mean free path* of dislocations and their spacing—which leads to a constant strain-hardening rate (12), as it has been conventionally identified with ‘stage II’ strain hardening. In our view (Section 2.1.1), an athermal hardening rate is not a ‘stage’ but a *mechanism*, which is operative as one component of strain hardening throughout the entire regime. In this sense, then, the principle of similitude (between Λ and l or λ) holds throughout.

However, this requires defining the mean free path as just the quantity that enters the athermal hardening rate—as in (14)—not as a phenomenological expression for the entire hardening rate—as in (12). (Kuhlmann-Wilsdorf [8,9] did the latter—and also incorporated the flow stress scaling [Eq. (3)] into the meaning of ‘similitude’.) In our view, then, it is the ‘dynamic recovery’ rate $d\rho_r/d\gamma$ that breaks the similitude: it cannot depend on l and λ alone. We also find that the cell wall *thickness* does *not* scale with the other length parameter.

The challenge of ‘dynamic recovery’ modeling is to achieve it within the context of similitude for the flow stress (determined by a single structural parameter). Significant insight can be gained, in this context, by observing that—only when dynamic recovery is important—there are significant *transients* upon changes in temperature or strain rate (Fig. 16). Any observed transient requires an additional independent state parameter—which attains its own steady state at the end of the transient, and is in continual steady state along a monotonic stress–strain curve. One may say that the monotonic stress–strain curve is itself a transient (as it is regarded in the creep literature): namely, for the *first* state parameter.

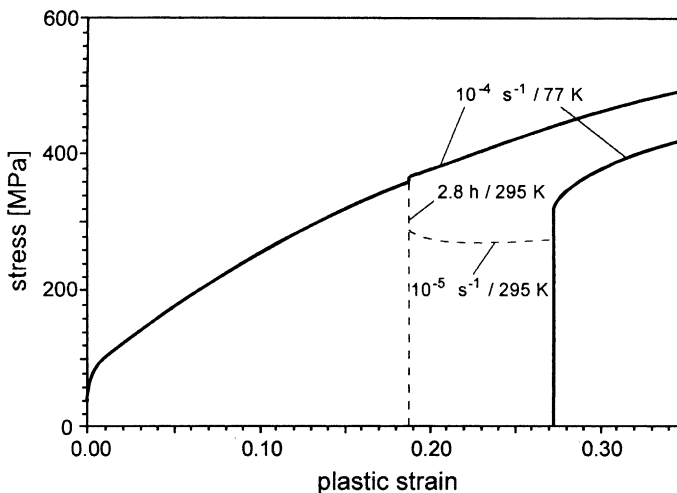


Fig. 16. Two texture free Cu-samples, produced by powder compaction, were deformed in compression at 77 K to an identical flow stress level. They were then both kept at room temperature for about 2 h; where one of them was deformed by about 5% strain at a low rate, and then for both samples deformation was continued at 77 K. The fig. is an illustration of the transients occurring after temperature changes and also of the work-softening phenomenon.

3. Analysis of mechanical behavior

The analysis to follow in this section is primarily based on stress–strain tests of polycrystals conducted, in the common way, by uniaxial deformation, primarily compression. Here are the directly obtained experimental quantities the normal stress σ (applied force divided by the actual cross section of a sample) and the longitudinal strain ε (logarithm of the ratio of the actual and the initial sample-length), which are used throughout this chapter. Physically they reflect the averages of the shear flow stress τ and the shear strain γ of all crystallites of a polycrystal, which can be converted into σ and ε by a suitable model of polycrystal plasticity, if the crystallographic texture is known. We generally use the Taylor model. For a comparison of results obtained in different testing modes (tension, compression, torsion), also von Mises' equivalent stress and equivalent strain are discussed, only to show their shortcomings..

3.1. Time recovery versus strain softening

The term 'dynamic recovery' is generally used for 'Stage III' processes: a decreasing hardening rate, strongly dependent on temperature and strain rate. One interpretation of this process, since Bailey [86] and Orowan [87], has been that it is due to recovery (in the classical sense, dependent on *time*). In superposition with the hardening component (dependent on plastic *strain*), the resulting net change of the flow stress $d\sigma$ at a given temperature is

$$d\sigma = \Theta_0 d\varepsilon - R_t dt \quad (16)$$

and dividing by $d\varepsilon$ leads to the net hardening rate with $\Theta \equiv d\sigma/d\varepsilon$ (the counterpart to $\theta \equiv d\tau/d\gamma$)

$$\Theta = \Theta_0 - R_t/\dot{\varepsilon} \quad (17)$$

If Θ approaches zero, deformation takes place under constant stress and constant strain rate i.e. under steady-state conditions. In this limit Eq. (17) degenerates to $\dot{\varepsilon} = R_t/\Theta_0$. Since, as will be seen, Θ_0 can be considered constant, R_t is expected to depend on flow stress by a power relationship just like the steady-state strain rate does. Thus, in the framework of this approach, the flow stress enters the above equation in the form

$$\Theta = \Theta_0 - R_{t0}(T)\sigma^n/\dot{\varepsilon} \quad (18)$$

where R_{t0} is a constant.

This equation might be adequate for the description of creep at high temperature with the power n close to $n=3$ [88–90]; in two fundamental aspects, however, it is in serious conflict with the observed work hardening behavior in the temperature regime discussed here. First, the equation demands a strong downward curvature in

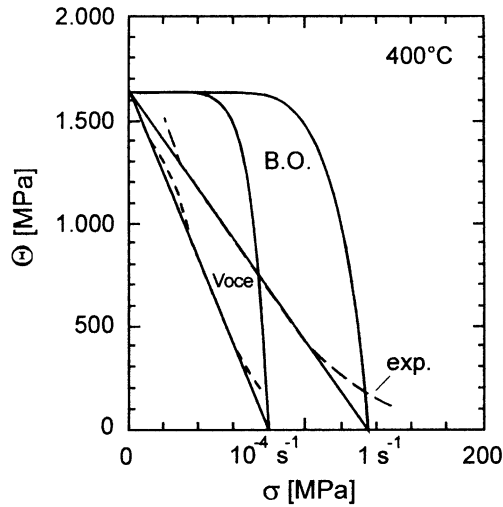


Fig. 17. Work-hardening rate versus flow stress. Curves for copper polycrystals (see Fig. 20) deformed at 400 °C at two different strain rates are compared with theoretical curves according to Eqs. (18) and (19).

plots of Θ vs. σ , implying an extended hardening range followed by a very rapid approach towards steady state, while in reality the transition is rather smooth in correspondence with the more linear decrease of the Θ – σ -curves. Second, the strain-rate dependence of the hardening coefficient is considerably weaker than suggested by Eq. (18) as becomes evident already from inspection of the strain-rate effects on the work-hardening characteristics presented in the foregoing chapter (Fig. 3).

These drawbacks of Eq. (18) for covering the general features of hardening behavior are demonstrated in Fig. 17 where theoretical Θ – σ curves, calculated with the help of Eq. (18), are compared with those actually measured on Cu-polycrystals at 400 °C for two different strain rates. The experimental stress exponent varies with temperature and is rather large, namely about $n=8$ at higher and $n=35$ at lower temperature. These values inserted into Eq. (18) lead to an almost step-like decrease of Θ with σ ; we used a case with $n=8$ for the demonstration at 400 °C. The actually observed curves decrease in a rather smooth fashion, namely, almost linearly with stress. The straight lines are linear approximations of the experimental curves (dashed) and follow an equation of the form [21,26–28]

$$\Theta = \Theta_0 - R_d \sigma / \dot{\epsilon}^{1/n} \quad (19)$$

where R_d and n vary with temperature but are independent of stress and strain rate.

Fig. 18 compares experimental stress–strain curves at room temperature and 400 °C with those computed on the basis of Eqs. (18) and (19) by integration. The integration constants were chosen such as to match the experiments in the range of low strains around $\epsilon=2\%$. In correspondence with Fig. 17 also in this presentation the transition from the steep initial part to the flat course at large strains is much too sharp when Eq. (18) is used, in particular at low temperature. The transition

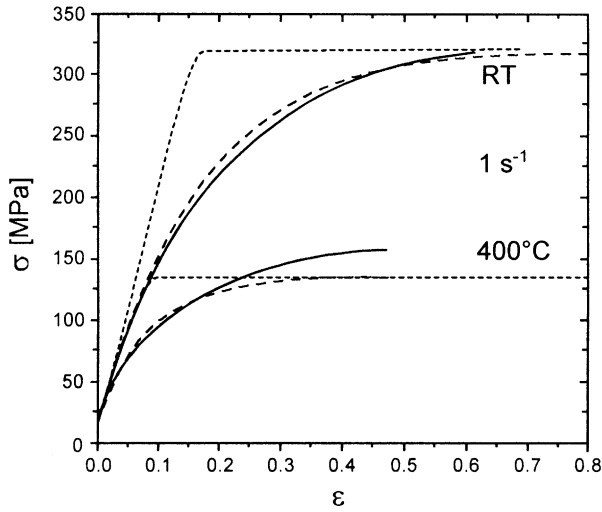


Fig. 18. Stress–strain curves at room temperature and 400 °C, calculated using Eqs. (18) (dotted) and (19) (dashed), are compared with experiments (solid) (see Fig. 19).

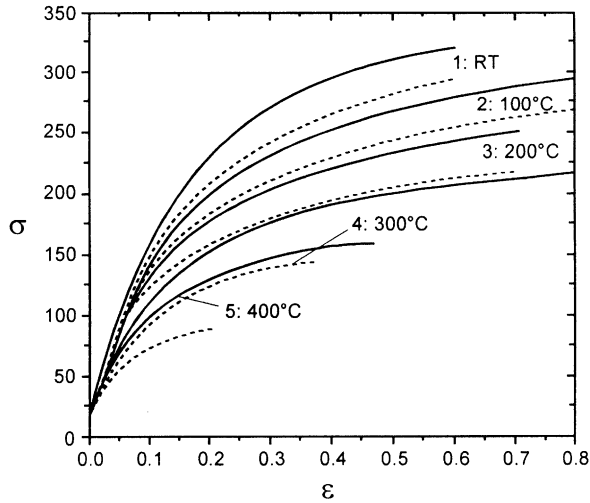


Fig. 19. σ – ε -curves of Cu-polycrystals for a wide range of temperatures and two strain rates, 1 s^{-1} solid and 10^{-4} s^{-1} dashed, from [47].

becomes smoother at the higher temperature, although it still remains too abrupt. This is contrary to the generally observed trend that the transient increases in its length i.e. becomes smoother with decreasing temperature. Eq. (19) does not suffer from these deficiencies but is in excellent agreement with experimental findings, where σ – ε curves are even more accurately approximated than Θ versus σ .

As the obvious result of this comparison, we like to emphasize that certainly for low and medium temperature Eq. (19) describes observed behavior very well, while

Eq. (18) totally fails even in covering general tendencies. Although both relationships lead to practically the same steady-state law at $\Theta = 0$, their physical basis is totally different. Dynamic recovery at low and medium temperature is thermally assisted strain softening governed by the glide of dislocations under the action of the applied stress; it is not time recovery driven by the flow stress and involving dislocation climb by some kind of diffusional processes. Merely from the analysis of steady state behavior it is difficult to distinguish between these two concepts. The flow-stress dependence of the work hardening coefficient, however, i.e. the approach towards steady state, allows such a distinction.

The general conclusions of our analysis are strongly supported by the experimental observations that were reported in Fig. 16: in a sample strained at 77 K, straining at room temperature produces remarkable softening, manifesting itself in a considerable reduction of the subsequent flow stress at 77 K; merely keeping the sample at room temperature for the same amount of time does not produce this result.

3.2. Scaling relationships for flow stress and work hardening

If Θ_0 were strictly a constant and if the hardening rate actually decreased with flow stress in a strictly linear fashion as proposed by Eq. (19) then, evidently, Θ – σ -curves in the whole range of temperatures and strain rates where this scheme applies would form a one-parameter set, in which the Θ -axis does not need any scaling except for the T -dependence of the shear modulus, but a scaling of the σ -axis would condense the whole set into a single curve. The stress scaling parameter can be the stress level at any arbitrarily chosen, but fixed value of Θ . In reality, strain hardening does not meet these idealized conditions and thus the question arises whether the above scaling rules hold also for experimentally observed Θ – σ dependencies. It will be seen that this is indeed the case for the overwhelming part of the hardening curves. To show this, Cu polycrystals shall serve as an example.

Fig. 19 displays some of the stress–strain curves measured in compression between room temperature and some of the 400 °C at four different strain rates between 10^{-4} and 1 s^{-1} [47]. In Fig. 20, the corresponding curves for the normalized strain hardening coefficient are plotted: Θ/μ is given as function of the normalized flow stress σ/μ . The curves are not straight, they rather exhibit tails with an upward curvature at the low and the high stress end. The low stress regimes are not particularly important since they correspond to only a very small portion of the stress–strain curve, at the beginning around yielding up to strains less than 2% and can be ascribed to the effect of grain boundaries on dislocation storage (see Section 3.6) The tails at high stress span also a rather low stress range but corresponding strains are rather large.

In any case, the curves in Fig. 20 look like a single-parameter set, for the most part, much could thus be gained by scaling them. We shall use the (straightish) middle part of the curves in Fig. 20, which are assumed to be typical of stage III behavior. It turns out that it is possible to scale the Θ -axis by a single value, Θ_0 , independent of the strain rate, and dependent on temperature only through the shear modulus, μ ; this simplifies the derivation of a scaling factor for the stress axis.

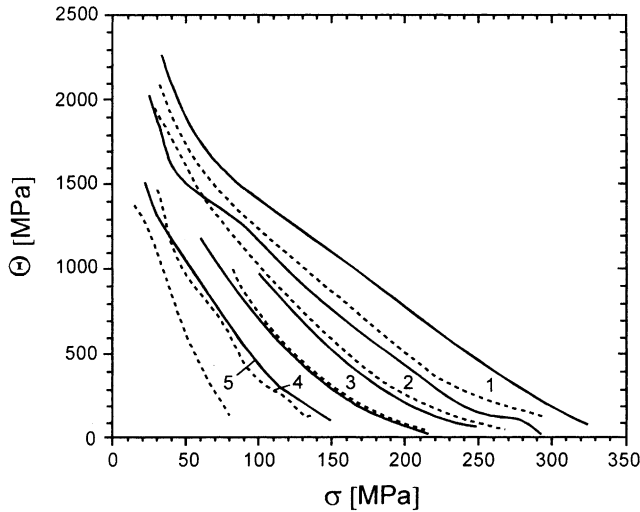


Fig. 20. Θ – σ -curves corresponding with the stress–strain curves in Fig. 19 for two strain rates (solid and dashed lines).

In principle this scaling stress could be chosen as the stress level for any Θ/μ -value of this regime and a convenient value would be just in the middle, i.e. at $\Theta_0/2$ [24]. An alternative procedure, which we have repeatedly employed in the literature, is to approximate each Θ/μ – σ/μ curve by a tangent to the rather straight middle part with a fixed value for the intercept on the ordinate (or at the yield stress), at Θ_0/μ . The intercept on the σ -axis is then used as the scaling stress. Inasmuch as this straight line—if it held true for the entire regime—would correspond to the Voce law

$$\Theta = \Theta_0 \left(1 - \frac{\sigma}{\sigma_v} \right) \quad (20)$$

we identify this particular scaling stress as σ_v . (Please note: the subscript in σ_v is not to be confused with the roman numeral for 5.)

Fig. 21 displays replots of the curves in Fig. 20, but now in a scaled form as Θ/μ versus σ/σ_v . It is seen that all curves fall into a very narrow band where the width of the band is identical with the experimental uncertainties of an individual curve. The set of curves for all temperatures and strain rates forms a single master curve for the dependence of the hardening rate on the flow stress.

To re-iterate: the Voce law does not fit the stress–strain curves over the entire regime; it does give a reasonable fit over a significant range of Θ/Θ_0 and—more importantly—provides a convenient method for arriving at a scaling stress. A general form for the hardening law would then look like

$$\frac{\Theta}{\Theta_0} = E \left(\frac{\sigma}{\sigma_v} \right) \quad (21)$$

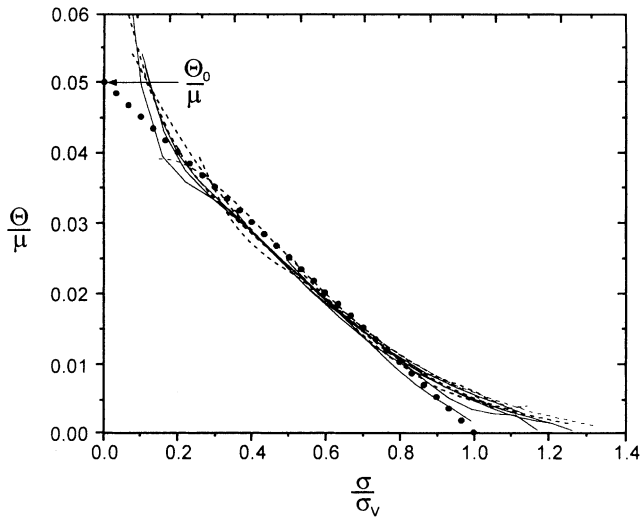


Fig. 21. Normalized Θ – σ -plots for five temperatures from RT to 400 °C at the two strain rates, 10^{-4} and 1 s^{-1} . The dotted line is the Voce approximation with $\Theta_0/\mu = 0.05$.

The arbitrary function $\mathbf{E}$ must, in principle, be determined for each case. To the extent that it has any generality for a wide set of temperatures and strain rates (and perhaps even for a whole class of materials), it may be called a *master curve*.

Phenomenological approximations to this function, which differ from the Voce law, have been proposed. As an example, a particular one that has worked well in some cases is [91]

$$\frac{\Theta}{\Theta_0} = \left(1 - \frac{\sigma}{\kappa\sigma_v}\right)^\kappa \quad (22)$$

with the adjustable parameter $\kappa \approx 1.3$. It is shown in Fig. 22, along with the experimental curves of Fig. 21.

In summary, there is no doubt that the Θ – σ curves for different temperatures and strain rates form a one-parameter set of curves. Thus, the hardening behavior can be predicted from a knowledge of only two parameters of the master curve where one is the characteristic strain hardening value Θ_0 the other the ‘scaling stress’ chosen here as σ_v . Θ_0/μ is a true constant while the scaling stress $\sigma_v = \sigma_v(\dot{\epsilon}, T)$ —which is also proportional to $\mu(T)$ —depends in addition on strain rate and temperature through thermal activation. Details of the nature of the recovery mechanisms can be deduced from the temperature and rate dependence of σ_v , which will be the focus of the next section.

3.3. Effect of rate and temperature on strain hardening

In the 1960s and 1970s, in many investigations on single crystals, detailed evaluations of the effect of temperature and strain rate on work hardening were carried out

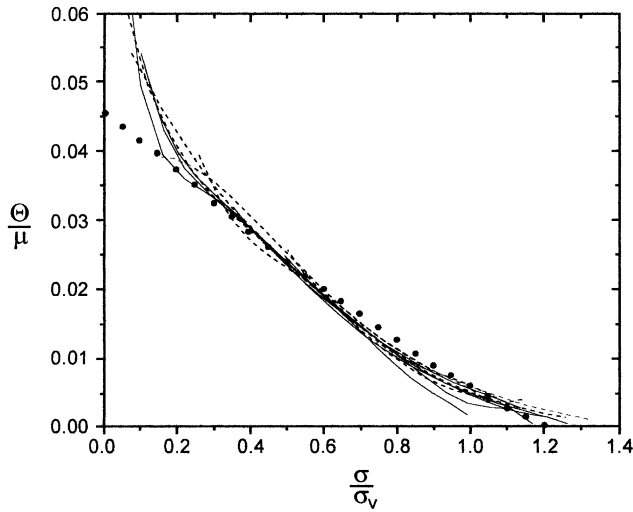


Fig. 22. The work hardening coefficient as function of the flow stress of Fig. 21 is approximated (dots) by Eq. (22) using the values: $\Theta_0/\mu = 0.045$ and $\kappa = 1.3$.

using the so called τ_{III} -analysis introduced by Haasen, Seeger and others [55,56,92]. Here, τ_{III} marks the stress level at the beginning of stage III and the analysis focuses on the effect of thermal activation on τ_{III} . Mecking and Lücke [22–24] have shown that the formalism used in the τ_{III} -analysis does not only hold for the beginning of stage III but can be extended to the whole hardening curve since the relative rate and temperature sensitivity of the stress for any fixed value of θ/θ_{II} was found to be a constant for a given rate and temperature, i.e. not to vary along a hardening curve. This rule was found to hold not only for single crystals but for polycrystals as well and it can be written as

$$\left. \frac{\partial \ln \sigma}{\partial \ln \dot{\epsilon}} \right|_{\theta/\theta_0, T} = \frac{1}{n} \quad \text{and} \quad - \left. \frac{\partial \ln \sigma}{\partial \ln T} \right|_{\theta/\theta_0, \dot{\epsilon}} = \frac{1}{n^*} \quad (23)$$

since θ_{II} is equivalent to Θ_0 introduced in Fig. 22. Thus the above equations are a specific way of expressing the scaling rule derived in the last section; namely that $\Theta/\mu - \sigma/\mu$ plots represent a one-parameter set of curves of a specific form in which only the σ -axis requires scaling by a characteristic stress level whose rate and temperature dependence is then characteristic for that of the whole hardening curve. The characterization by a certain level of Θ/Θ_0 has the advantage of being more well-defined than ‘the beginning’ of stage III (which, in polycrystals, practically coincides with the yield stress). The Voce stress, σ_v , introduced in the last section, is another well-defined scaling stress.

The fact that the derivatives defined above do not vary with deformation (i.e. with the value of Θ/Θ_0) points to a striking similarity with the so called Cottrell–Stokes law [93], according to which the relative rate- and temperature-sensitivity of the flow stress for a given structure (i.e. a fixed dislocation density ρ) do not vary with

deformation (i.e. do not vary as that density changes). The corresponding relationships are

$$\left. \frac{\partial \ln \sigma}{\partial \ln \dot{\epsilon}} \right|_{\rho, T} = \frac{1}{m} \quad \text{and} \quad - \left. \frac{\partial \ln \sigma}{\partial \ln T} \right|_{\rho, \dot{\epsilon}} = \frac{1}{m^*} \quad (24)$$

where the values for m and m^* are usually determined from incremental changes of strain rate (m) and temperature (m^*).⁴ The actual values for n as those for m are found to vary roughly linearly with $1/T$ but only very weakly with $\dot{\epsilon}$, while on the contrary n^* as well as m^* depend only weakly on T . Note that $n/n^* = Q/kT$ is the normalized apparent activation energy for dynamic recovery while m/m^* is that for plastic flow at constant structure (to be precise: Q is the activation enthalpy)

In the conventional τ_{III} -analysis the rate and temperature dependence of τ_{III} is evaluated on the basis of an Arrhenius equation with a logarithmically stress dependent activation energy which in terms of σ_v can be written as

$$\Delta G = A \cdot \ln(\sigma_{v0}/\sigma_v) \quad (25)$$

where A is assumed to be a material constant [40, 94], which depends on the relative stacking fault energy. If (25) were true, experimental data would fall on a straight line in a diagram of $\ln \sigma_v$ vs. $\Delta G = kT \ln(\dot{\epsilon}_0/\dot{\epsilon})$.

Fig. 23 displays a set of data on copper polycrystals in this way, with $\dot{\epsilon}_0 = 1 \text{ s}^{-1}$; that is, just with $\dot{\epsilon}$ measured in units of s^{-1} . There are two things wrong with such a

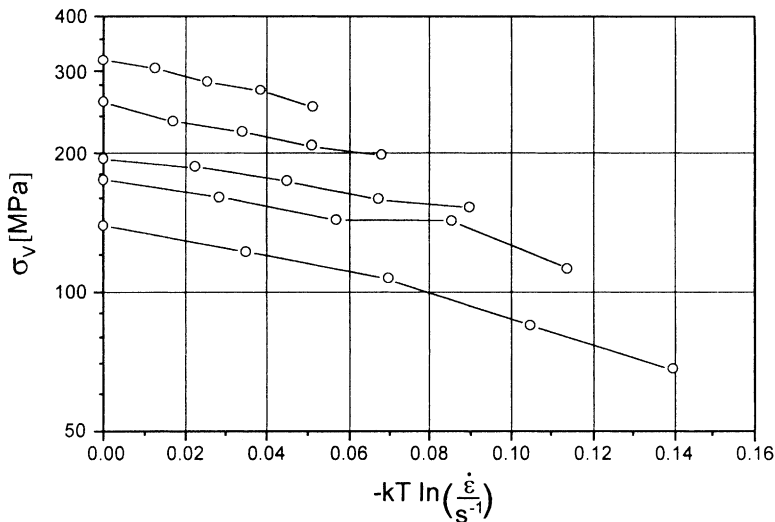


Fig. 23. $\text{Log}(\sigma_v)$ vs. $-kT \ln(\dot{\epsilon}/\text{s}^{-1})$ for Cu-polycrystals. Points for the same temperature are connected. (Note that, for any $\dot{\epsilon}_0 \neq 1 \text{ s}^{-1}$, no point lies on the ordinate.)

⁴ Note that m is here chosen, like n , to represent the stress exponent and not, as often, its inverse, the rate sensitivity.

plot. First, the value of $\dot{\epsilon}_0$ in any physically based Arrhenius equation is never 1 s^{-1} ; it is in fact not very well known, except for a general order of magnitude. A good way to choose it is such as to unify the experimental T - and $\dot{\epsilon}$ -dependence provided the order of magnitude comes out as expected. The second shortcoming, already inherent in Eq. (25), is that all flow stresses are expected to be proportional to the shear modulus—which itself depends on temperature in a non-trivial way; the same effect leads to a proportionality of ΔG to μb^3 [3,25,95, Section 2.3.2]. Finally, since it is already evident in Fig. 23 that the logarithmic dependence of the activation energy on stress is not exact over the whole stress range since in particular at low stresses the lines are not straight.

A more general form will be employed in the following, with an unspecified functional dependence of the activation energy on stress:

$$\frac{\Delta G}{\mu b^3} \equiv \frac{kT}{\mu b^3} \ln \frac{\dot{\epsilon}_0}{\dot{\epsilon}} = g \left(\frac{\sigma_v}{\sigma_{v0}} \frac{\mu_0}{\mu} \right) \quad (26)$$

If, as required by (26), $\Delta G/\mu b^3$ is a sole function of σ/μ then a value $\dot{\epsilon}_0$ must exist that brings all data of Fig. 23 on a unique curve in a plot of σ/μ versus $kT/\mu b^3 \ln(\dot{\epsilon}_0/\dot{\epsilon})$. The optimum value for $\dot{\epsilon}_0$ that fulfills this requirement within the lowest scatter is $\dot{\epsilon}_0 = 10^7 \text{ s}^{-1}$, as determined by trial and error (and in agreement with the order of magnitude expected from dislocation theory). The corresponding plot of the data of Fig. 23, displayed in Fig. 24, clearly shows that Eq. (26) describes the experimentally determined σ_v -values with rather high accuracy. Then the curve followed by the data points is the actually measured dependence of the normalized activation energy on the normalized stress σ_v/μ .

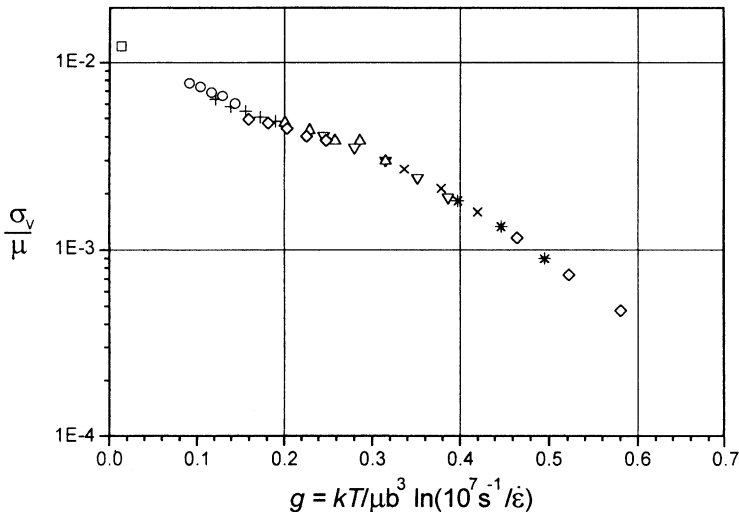


Fig. 24. A plot of $\log(\sigma_v/\mu)$ vs. $kT/\mu b^3 \ln(10^7 \text{ s}^{-1}/\dot{\epsilon})$, verifying the validity of Eq. (26). Here, compared to Fig. 23, additional data for higher temperature are included [47].

For completeness, Fig. 25 plots the same stress data as a function of temperature only, for several strain rates. It is this ‘first-order’ diagram that has been universalized in Fig. 24 by incorporating the strain rate in accordance with an Arrhenius law of thermal activation.

Once the dependence of the strain-hardening rate on the flow stress on the one hand, and the dependence of the scaling stress on temperature and strain rate on the other, have been determined experimentally, they can be used to predict stress–strain curves for any desired temperature and strain rate (within the range of validity of the relations). The most important result of the analysis of the last two sections is that the complete information is concentrated in only two master curves, $\Theta = \Theta(\sigma/\sigma_v)$ and $\sigma_v = \sigma_v(T, \dot{\epsilon})$; these are displayed in Figs. 21 and 24.

Much as in the case of the hardening law, so in the thermal activation description, the plot that unifies the data is sometimes approximated as a straight line—and here, too, this does not hold over the whole regime. Wherever it is valid, Eq. (25) holds, and it can be rewritten as

$$\frac{\sigma_v}{\mu} = \frac{\sigma_{v1}}{\mu_1} \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_1} \right)^{KT/A} \quad (27)$$

where σ_{v1} , $\dot{\epsilon}_1$, and μ_1 are a set of values in the region where the straight-line approximation holds. Inverted, this is the equation for power-law creep—with the important proviso that the creep exponent n depends in a continuous way on the temperature. This statement is entirely equivalent to an activation energy that depends in a continuous fashion on stress. The fact that the data in Fig. 21 do not lie on a straight line could be interpreted as a temperature dependence of A [92] (due to

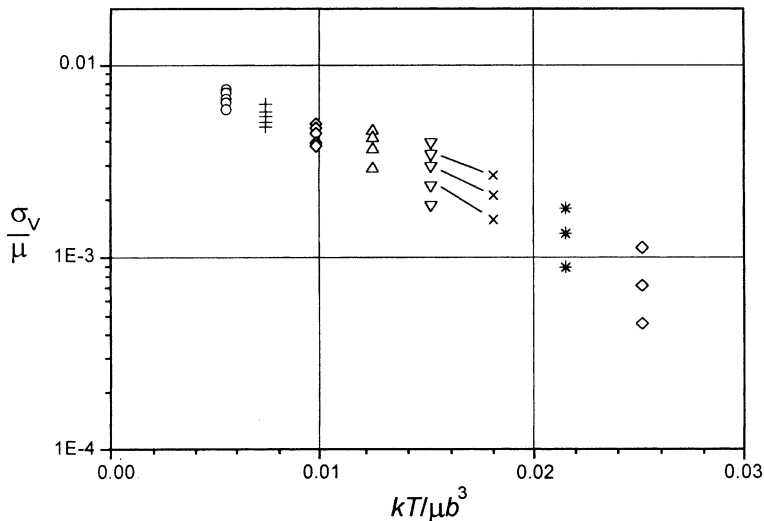


Fig. 25. Same data as in Fig. 24 plotted as a function of the normalized temperature for 3–5 strain rates between 1 s^{-1} and 10^{-4} s^{-1} .

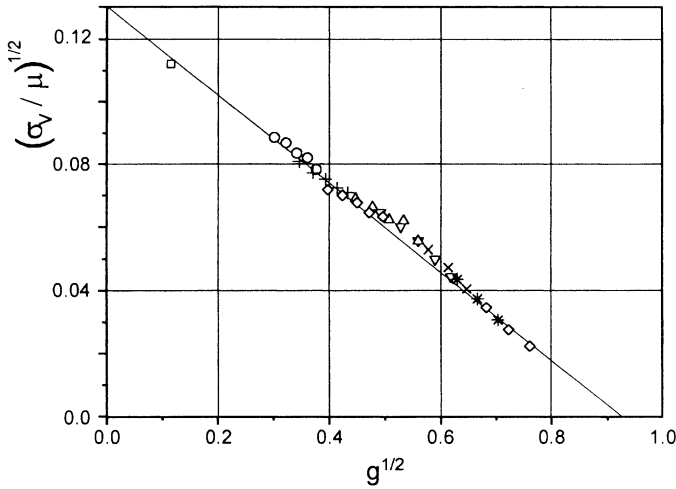


Fig. 26. Same as Fig. 24 but with coordinates according to Eq. (29), square root of σ_v/μ versus square root of g .

a temperature dependence of the stacking fault energy in addition to its implied temperature dependence through μ). At this point, the description (27) loses its elegance.

A systematic search for other algebraic approximations to the data (for many materials) has led to the conclusion that the phenomenological equation [25]

$$\Delta G = F_0 \left\{ 1 - \left(\frac{\sigma}{\sigma_0} \right)^p \right\}^q \quad (28)$$

where F_0 is the total activation energy at zero stress and σ_0 the maximum stress at zero temperature, fits these data best with $p = 1/2$ and $q = 2$. It is shown in Fig. 26 for the same copper data.

To the extent that a straight line is accepted as a good representation of these data,⁵ the relation

$$\frac{\sigma_v}{\mu} = \frac{\sigma_{v0}}{\mu_0} \left\{ 1 - \left(\frac{1}{g_0} \frac{kT}{\mu b^3} \ln \frac{\dot{\epsilon}_0}{\dot{\epsilon}} \right)^{1/2} \right\}^2 \quad (29)$$

which includes the proper normalizations and the abbreviation $g_0 \equiv F_0/\mu b^3$, can be used as a more accurate (though less convenient) analytical description than (27) of the function $\sigma_v(T, \dot{\epsilon})$. However, neither one is expected on the basis of any first-principles theory: only the parameter combinations embodied in the coordinates are. Since an accurate and general form for these dependencies is not ever likely to be

⁵ It is conceivable that the slight ‘bump’ in the middle of the plot is due to dynamic strain aging in this OFE copper.

found we continue to suggest as the best method for constitutive descriptions to employ two ‘master curves’ for each material, namely one for the dependence of the hardening coefficient on the flow stress and the other for the dependence of the scaling stress on temperature and strain rate.

The prime advantage of a well fitting equation like (29) is that the parameters in it (which depend on material) can be determined in a reproducible way; for example by extrapolating to the axes in Fig. 26. One must bear in mind, however, that the extrapolated values themselves, such as g_0 , depend on the particular exponents chosen for the plot and really should not be interpreted as having a physical meaning beyond the fact that they describe the measured range well. We would also warn against searching for a ‘best fit’ by computer to (29), letting the three parameters float (or even five, if p and q were to be let go—always assuming that measured values are used for $\mu(T)$, or a separate best fit to modulus data). The value of $\dot{\epsilon}_0$ can be measured, to sufficient accuracy from overlapping data on the T - and $\dot{\epsilon}$ -dependence. The value of g_0 is subject to the constraint that it should be the same for any particular material. Only the value of σ_{v0} is ‘adjustable’ to a particular set of test samples.

3.4. Material scaling

3.4.1. Pure fcc metals

In a study of work hardening in single crystals, of various fcc metals, all with the same orientation for multiple slip, Mecking et al. [40] have found, that appropriate scaling of the stress and of the temperature unifies the τ – T -dependence of all investigated materials into a single master curve. The analysis, however, was incomplete in so far as the rate dependence could not be included because all data were for only one strain rate except for Al. In what follows we will process data on polycrystals by applying a similar scheme as for the single crystals, we will include, however, the rate dependence from the beginning.

It has to be noted that a different extrapolation routine was used in the single crystal study to determine the stress at $\theta=0$. In the present study, the values for the Voce stress have been corrected for non-dislocation contributions to the flow stress. To this effect the σ_v -data have been determined from plots of θ versus $\sigma-\sigma_Y$, where the σ_Y -values were found by an optimization procedure: each θ – σ curve was shifted along the stress axis until the tangent of the (straight) middle part intersected the abscissa at θ_0/μ , a common value for a whole set of curves for one material; the tangent then intersects the new stress axis at σ_v . Generally the corrections were quite small—within the size of the symbols of the graphs for σ_v —and thus they are of no particular relevance within the present context.

Fig. 27 is a replot of Fig. 24 for Cu, where data for Ag-, Ni- and Al-polycrystals are added, obtained at different temperatures for various strain rates (data for Ag from [96] full points, [97] crosses and [98] open square). Using the same value for $\dot{\epsilon}_0$ as for Cu in Fig. 24 makes the data gather on separate curves, one for each material. Each individual curve, then can be interpreted as the σ_v – ΔG -dependence for that material. We have presupposed already in Fig. 27 that $\dot{\epsilon}_0$ is an universal constant for all the considered materials, which might not be exactly true but the data base for

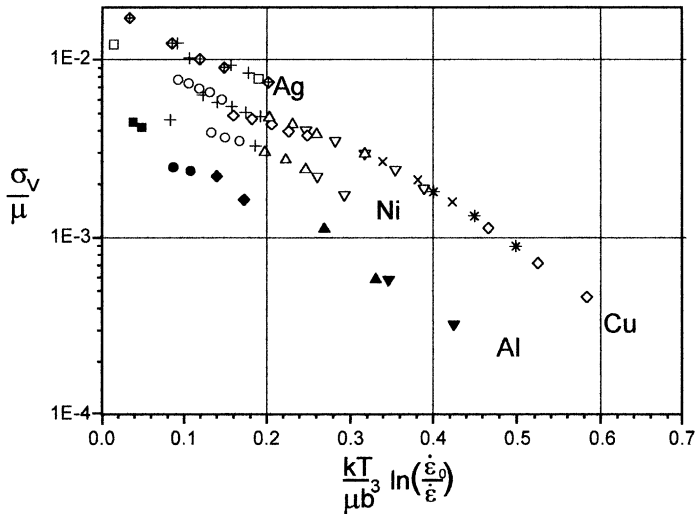


Fig. 27. $\text{Log}(\sigma_v/\mu)$ vs. $g = (kT/\mu b^3) \ln(\dot{\epsilon}_0/\dot{\epsilon})$ for Ag-, Cu-, Ni-, and Al-polycrystals [27,47,48]. The symbols refer to a certain temperature except for Ag [96–98] where they denote their source (see text).

Ag, Ni and Al is not sufficiently complete to prove the accuracy of this assumption. If, however, $\dot{\epsilon}_0$ would be material dependent, this would not influence the general result of this investigation and also would have practically no effect on the validity of the final conclusions.

Fig. 28 exhibits the same data as Fig. 27 by employing, on the basis of Eq. (29), coordinates which were also used in Fig. 26. It is seen that, as well as for Cu, also the curves for the other metals are straight over a considerable range and the intercepts with the axes determine the values for g_0 and σ_{v0} in Eq. (29) which are characteristic for each material. It appears that a correlation exists between the two intercepts, so that this is actually a one-parameter set of curves (see also Fig. 30). We have interpreted both in terms of the normalized SFE [40]. If this were generally applicable, one could have a single master curve (or even a closed-form equation) for all fcc materials. This is shown in Fig. 29, where also the values obtained for g_0 and σ_{v0} are listed together with the reduced stacking fault energy $\chi/\mu b$.

As already pointed out at the beginning of this chapter; the real physical quantity to be considered is the resolved shear stress τ_v , since the normal stress σ_v is influenced by the Taylor factor M . This is not known exactly since the textures have not been measured for all materials considered in Figs. 27–29. So far we have ignored in the above evaluation possible differences of M , implicitly assuming, that fluctuations from one material to the other did not vary systematically with the sequence of the stacking fault energy. In view of the conventional thermomechanical treatment of the investigated samples, we assume that actually existing textures were rather weak. Therefore it seems safe just to use the Taylor factor $M \sim 3$ of a random orientation distribution, for converting σ_v into $\tau_v = \sigma_v/3$ (see also Fig. 32). As will be seen in Section 3.6, by straining in different deformation modes, the texture evolves in such

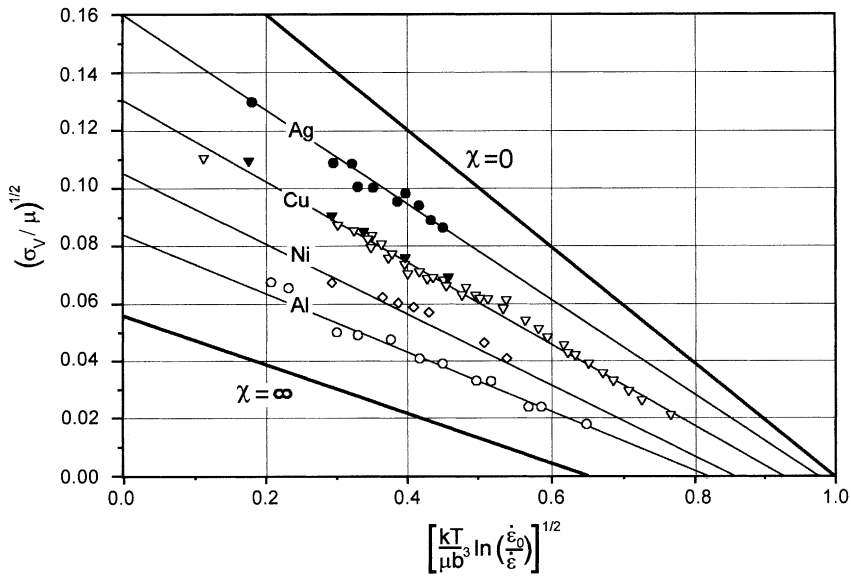


Fig. 28. Same as Fig. 27 but $\sqrt{\sigma_v/\mu}$ versus $\sqrt{g}$ according to Eq. (29), where the solid triangles for Cu have been added. They are from the torsion tests reported in Fig. 39: applied shear stress has been converted to normal stress with the help of the von Mises relationship. Here like in preceding figs. at intermediate g -values the data exhibit a small hump which is probably due to strain aging caused by impurities. The two border lines for the values zero and infinity of the stacking fault energy are obtained by the extrapolation method displayed in Fig. 30.

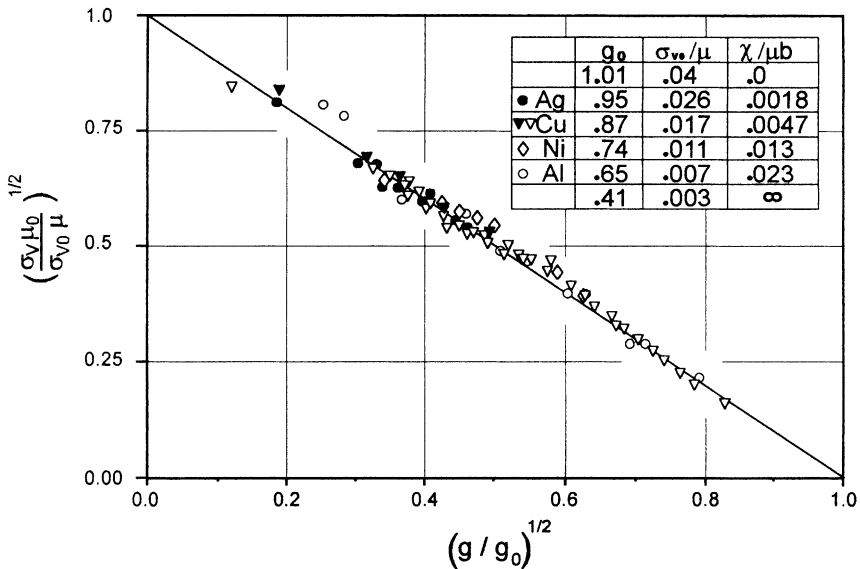


Fig. 29. Same as Fig. 28 but coordinates reduced by the intercepts σ_{v0}/μ_0 and g_0 . The values used are given in the insert together with the values of the relative stacking fault energy from [100], merely for Ni from [101].

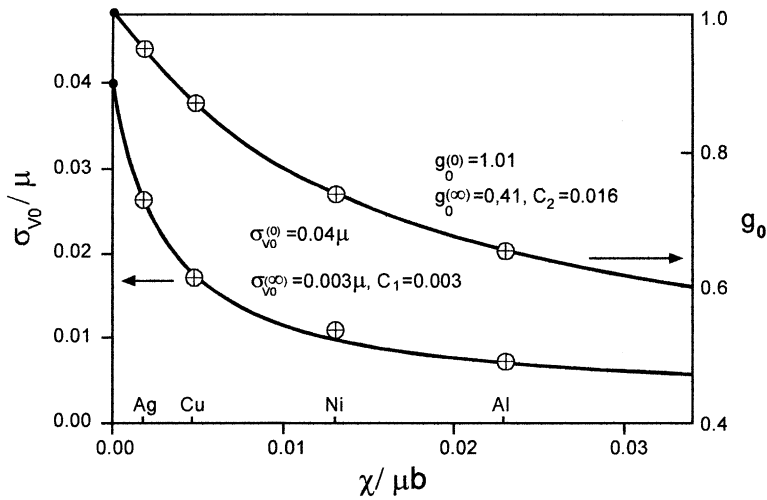


Fig. 30. The data listed in Fig. 29 plotted as σ_{v0} and g_0 versus the relative stacking fault energy $\chi/\mu b$. The dashed lines are least square fits on the basis of the formula described in the text. The optimum values of the fitting parameters are given in the figure.

a manner that the corresponding Taylor factors differ by less than 20%. This is also evident in Fig. 28, where the data for Cu obtained in compression and in torsion tests are virtually the same. Occasionally, however, larger differences have been reported such as in Ref. [99], where the Voce stresses obtained in tension were about 50% higher than those for Cu in Figs. 27 and 28, but they extrapolate to the same value of g_0 .

From these results, it would appear safer to relegate σ_{v0} to a quantity that needs to be determined in each case. The value of g_0 , on the other hand, may well be a constant for each material. This would seem reasonable in view of the fact that it relates to a local activation process; it is expected to depend on physical properties like the SFE but, unlike σ_{v0}/μ , not on microstructure, texture, etc.

The materials of the present analysis cover a wide range of stacking fault energies [100,101] and thus the results may be utilized to predict the master curve for any other fcc material with known stacking fault energy by a suitable method of inter- and extrapolation resp. For this purpose the values of the table inserted in Fig. 29 are displayed in Fig. 30. The observed dependencies of σ_{v0}/μ and of g_0 on the relative stacking fault energy, $\chi/\mu b$, are qualitatively the same for both cases. They start out with a rather rapidly decreasing branch and then flatten gradually towards a horizontal course at large values of $\chi/\mu b$.

At least qualitatively, such asymptotic behavior is expected in view of the atom arrangement in extended dislocations: above a certain level of $\chi/\mu b$ the width of the stacking-fault ribbon falls below one atomic distance, from where on the atom arrangement in the dislocation core remains practically unaffected by a further increase of $\chi/\mu b$. Thus we assume that in Fig. 30 σ_{v0}/μ as well as g_0 gradually vary with $\chi/\mu b$ between the two limits: $\sigma_{v0}(0)/\mu$ and $g_0(0)$ for $\chi \rightarrow 0$ and the asymptotes

for $\chi \rightarrow \infty$ at $\sigma_{v0}(\infty)/\mu$ and $g_0(\infty)$. We are not aware of any theoretical method to estimate any of these constants, they rather must be determined experimentally.

The types of formula which capture the described interdependencies are

$$\frac{\sigma_{v0}(0)}{\mu_0} - \frac{\sigma_{v0}(\chi)}{\mu_0} = \left[\frac{\sigma_{v0}(0)}{\mu_0} - \frac{\sigma_{v0}(\infty)}{\mu_0} \right] \left[1 + c_1 \frac{\mu b}{\chi} \right]^{-1}$$

and

$$g_0(0) - g_0(\chi) = [g_0(0) - g_0(\infty)] \left[1 + c_2 \frac{\mu b}{\chi} \right]^{-1}.$$

They were found by a search for presentations in which the data follow straight lines, which is the case in plots of $1/[\text{const.} \cdot \sigma_{v0}/\mu]$ —and also $1/[\text{const.} \cdot g_0]$ —versus $\mu b/\chi$ if the empirical constants are adequately chosen. By eliminating χ the above equations provide an analytical expression for the interdependence between the material constants σ_{v0} (representing τ_{v0} and g_0 , which are not independent quantities, rather they are interconnected by the stacking fault energy as the only (though hidden) parameter. The utilized equations are purely heuristic and have been introduced here for the mere purpose of determining the functional dependence of σ_{v0} on χ , and of g_0 on χ , in a reproducible way.

The parameters in the above functions were fitted to the data in Fig. 30, with the help of a least-square routine, and the obtained curves are drawn as solid lines in the figure, where also the optimum values for the constants are given. With their help any $\sigma_{v0}/\mu - g_0$ pair and the corresponding master curve in Fig. 28 can be determined for any value of $\chi/\mu b$. The master curves for $\chi=0$ and $\chi=\infty$ found in this way are inserted in Fig. 28; they demarcate in $\sigma_v - g$ -space the territory where the master curves of all fcc metals (with any possible stacking fault energy) will be located.⁶

According to this evaluation is the stacking fault energy the material parameter that controls the level of normalized stresses, to which a material can be work hardened at a given temperature and strain rate. Thereby the differences between different materials can differ by more than one order of magnitude, as seen from the ratio of the limiting values of the Voce stresses at zero temperature, $[\sigma_{v0}(0)/\mu_0]/[\sigma_{v0}(\infty)/\mu_0] = 13$. Thus the stacking fault energy is crucial for the performance of a material in practical applications. E.g. due to its low SFE, cold worked Ag is about four times stronger than cold worked Al, although their elastic moduli are virtually the same. The $\chi/\mu b$ -values of many pure fcc metals are close to or even larger than that of Al [102] and so the corresponding master curves will gather between the Al-curve and the lower bound in Fig. 28.

3.4.2. Single-phase alloys, other lattice structures

At the end we like to have a quick look on the implications of this analysis for other material classes, bcc and hexagonal materials, by a short summary of the more

⁶ At low temperatures the master curves for low stacking-fault-energy materials may be somewhat affected by mechanical twinning. It occurs in Cu below about 100 K and in Ag below about 400 K at larger strains.

comprehensive description in [103] Solute additions have an impact on bulk parameters. While generally the shear modulus and the Burgers vector are changed only moderately by alloying, variations of the stacking fault energy can be very large. This effect is made use of in many technical alloys such as Cu-alloys (particularly α -brass and Cu–Al) [104], Ni–Co alloys [101] and austenitic steels (stainless in particular) [105]; they all have very low stacking fault energies. What matters in these systems is the indirect effect of the alloy additions: they increase the work hardening capacity and, in this way, generally improve the strengthening as well as the forming behavior in deep drawing of a material. They lead to an upward shift of the master curves which are located somewhere between the upper bound and, let's say, the Ag-curve in Fig. 28. The lowering of the stacking fault energy in combination with the increase of the general strength level promotes twinning as an additional deformation mode. Though this seems to have only a minor effect on the stress–strain curve [104] the influence on the microstructure and the texture is always quite large [66,106].

Of course, in many solid solutions the main effect of alloying is not the indirect one by the impact on the bulk parameters [107]. If the solute atoms possess a quite different size or a quite different chemical nature, as compared with the host atoms, they present strong local obstacles to dislocation motion and then they cause directly an increase of the yield strength. It has been found that, in many alloys, solute concentrations influence the entire stress–strain curve; for example in a multiplicative rather than an additive way [108]. Thereby the combined effect of the direct and indirect contributions may play a special role. The influence of solutes is particularly strong in the temperature regime of strain aging due to segregation on the moving and also on the accumulated dislocations (the origin of a higher yield stress and of a higher hardening rate). These effects become important at temperatures much lower than that for bulk diffusion; their defining characteristic is a negative contribution to the strain-rate sensitivity.

In many aspects *bcc metals* behave like fcc metals, provided the temperature is high enough, so that the strength is governed by dislocation–dislocation interaction just like in fcc metals. Pure bcc metals generally also possess rather high stacking fault energies and therefore the work hardening is in many cases comparable to that of Al in terms of reduced stress and reduced temperature. A totally different situation arises at low temperature where the lattice resistance is high. Within our framework, there would be two effects: (a) the “Peierls stress” must be subtracted from the observed flow stress to obtain the dislocation–interaction contribution to the flow stress; (b) the dislocations are straighter, since the edge parts move out rapidly, and therefore the areal-glide mechanism that forms the basis of the statistical dislocation-storage processes (Section 4) will be less effective; it may be possible to describe this simply by a lower value of θ_0 [109]. On the other hand, Kubin et al. [110] have postulated on the basis of dislocation-dynamics simulations, that strain hardening in bcc metals follow an entirely different mechanism.

Hexagonal metals represent a special case. Due to the low symmetry, the hexagonal lattice does not provide a sufficient number of crystallographically identical slip systems to fulfill the von Mises criterion for polycrystal deformation. In order to generate a set of five geometrically independent slip systems several different crystal-

lographic slip modes, with often quite different strength, must be activated. Therefore, the strength of polycrystals is generally controlled by the hard modes. The situation is opposite in unconstrained deformation of single crystals, where normally only soft modes are activated. Basically the same mechanisms as in cubic metals act in hexagonal material in a similar way; they are, however, not so easy to analyze. The individual slip modes can respond, quantitatively as well as qualitatively, quite differently to changes of external or internal parameters [107]. Furthermore, due to the high internal stress levels set up in polycrystals, twinning as an additional deformation mechanism comes into play in hexagonal metals much more frequently than in cubic metals [111,112].

3.5. *Single-crystal effects*

The division of stress–strain curves into different stages is based on extensive studies of single crystals deformed in single slip. The distinction of these stages has proven to be very useful since they each depend in a characteristic way on deformation conditions (temperature, strain rate) and also on crystal orientation. However, in polycrystals, stage I is absent and stage II often degenerates into a low-strain limit that is athermal (though not straight) as in single crystals. One of the questions we wish to address here is to what extent a distinct, linear stage II exists in single crystals, and whether this may be related to the presence of stage I.

Single crystals, deforming by single slip, start after yielding with a regime of low hardening rate, stage I, which is terminated by a steep increase of the hardening coefficient to level almost one order of magnitude higher than in stage I, actually, a factor of 4–5 (see Fig. 31). The hardening coefficient of stage II, θ_{II} , is athermal i.e. it is practically not influenced by temperature and strain rate, but it is sensitive to crystal orientation. There seems to be an interdependency between the extension of stage I and the slope in stage II in the sense that θ_{II} becomes the higher the shorter stage I is. The termination of stage I has been observed to be connected with the onset of considerable dislocation activities in secondary slip systems [46,74,75,113] although the contribution of secondary slip to the macroscopic strain remains negligibly small [22]. The interdependence between the extension of stage I and the hardening rate in stage II can be made out already from the very first Fig. 1 of this article, where stress–strain curves for Cu single crystals of various orientations are presented [17]. The length of stage I is smallest and at the same time the stage II slope is highest in orientations near the symmetry line (001)–(111) where a strong tendency for secondary slip is to be expected. No stage I at all develops in symmetrical orientations with multiple slip from the beginning of deformation which are known to exhibit the highest hardening rates.

Examples of these interrelationships are given in Fig. 31 by a set of θ – τ -curves for Ag single crystals of 3 different orientations for various temperatures [22,29,114]. The single slip orientation exhibits at low stress a steeply increasing branch, followed by a more smoothly decreasing branch at high stress with a maximum in between. The maximum corresponds to an inflection point in a stress–strain plot, and it is seen that at low temperature, where the maximum becomes rather flat, a linear part would be a good approximation for a wide range of the stress–strain curve in the vicinity of the inflection point.

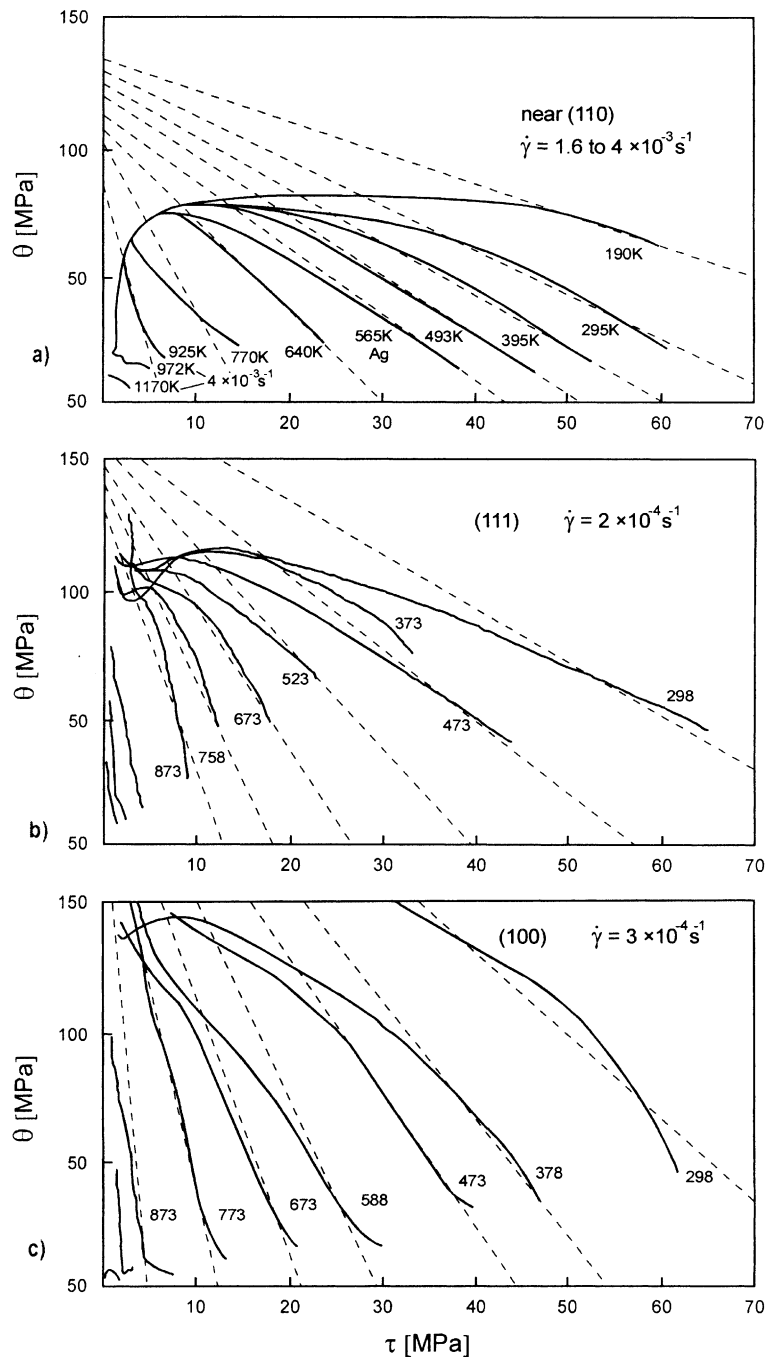


Fig. 31. θ - τ -plots at different temperatures for Ag single crystals of three different orientations, from top to bottom: single slip orientation near (110) initially, (111)-orientation and (100)-orientation.

The multiple slip orientations have a quite different appearance [115]. The curves for the (100) orientation decrease continuously from a high θ -level which is not affected by temperature within the limits of experimental accuracy. The (111)-orientation occupies an intermediate position. Some curves exhibit a flat minimum followed by a flat maximum, some show a continuous decrease. These qualitative differences are ascribed to a scatter of crystal orientations around the ideal (111)-position, leading initially to single slip or multiple slip depending on the actual deviation of a crystal orientation from the (111) direction. Apart from these details the initial part is insensitive to temperature also here.

In summary the common characteristic of all the θ – τ -curves is that they start out with an athermal part, which increases with stress when deformation is by single slip but which decreases when it is by multiple slip. A region of constant hardening rate is hard to differentiate in any of the curves in Fig. 31, with the exception of those for the single slip orientation at low temperature below room temperature.

With respect to stage II, we thus conclude that it may often be no more than an inflection point on the stress–strain curve between stages I and III; its value depends on stage I properties. However, at the low temperatures, the extent of a flat region in the θ – τ diagrams is so extensive that a value of θ_{II} can be reliably identified.

While the curves of the different orientations exhibit large differences at the beginning of deformation in the athermal stage II, they develop close similarities in the succeeding temperature dependent stage III, where all orientations approximately follow Voce-type behavior, in the sense that in this regime each curve develops an approximately straight portion for every temperature. It is seen that in this regime the experimental curves can be reasonably well approximated by a set of straight lines, which are drawn in such a way that the intercept with the θ -axis is at a fixed value of θ_0/μ which was chosen as 1/200 for the single-slip-, 1/150 for the (111)- and 1/100 for the (100)-orientation. Note that these values of θ_0 are all significantly higher than those for θ_{II} .

The T - and $\dot{\epsilon}$ -dependence for each orientation can now be characterized by that of the scaling stress τ_v at $\theta=0$ and this is plotted in Fig. 32 as the normalized stress τ_v/μ versus the normalized activation energy g , using the square roots according to the empirical Eq. (29). It turns out that the τ_v - values for the single-slip- and (100)-orientation are quite similar but that the (111)-orientation shows some differences in the general level of τ_v .

With respect to polycrystal hardening we conclude from these observations that texture can affect τ_v on a grain level, at the most, within the range of the scatter band in Fig. 32. In view of the much wider distribution of crystallite orientations even in strongly textured polycrystals as compared to single crystals, however, we assume that the variations of τ_v due to variations of texture are very small in reality. When large differences of σ_v are observed, they generally are to be ascribed to differences in Taylor factors rather than in resolved shear stresses, as will be discussed in more detail in the next section. Another point is that the differences between the behavior of different crystal orientations are largest at the beginning of deformation where stage II processes dominate. We conclude from this observation that also in polycrystals variations in microstructure (texture, grain size and annealing state)

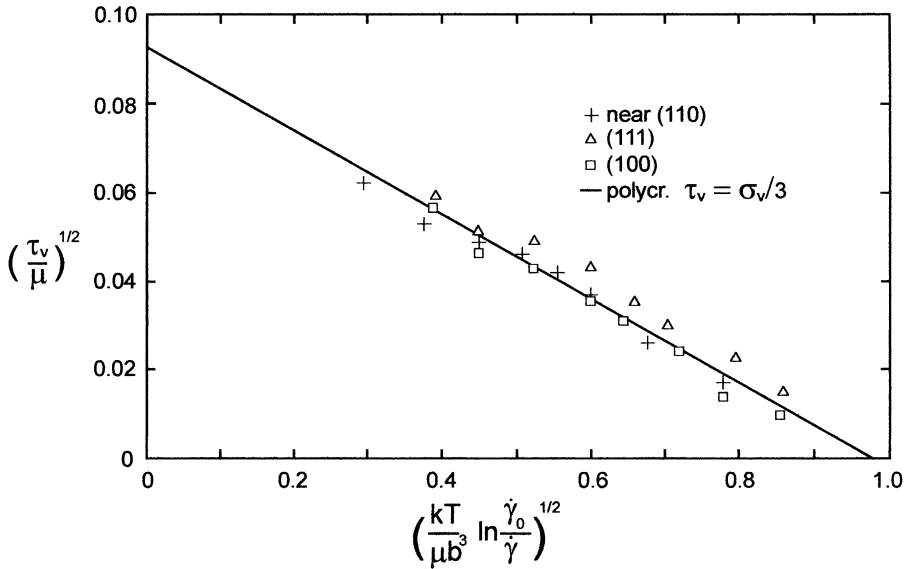


Fig. 32. Square root of τ_v / μ versus square root of $\Delta G = kT \ln(\dot{\gamma}_0 / \dot{\gamma})$ for the τ_v values of Fig. 31 with $\dot{\gamma}_0 = 3\dot{\epsilon}_0 = 3 \cdot 10^7 \text{ s}^{-1}$. For comparison with the polycrystal data the straight line for Ag of Fig. 28 is inserted in the above diagram, using $\tau_v = \sigma_v / 3$. The agreement with the single-crystal data is very good, confirming the value for g_0 given in the insert in Fig. 29 for Ag.

affect the hardening coefficient $d\tau/d\gamma$ much more at low strains than at high strain, since they influence the stage II processes much more than the stage III processes.

3.6. Polycrystal effects

The interpretation of polycrystal hardening on the basis of single crystal behavior needs two additional microstructural elements to be considered, namely the average grain size and the distribution of grain orientations i.e. the texture. It is well known that, at least for pure fcc materials, the dominant influence is by the texture rather than by the grain size. The main reason lies in the direct effect of the Taylor factor on stress and strain, which can be quite large [116], but also the indirect effect of texture on the hardening parameters has been assumed to be important [117]. In the following we will therefore treat first the texture effects, before considering modifications of flow stress and hardening behavior due to grain size.

If a strain increment $\delta\epsilon$ is imposed on a grain it responds with a certain shear increment $\delta\gamma(\mathbf{g})$ which is here meant to be the sum of all incremental shear contributions of the various slip systems which are activated in that grain and which depend on the position $\mathbf{g}$ of the grain orientation in orientation space. The ratio of these two quantities is the Taylor factor [118] $M(\mathbf{g}) = \delta\gamma(\mathbf{g}) / \delta\epsilon$. In the Taylor model the strain increment $\delta\epsilon$ is the same for all grains so that

$$M \cdot \delta \varepsilon = \delta \gamma \quad (30)$$

and because equality of external and internal work demands $\tau \delta \gamma = \sigma \delta \varepsilon$

$$M \cdot \tau = \sigma \quad (31)$$

(For multiaxial deformation one needs a tensor formulation [50].) Here τ is the average shear stress needed for crystallographic slip; M and $\delta \gamma$ are averages over all grain orientations [119,120] which, of course, depend on the texture of the material. Variations in M can be quite large as seen from the following example. For a (100)-oriented grain deformed in tension $M(100)=2.45$ while it is 50% higher for the (111)-orientation with $M(111)=3.67$. Drawn fcc material generally contains fiber textures with (100) and (111) being the fiber axes. The relative intensity of these two components depends on the thermomechanical history (and also alloying additions) so that by suitable control of processing parameters material with a strong (100), a strong (111) texture or any mixture of these two can be produced [121] for which quite different hardening curves are to be expected.

We have seen in the section on single crystals that $d\tau/d\gamma$ - τ -curves of (100)- and (111)-oriented single crystals exhibit some differences, particularly in the low-strain regime. These are, however, much lower than the differences between $d\sigma/d\varepsilon$ - σ -curves of the same crystals due to their different Taylor factors as demonstrated in Fig. 33, and our conclusion is: the often observed differences in hardening curves of

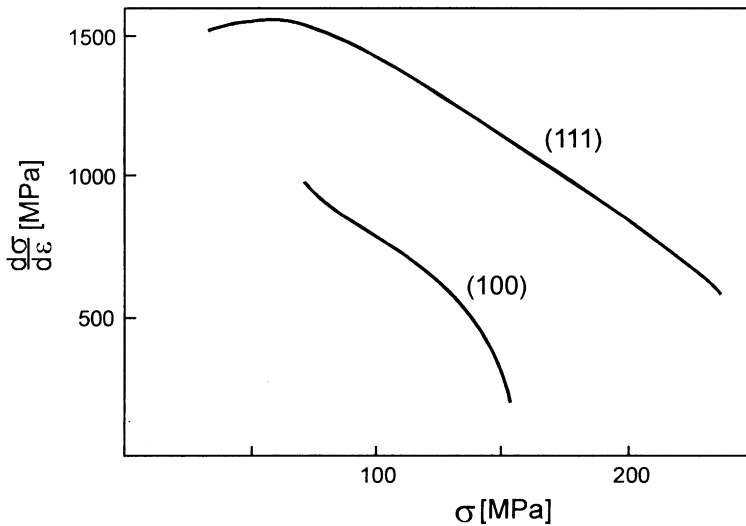


Fig. 33. $d\sigma/d\varepsilon$ - σ -plots for Ag single crystals with (100)- and (111)-orientation deformed at room temperature. In the corresponding $d\tau/d\gamma$ - τ -curves in Fig. 31(b) and (c) the hardening of (100) appears to be the stronger one at low strains, but since $d\sigma/d\varepsilon = M^2 d\tau/d\gamma$ and $\sigma = M\tau$, (111)-hardening is always the strongest in the above presentation. The two curves represent the expected range for textured polycrystals with different weightings of these two components.

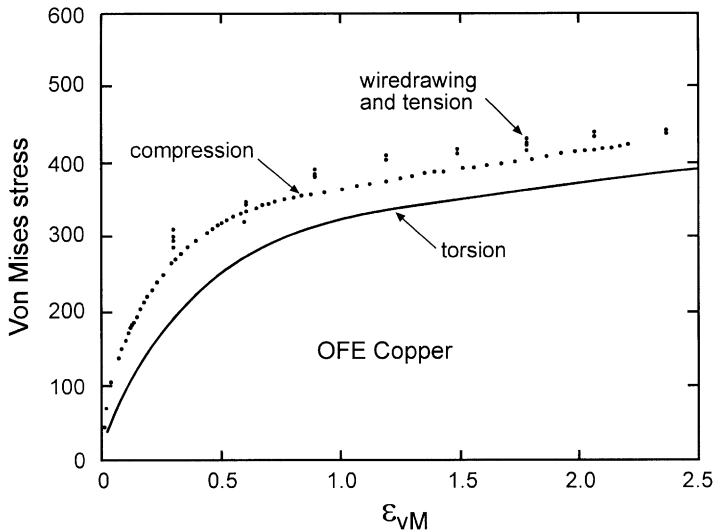


Fig. 34. Von Mises stress versus von Mises strain for OFE copper of medium grain size obtained in torsion tension and compression [122].

polycrystals, with the same composition and average grain size, are most likely caused by differences in Taylor factor due to differences in texture. As compared to this texture effect other possible influences on hardening behavior seem to be of second order and thus they can be neglected in most cases.

With this result in mind let us consider the influence of the deformation mode on hardening. Stress strain curves generated in different testing modes are usually compared by using von Mises stress $\sigma_{v,M}$ and von Mises strain $\epsilon_{v,M}$ as in Fig. 34, where for OFE copper curves are presented which were obtained in torsion and in the two uniaxial modes, tension and compression [122]. They exhibit substantial differences and, as will be seen, for the most part these can be ascribed to differences in texture evolution for the various modes.

Texture evolution and resulting changes of the Taylor factor are taken into account in the calculated curves in Fig. 35. They have been generated on basis of the Taylor model from the unique $d\tau/d\gamma - \tau$ -relationship, displayed in Fig. 36, by taking into account the continuous change of the Taylor factor as the simulated texture evolves differently along the different strain paths [118] (for a detailed description of the method for determining the optimum $d\tau/d\gamma - \tau$ curve the reader is referred to [116]). Although under these circumstances the relations between $\tau = \sigma/M$ versus $\gamma = \int M d\epsilon$ are the same for all the testing modes, in von Mises coordinates the calculated stress-strain curves in Fig. 35 spread apart just by the same amount as the experimental ones in Fig. 34. In the main this result has been confirmed by Kopacz and coworkers [123] in a similar study where Taylor factors determined from experimental textures, measured along the various strain paths, were used instead of computer simulation.

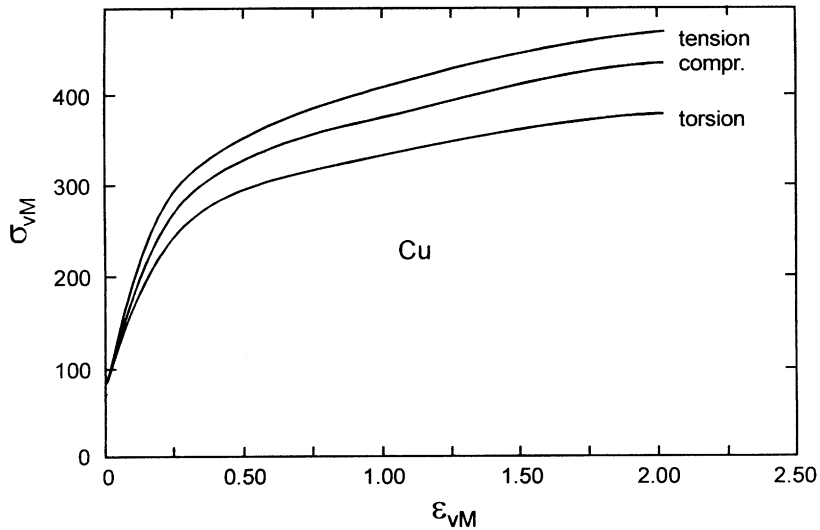


Fig. 35. Von Mises stress versus von Mises strain as calculated from the unique hardening relationship $\theta = d\tau/d\gamma$ versus τ of Fig. 36 by taking into account the continuous variation of the Taylor factor as the texture evolves differently along the different strain paths in torsion, tension and compression.

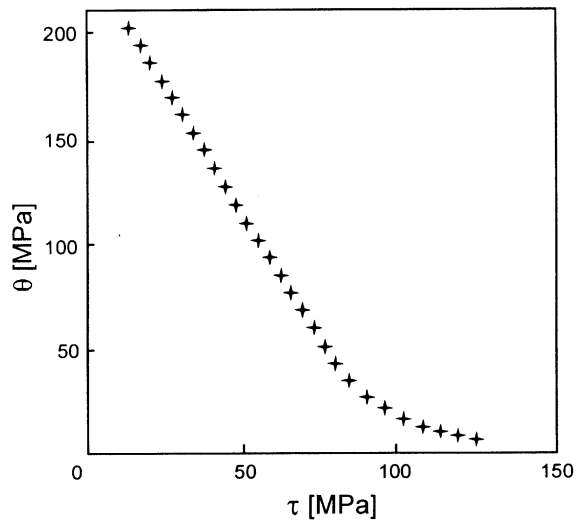


Fig. 36. The $\theta = d\tau/d\gamma$ – τ relationship that has been used to derive the stress/strain curves for the different deformation modes in Fig. 35. In Fig. 38 it is compared with the corresponding hardening curve of a rolled single crystal. (Note: compared to the original curve in Ref. [122] the origin of the abscissa is shifted by 10 MPa in order to allow the comparison with the single-crystal study, where also no corrections had been made for non-dislocation contributions to the yield stress.)

This comparison gives clear evidence that the variations of hardening behavior observed in various testing modes, to a large extent, can be accounted for by specific differences in texture evolution, i.e. the evolution of Taylor factors, although some deviations remain. These, possibly, are an indication of an indirect effect of the deformation mode on hardening parameters on grounds of the following arguments. In a polycrystal the selection of active systems varies from grain to grain [118,119] so that the evolution of the dislocation structure depends on the orientation of a grain [124]. Therefore the hardening is certainly not precisely the same for every grain in a polycrystal. In addition, the selection of active slip systems depends also on the prescribed strain path (Tome et al. [117]) so that the employed deformation mode might have an effect on hardening on a grain level that superimposes on the direct texture effect.

In a pure shear test, as in torsion, the number of active slip systems per grain, in the average, tends towards 2, to the same extent as a characteristic texture evolves with strain, the corresponding number is slightly below 3 in rolling, a little bit above 3 in axisymmetric tension and about 4 in compression [125]. Although it is not clear how the multiplicity of slip influences the resultant hardening, it would be very surprising if there were no effect at all, given the importance of local relaxation processes discussed in Section 2 and the orientation dependence of single-crystal stress–strain curves displayed in Fig. 31. The difficulty is that a correct evaluation requires the quantitative separation of the direct effect of texture which, in most cases, is much larger than the indirect ones, as exemplified in Fig. 33.

A similar situation arises in investigations of the dependence of strength and hardening on *grain size*. There are many effects of grain boundaries on plasticity. The primary one is that they serve as the mathematical boundaries where compatibility between neighboring grains must be enforced [50,126,127]. If this necessity is taken to be paramount (and more important than the enforcement of equilibrium) the Taylor model ensues. Any dislocations piling up at the boundaries before full-scale plasticity has been reached, serve to raise the stresses on other slip systems and thus lead to the ‘polyslip’ [126,128] characteristic of the Taylor model; they cannot then be used again as a cause for an increase in the yield stress of the Hall–Petch type [128].

As has been shown in Section 2.3.3, the capture probability of a piece of dislocation can be described by a mean free path, Λ . Grain boundaries could, in addition to their effect on the Taylor flow stress, influence strain hardening by an additional contribution to the storage rate, for example according to

$$\frac{d\rho}{d\sigma} = \frac{1}{bD} + \frac{1}{b\Lambda} - k(\dot{\epsilon}, T)\rho \quad (32)$$

where D is the grain size and the second term is proportional to the square root of the dislocation density [38,42,129] This relation leads to an up-shift of the data in a diagram of $\sigma\Theta$ versus σ [29] as has been verified on Nickel polycrystals [83], Fig. 15, (and has been used, also on different grounds, by Acharya and Beaudoin [130]).

It is obvious that, when the rate of strain hardening is greater for smaller grain sizes, so is the flow stress compared at constant levels of strain. Although strain is not a proper state parameter, it has been used to plot data according to Hall–Petch in the absence of a sharp yield drop (which is typical for Hall–Petch materials). This trend has been observed—although the slope is an order of magnitude lower than typical Hall–Petch slopes [83,128,131,132]—and also an exact straight line is generally not observed. Therefore it is unlikely that the same mechanisms are controlling in both cases. It has even been argued [126,133] that in fcc polycrystals the main effect is by texture since annealing treatments, employed for producing variations of grain size, usually are accompanied by texture differences.

So far the experimental findings are not cleared up sufficiently, so that we have to leave it open to what extent observed phenomena such as, the change of yield stress with grain size and cross over of stress–strain curves in certain regimes of grain size, are texture effects or intrinsic work hardening effects.

We like to close this discussion with some general comments on the possible role of interfaces on dislocation accumulation in polycrystals. Besides the preexisting grain boundaries new interfaces may build up during deformation since grains break up into regions where different slip systems are active and where thus the lattice rotations are different [134]. This kind of heterogeneity is already a natural constituent of the Taylor model [135] since, due to the abundance of equivalent slip systems in fcc systems [118,119], a prescribed strain can be reached by various sets of slip systems which are all equivalent if the critical resolved shear stress is the same for all of them. However, there may be fluctuations in the locally acting stress across a grain due to the interaction with its immediate neighbors, resulting in a specific choice of the combination of slip systems in the sphere of influence of each specific neighbor. So even if the strain were the same everywhere, as demanded by the Taylor model, the lattice rotations do not need to be the same, so that a grain may break up into several orientations separated by subboundaries which may eventually evolve into large-angle boundaries at large strain. Although the wall-like dislocation arrangements which build up the orientation differences have no long range stress field and thus, in a first approximation, do not contribute to the flow stress, they may influence dislocation storage just like grain boundaries do [128,130]. Not much is known quantitatively about the breaking-up of grains into a mosaic of different orientations, it is conceivable, however, that this process is suppressed in small grains and thus only occurs in sufficiently coarse-grained material. Therefore a critical grain size may exist from where on the evolution of a mosaic structure becomes possible, resulting in a discontinuity of the frequency of interfaces which in turn would become manifest in peculiarities of hardening behavior at grain sizes just near the critical one. Indeed some strange effects such as crossovers of stress–strain curves have been observed in a grain size regime around a few microns [131,136]. Unfortunately, however, detailed texture studies have been made only occasionally in this context so that, in principle, the origin of these effects remains unexplained.

Finally, there is the possibility that the whole mechanism of plastic deformation changes under some extreme conditions of texture and grain size. It is possible that polycrystals with extraordinarily large grains exhibit different mechanism. However,

one must keep in mind that, when only a few grains are present in each cross section, heterogeneity of strain may build up across individual grains and regions may exist where only single slip will be activated. Then polycrystal averaging is not appropriate; only finite-element simulations are expected to give agreement between theory and experiment for specific grain assemblies. Another example holds for grain sizes that are very small in at least one dimension [137,138].

A new class of materials which must be considered in relation to grain size effects are nanoscale materials which may be either single phase or two phase [139,140]. Depending on the method of fabrication these may be layered structure, multi-filamentary composites, fine-scale rod-shaped particles embedded in a plastic matrix or ultrafine-scale equiaxed grains. Many attempts have been made to deduce the plastic behavior of these materials from hardness and compression tests on bulk material but in addition tests of free standing films have been performed and in a few instances direct observations have been conducted by TEM [141–143].

A number of authors have examined properties as a function of some controlling length scale such as layer spacing and observed deviations from Hall–Petch behavior when the scale is reduced to about 20–50 nm. In a few cases negative Hall–Petch slopes have been reported in this range [144] (which often originate from processing flaws [145]). The results strongly suggest that, when the scale of the microstructure is reduced to about 50 nm, the mode of plasticity is changed so that the interfaces, either in the form of grain boundaries or interphase boundaries, dominate the plasticity [146]. They are the sources and sinks for the dislocations involved in plastic deformation and it is the local forces and configurations at the interfaces which control the flow processes. The slip distances are so small that few intersections occur between dislocations, and dislocations are emitted from one interface and incorporated into the adjacent one. Thus the dislocation storage is at the interfaces and in essence energy is stored as additional surface area created by the flow processes. However, the surfaces created may have high local elastic stresses in the case of two phase materials.

In view of the scale of the microstructure it is also possible that local diffusional processes can contribute to plastic flow and this may account for the negative Hall–Petch slopes which have been reported (however porosity and/or flaw density need to be considered before data are evaluated quantitatively [145]). As many of these fine-scale materials have strength levels of 1–5 GPa they are of considerable technological importance and present a fruitful area for future application of the analysis of the temperature and strain-rate dependence of the flow stress and the work-hardening behavior of these types of new materials. Moreover, study of temperature and strain-rate dependence may provide critical insight into mechanisms of plasticity.

3.7. Stage IV (and V)

Essential features of work hardening are sketched in Fig. 37 as schematic θ – τ curves for the two limiting cases, namely, polycrystals and single crystals deformed in single slip. For polycrystals the rapidly decreasing low-stress end is affected by certain elements of the initial microstructure but after a small amount of strain it

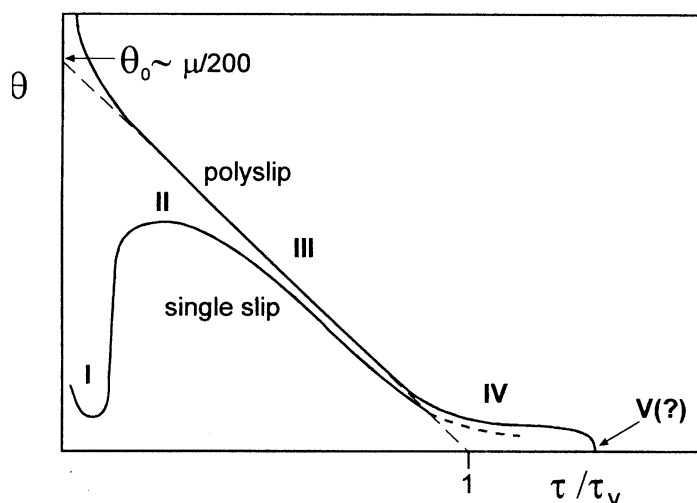


Fig. 37. Schematic sketch of hardening coefficient versus flow stress illustrating the hardening stages for polycrystals in comparison to those for single crystals deformed in single slip.

usually merges into an almost linearly decreasing branch. The behavior can be idealized by a straight line (dashed in Fig. 37) whose intercepts with the axes are the athermal hardening level θ_0 and the temperature and strain-rate dependent Voce stress τ_v . In most cases, single crystals deformed in polyslip show quite similar curves as polycrystals while, in contrast to these, single crystals deformed in single slip behave quite differently. They start out with very low hardening in stage I before the hardening coefficient rises in a burst-like fashion to the high athermal level in stage II from where on it decreases in stage III in a similar way as with polycrystals.

The continuous decrease of hardening in stage III suggests a steady-state limit of the flow stress at τ_v , but a steady state is never established in reality; at least not at low and medium temperature. As stage III appears to approach a saturation level the new stage IV intervenes when the hardening coefficient has dropped to a particular level much below θ_0 . The concept of a stage IV to follow at large strain on the established stages I, II and III was elaborated by Aernoudt et al. [10] as part of the stress–strain curve at a constant hardening rate. It was later broadened [147,163] to include a further decrease in hardening rate by one law or the other— or a renewed increase in hardening rate. The latter was the case for the first work in which the terminology “stage IV” was introduced [148] although, according to Rollett [149], this typically happens only for two-phase alloys.

Due to the low hardening levels in stage IV with $\theta < \tau$, it is difficult to track the hardening mechanisms because they must compete with geometrical and structural instabilities including damage. In addition, a considerable enlargement of surface and interface area and a corresponding increase of internal energy comes along with strain levels where stage IV usually begins [150]. It has been suggested [151] that stage I marks a lower bound of work hardening corresponding to a minimum level of dislocation storage, which is always present, but which is masked by the high

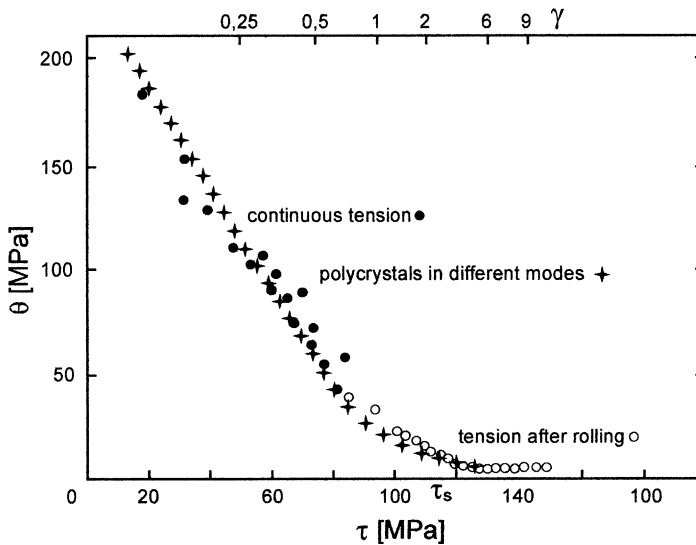


Fig. 38. θ - τ plot for Cu single-crystal orientation (110)[112]—i.e. (rolling plane)[rolling direction]—rolled to a shear $\gamma=9.5$. The data points are from the original diagram of the diploma thesis of Rudy [156]. They were obtained for lower strains up to necking in a continuous tensile test and from there on by measurements of the flow stress in tension on samples cut at various degrees of thickness reduction from a rolled crystal. For comparison, the θ - τ curve of Fig. 36, abstracted from the evaluation of the polycrystal curves in Fig. 34, is inserted and agrees surprisingly well with the single-crystal data.

storage rate of the stage II processes and, therefore, shows up as stage IV when athermal storage of dislocations is essentially balanced by dynamic recovery at the end of stage III. In addition, depending on deformation conditions, certain large-strain effects may be superimposed, such as in axisymmetric tension of bcc material, where grain curling in connection with the formation of non-recoverable dislocation arrangements has been assumed to make a considerable contribution [152].

Stage IV has been found in many pure metals and alloys in polycrystals under many deformation modes, in wire drawing as well as in rolling and in torsion [13,57,149,153–155], and also in single crystals [128,156] as seen in Fig. 38 from the hardening curve obtained by rolling of a single crystal with the brass orientation (110) [112]. (The advantage of this orientation is that it does not change neither in rolling nor in tension (along [112]), that in both cases exactly the same slip systems are activated and that it does not develop shear bands [157,158].)

The phenomenology of stage IV, however, has not reached the stage of generally accepted overarching observations as for the stages II and III. And while a number of mechanisms have been suggested that would lead to stage IV, none has emerged as widely accepted; we will mention some of them here and in Section 5.8.

Finally, there is a certain arbitrariness about the separation between stages III and IV. Generally, a Voce law is a good description of hardening only at medium stresses in the middle of stage III. In principle, it would be possible to count as “stage IV” all (upward) deviations from linearity in θ versus τ at larger stresses [149]. We

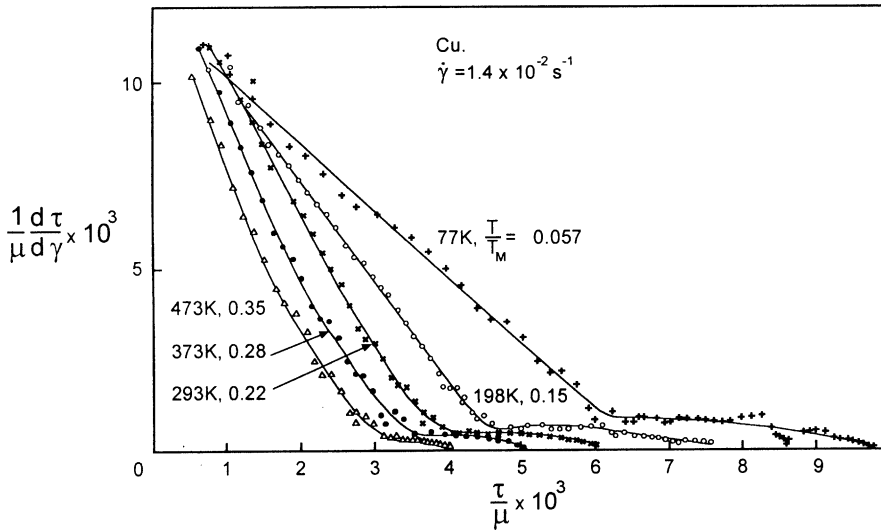


Fig. 39. Work hardening coefficient versus flow stress in terms of applied (not resolved!) shear stresses and shear strains as observed by Alberdi [57] in torsion tests on Cu polycrystals. (A practically identical set has been published by Zehetbauer and Seumer [159].)

will follow here a different convention: we define the beginning of a domination by stage IV processes by the appearance of a more-or-less sharp kink in the θ – τ diagram as seen in Fig. 38 (and Fig. 39 for a number of different conditions).

In general, we presume that here, as in the previous stages, it is really processes that one should identify: they are likely to take place all along the stress–strain curve.

In particular, torsion tests have been employed for studying material behavior in detail since they allow straining without any relevant changes of sample dimensions, without necking, barreling or friction effects, and without creating new surface area (though the increase of interfaces is as substantial as in any other testing mode). Fig. 39 shows a set of stress–strain curves observed in [57,159.] by torsion of Cu polycrystals at various temperatures.

In terms of experimentally observed θ – σ dependencies the phenomenology of stage IV appears to be relatively simple. If we use as the defining feature for the beginning of stage IV the kink in the θ versus τ diagram and we define the hardening level at this kink as θ_{IV} , then as seen from Figs. 38 and 39, there exists a relation between this parameter and the stress at this point which is approximately equal to the stage-III-parameter τ_v :

$$\theta_{IV} = c \cdot \tau_v \quad (33)$$

This empirical relation has been confirmed for many materials for different deformation modes [149,153]. In $d\tau/d\gamma$ – τ -plots (with γ being the total crystallographic shear and τ the resolved shear stress) the proportionality constant c is in

the range between 0.05 and 0.1. These values have to be multiplied by the respective Taylor factor M depending on the employed testing method and the given texture: for a textureless fcc material with $M=3$ for uniaxial deformation the range of the constant is $0.15 < Mc < 0.3$ and with $M=1.8$ for torsion it is $0.09 < Mc < 0.18$ in terms of applied shear stress and shear strain. The line drawn in Fig. 39 corresponds to $Mc=0.12$ and it is seen that the intersection with the experimental curves marks the bend-over from stage III to stage IV quite well. This order of magnitude means that stage IV is not observable in tensile tests where necking is initiated at the Considere point ($\Theta=\sigma$) well before the beginning of stage IV. We also like to emphasize that stage IV (and V) in the present terminology must not be confused with stages IV and V in the notation of Haasen and coworkers [160]; they refer to hardening behavior observed by Siethoff et al. [161] for high temperature deformation of semiconductors Ge and Si at much higher hardening levels than those corresponding to the above relationship for stage IV.

When Eq. (33) holds, this means that the temperature and strain-rate dependence of stage IV is basically the same as that of stage III: no new thermal activation is called for. Also any dependence on stacking fault energy is dominated by stage III processes since the slope in stage IV stays in a fixed relationship to the scaling stress for stage III.

Stage IV in our nomenclature is only significant at low homologous temperature (less than about half the melting temperature). As is evident in Fig. 39—and also a general observation—the situation $\theta=0$ may be reached eventually at higher stress. This is sometimes described by a continuously decreasing hardening rate in stage IV, and sometimes by a separate stage V. Note again different terminology in the papers of Haasen et al. [160] The latter method has the advantage that a different temperature- and rate-dependence may be introduced for this stage.

From the evaluation of data on the stored energy also Seeger [162] concludes that stage V is an independent stage of deformation. Zehetbauer and coworkers [159,163] consider the slope in stage IV as constant and ascribes all downward deviations to a new stage V. From measurements of the electrical resistivity and of the rate sensitivity of the flow stress they conclude that stage V is connected with point defect generation and absorption, producing a steady-state situation at $\Theta=0$. In this context it is a puzzling observation that the mobility of deformation-induced point defects seems to be of no importance. For example, there is an annealing stage [164] in Cu at $T=230$ K (and in Al at $T=200$ K), which is generally interpreted as being caused by vacancy migration; a change of stage IV behavior, however, does not show up in this temperature range as seen from Fig. 2 where no discontinuity occurs between the curves at 293 and 198 K.

Aernoudt et al. [165] assume that Θ decreases continuously in stage IV and that it is terminated by damage accumulation, eventually leading, at the end of stage IV, to failure and a corresponding drop of Θ . In their concept stage V is assigned to damage processes like void formation, initiation and propagation of cracks. This conclusion is supported by the result of torsion tests with superimposed hydrostatic pressure [166], such as reproduced in Fig. 40, showing an elongation of stage IV to a considerably higher stress level (and even more remarkably higher strain) while its general course remains unaffected.

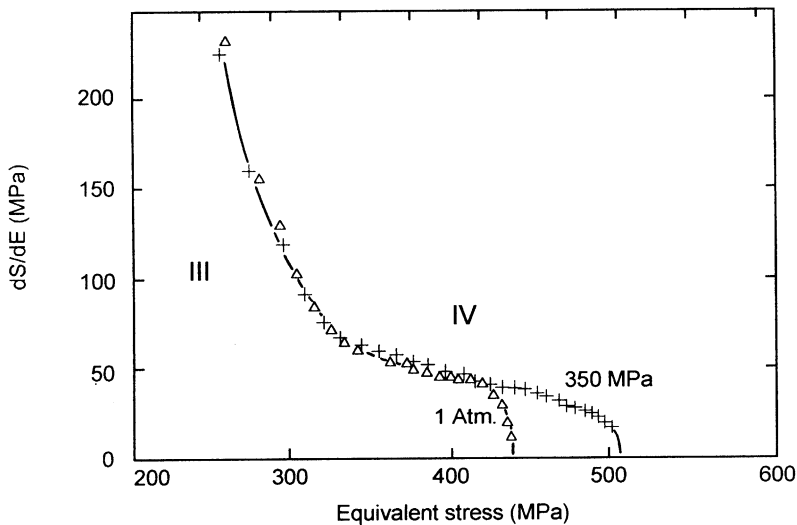


Fig. 40. Influence of hydrostatic pressure on hardening behavior of Cu in torsion [166]. Hydrostatic pressure of 350 MPa postpones the end of stage IV to an almost 70 MPa higher stress corresponding to the enormous increase in strain of $\Delta E = 3.0$. The effect of hydrostatic pressure is possibly reason for an influence of the deformation mode on the length of stage IV, e.g. no hydrostatic stress component builds up in free-end torsion but it can be quite large in forming operations such as swaging, rolling, drawing and extrusion depending on tool geometry and friction conditions.

An important technological aspect of stage IV behavior is that the hardening level as well as the length of stage IV increase with decreasing temperature. While σ_s , the stress where Θ drops to zero, and the Voce stress σ_v , which marks the end of stage III, both attain similar values at high temperature, they become quite different when the temperature is sufficiently low. Therefore, by straining in stage IV at low enough temperature material can be hardened to strength levels far above σ_v [167].

TEM observations on dislocation structures have been numerous, though systematic studies on structure evolution are rather rare [168,169]. Yet the generally accepted picture seems to be that recovery processes of stage III type move along with cell formation, a steady sharpening of cell walls and a continuous cleaning up of the cell interior. As deformation proceeds, subgrains form with sharp boundaries and an essentially dislocation-free interior. They remain roughly equiaxed and maintain a misorientation angle of about $1\text{--}2^\circ$. The dynamics of subgrain evolution seems to be linked with the characteristics of work hardening in stage IV, a point which has been elaborated in the models of [16,170].

Rollett et al. [149] conclude that the hardening rate in stage IV is governed by debris accumulation in the cell walls. They propose a simple model based on the idea that a certain fraction of those dislocations, that have participated in annihilation events by dynamic recovery, survives as debris of a dislocation dipole nature, in a process operating throughout stages III and IV in a similar fashion. While in the major part of stage III the net rate of dislocation accumulation is rather high as

compared to that of debris accumulation, the latter becomes dominant at the end of stage III in the new stage IV where athermal dislocation storage essentially is compensated by dynamic recovery, so that the remains become noticeable. The model predicts a constant hardening coefficient in stage IV which automatically obeys the above empirical rule and it generates the proper magnitude of the hardening coefficient if a fraction of 2–4% of the recovered dislocation length is considered as not recoverable, at least not by pure glide processes. Therefore, some kind of diffusion mechanism would be required to achieve a true steady state with no net storage of any dislocations at all, which is assumed to be possibly the case in stage V [149,164].

In a similar direction is the interpretation of Estrin et al. [170] who assume that dislocation accumulation in cell or subgrain walls is governed by a competition between dislocation storage and dynamic recovery and a continuous shrinking of the thickness of the cell walls. This shrinking is accounted for by the ad hoc assumption that the volume fraction of cells approaches exponentially a rate- and temperature-dependent asymptote. Their model leads to a hardening rate that decreases continuously with stress in a similar fashion as in stage III, but with a much lower slope.

Beaudoin and coworkers [130] have introduced the concept of slip plane lattice incompatibilities, which represent spatial fluctuations of the activity of slip planes leading to corresponding fluctuations in lattice orientation. The locally varying orientations are then separated by dislocation walls of “net dislocations”. In contrast to the statistically stored dislocations those in the walls are considered as not recoverable by glide processes (equivalently to the debris in Rolletts model) and to be the cause of stage IV hardening while the former ones are controlling hardening in the preceding stages. Observed characteristics of stage IV behavior could be well reproduced in a FE simulation..

As the most important outcome of this analysis within the present context, we like to point out some consequences of the found interdependencies between the work hardening parameters in the stages III and IV. Certainly the dependence of the hardening rate on flow stress is not only quantitatively but also qualitatively different in stages III and IV, since the scaling rules are different. This is evidence for a switch-over to another mechanistic law for structure evolution at the transition point from one stage to the other [152]. There are, however, strong interrelationships between the physical parameters that control actual behavior quantitatively.

Our main conclusion drawn from these findings is: Throughout all hardening stages at low and medium temperature hardening kinetics is governed by stress driven slip processes, which do not only control dislocation storage but the mutual annihilation and rearrangement of dislocations as well, and since they are active only during deformation they truly relate to dynamic recovery. We do not see a single phenomenon that calls for introducing diffusion as a controlling mechanism. In our opinion the main drawback of all models involving diffusion is that they rely on time effects in quite the same way as the Bailey–Orowan approach which is in serious conflict with fundamental experimental observation as demonstrated in the introduction to this analysis (Section 3.1). Merely stage V does not seem to fall into this scheme, but it is still an open question whether it is an independent stage at all,

which can be ascribed to special dislocation processes, possibly involving diffusion, or whether it requires consideration of damage and fracture mechanisms.

3.8. Assessment: what needs to be explained

A careful analysis of the data has led to the identification of some cardinal features of strain-hardening behavior in fcc metals (at least pure ones, for now)—emphasizing polycrystals, but taking in some features of single-crystal behavior as well. These were described primarily for tensile and compressive tests, in terms of the measured variables stress, σ , and hardening rate, Θ ; for the modeling we will use the shear stress, τ , and shear hardening rate, θ on a representative slip system (and leave the conversion to macroscopic measures to the reader, for each particular case).

One component of the strain-hardening rate is an athermal dislocation storage rate, giving a hardening contribution θ_0 that is, to a good approximation, constant and independent of material, and of the order $\mu/200$. It can be seen explicitly as an initial stage at low temperatures, but survives as low-strain limiting behavior to high temperatures.

When the hardening rate θ , in a continuous test, is normalized by θ_0 , and plotted against flow stress for a wide variety of strain rates and temperatures (including very low ones), a set of curves results that can be well idealized by a single-parameter set; it can be combined into a single master curve by normalizing the flow stress by a scaling stress, which may be taken to be stress at any constant value of θ/θ_0 (between, say, 1/4 and 3/4). Individual curves depart a bit from the master curve in the regime $\theta/\theta_0 < 1/4$.

A significant regime of this master curve can be well approximated by a linear decay (corresponding to the Voce law); the extrapolation of this regime to $\theta=0$ gives the ‘Voce stress’, τ_v , which is another measure of the ‘scaling stress’, which has proven especially useful.

A beginning of dynamic recovery in ‘stage III’ (at a stress $\tau_{III} > \tau_0$, the yield stress) can be identified only in cases where the difference between τ_0 and τ_{v0} is very large: in soft single crystals, and near 0 K.

The scaling stress τ_v (normalized by the shear modulus μ) is a function solely of the prescribed temperature and strain rate; when these two external variables are combined into an effective activation energy (normalized by μb^3), its dependence on the normalized stress collapses into a single (second) master curve, provided an appropriate value of the adjustable constant $\dot{\epsilon}_0$ is chosen (which is about 10^7 s^{-1} —and must be of that order to agree with the mechanisms known to operate in the thermal activation of dislocation motion).

This second master curve depends on the particular fcc metal investigated—in a sequence that follows the normalized stacking-fault energy, $\chi/\mu b$. (The extrapolation to the 0K Voce stress, τ_{v0} , may depend on the microstructure of the particular sample.)

Associated with dynamic recovery are ‘long transients’ upon a change in temperature and strain rate. Of special importance is the ‘inverse’ transient upon an

increase in temperature: it is work ‘softening’, with no instability despite the sustained negative hardening rate.

We find that “stage IV” processes still require much investigation: to establish the general experimental phenomenology for all cases; to confirm the structural aspects of dislocation arrangements; and then, to arrive at an understanding of which mechanisms are primarily active in these processes.

In the investigated range of temperatures and strain-rates, hardening-behavior is controlled solely by conservative glide of dislocations. With the possible exception of a stage V, diffusion appears to be not a controlling mechanism.

Finally: a problem that has not been addressed at all in this treatise (and hardly at all in the literature) is strain-hardening behavior upon a change in deformation path, which is of particular importance for understanding and simulation of the plastic anisotropy, i.e. of the yield stress and hardening-rate in deformed material as well as of plastic flow in all kinds of forming operations.

4. The ‘percolation model’ for flow stress and dislocation storage

Upon loading an annealed specimen, dislocations begin to move short distances at small stresses; eventually, long-range slip ensues—at a stress that we call the flow stress (Section 2.3.2). This is a property of the interaction of mobile dislocations with the initial obstacle structure and it is, by our choice, determined by glide processes only; i.e., it is two-dimensional.

Strain hardening is due to a change of flow stress with strain; i.e. due to a change in the obstacle structure (which would be observed after an intermediate unloading). It is appropriate to view this change in dislocation structure as a two-stage process. The first question is: what fraction of the previously mobile dislocations did get left behind in the specimen? And the second is: have these remnants actually formed a new arrangement that is, for example, stable against unloading and capable of resisting higher forward stresses? The latter almost always involves plastic relaxation in the third dimension—but the former is strictly a two-dimensional question. It is true that both processes may happen simultaneously; but they are logically separate. In addition—not surprisingly—it turns out that the initial storage process is dictated by geometry, and thus athermal, whereas the subsequent rearrangement may be significantly aided by thermal activation.

This partition of strain-hardening into two processes was also seen in the experimental results: while these two processes occur, in most instances, simultaneously during the straining of (pure fcc polycrystalline) metals, they are clearly separable as an athermal hardening component, and a thermally activated dynamic recovery component. We shall treat the modeling for these two mechanisms separately, in Sections 4 and 5. For the athermal hardening component, our main task is to explain its (surprising) order of magnitude, $\mu/200$. This must be understood in connection with the elasto-plastic transition. The mechanism of athermal storage should also, if possible, lay the groundwork for the heterogeneous three-dimensional dislocation substructure that develops with straining—which is important for the dynamic recovery mechanisms to be discussed in Section 5.

4.1. Dislocations under stress on a slip plane

Plasticity is often described as inherently a non-equilibrium phenomenon—perhaps because it is defined as irreversible deformation and occurs at a finite rate. From such a point of view, realistic models of plastic flow must be dynamic (and simulations are time sequences from some assumed initial configuration).

However, in most of the interesting regime of dislocation kinetics (and the only one we presume here), the finite rate is due to thermal activation of dislocations over obstacles (and the motion in between them is overdamped). Thermal activation proceeds from one metastable state to another one with lower energy, both of which are in mechanical equilibrium [25]. At least the configurations in stable mechanical equilibrium must therefore be known before thermal activation can be simulated. Often, a representative local interaction between mobile dislocation and obstacle is used. But in fact, a single, long dislocation in a crystal may be in equilibrium along its entire length—even when the applied stress is finite (so long as it is not too large). Then, the critical point along the dislocation, where thermal activation is most likely, is one of the local dislocation/obstacle interactions (the weakest on that ‘hard line’). A knowledge of all equilibrium configurations in a stressed slip plane is thus of fundamental interest.

We wish to emphasize, in this section, this dual view of dislocations in equilibrium and dislocations in motion. It is surprising how much can be learned for plasticity from equilibrium considerations alone. For example, in any slip plane, there is one stress above which no equilibrium configuration can any longer be found for an infinitely long dislocation line: this can be identified with the ‘flow stress’ for long-range slip. It corresponds to a topological flip-over of the connectivity between slipped areas versus unslipped areas in the plane. The term ‘percolation’ can legitimately be used for this critical topological situation, as well as for a dynamic event of beginning unlimited motion.

4.1.1. The limit of mechanical equilibrium in the slip plane

Kocks [20] introduced the notion of considering equilibrium configurations of dislocations in one slip plane under a variety of applied stresses—which means: under a variety of dislocation curvatures. To illustrate this concept, consider a ‘random pegboard’: say, a black wood-board with nails at random positions. Now place cylindrical objects of a particular (largish) size in all locations where they fit. Let us say these objects are foam and you have soaked them with white paint, which sticks to the board wherever a cylinder has been; this area is called ‘white’ (or later ‘soft’). The procedure can be simplified by moving everyone of the cylinders that has found a possible place around on the board to all the limits imposed by the neighboring nails. In this ‘experiment’, then, the inserting of a new cylinder corresponds to the activation of a source (considered easy), their round-about-translation to the movement of a dislocation under the applied stress until blocked. If you removed all the cylinders, there would be isolated white areas in a matrix of the (‘black’) background (Fig. 41a). At all the interfaces between black and white, there are dislocations in equilibrium under the given applied stress (even though the behavior is no

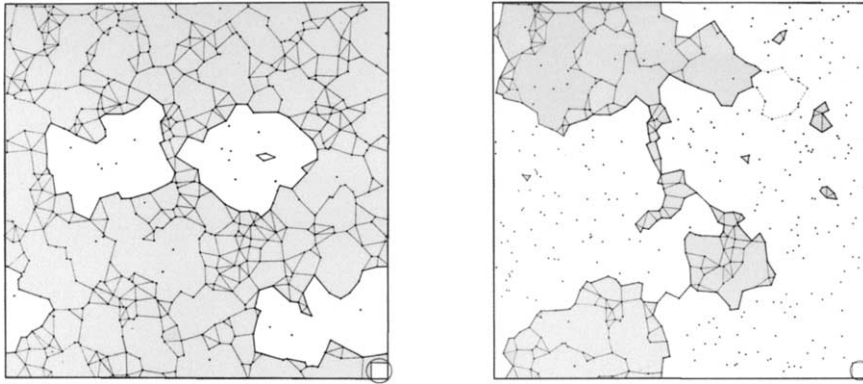


Fig. 41. 550 randomly placed strong point obstacles, connected by lines if they cannot be bypassed by a dislocation under the stress characterized by the circle in the lower right-hand corner (where the square contains in the average 1 point) [20].

longer strictly elastic). Note that between the point-like obstacles, the dislocations would be bowed 'out' ('convex' on a spreading loop); in the figure, the connections are shown as straight.

Now repeat the experiment with a thinner cylinder. This represents a smaller dislocation curvature, i.e. a larger stress. The white areas will be larger, and maybe a few of them touch. As you decrease the cylinder size, there comes a definite point at which the white areas all connect and the black areas are insular (Fig. 41b). In all these cases, the dislocations at the black–white interfaces are still in equilibrium.

Note that two important changes occur at the point when the topology changes. Firstly, so long as the white areas were insular, all dislocations that might have started were stopped in the slip plane; but when the white phase became continuous, any dislocation that would start at one arbitrary source can no longer be stopped (even though it leaves some parts of itself behind). This point corresponds to the limit of mechanical equilibrium (of an infinitely long dislocation line) in this slip plane. The other important change at the critical stress is that below it, all dislocations would run back upon unloading; above it, they remain around the 'hard spots', as 'concave loops': like Orowan-loops that tend to shrink under 'positive' stress (the same stress direction under which they have been formed).

The example we have used represents a particular case, based on three basic assumptions, which will now be justified as relevant. Firstly, the obstacles (the 'nails') were considered to be 'point'-like. This would be true, for example, for solute atoms on the atomic scale, or precipitates and inclusions on a microscopic scale. The interaction profile is, of course, extended; but the interaction is limited in extent in both directions in the plane [25]. A particularly important obstacle to dislocation motion is the interaction with other dislocations that thread the slip plane: the 'forest'. In a static situation, a single forest dislocation may, in the slip plane, be looked at as another point-like obstacle; in reality, it is not at a fixed position but flexes during the interaction with the moving dislocation. We will treat this flexing as a

detail of the interaction profile and thus regard forest dislocations as localized ('point-like') obstacles also.

Different obstacles differ in the strength of their interaction with a moving dislocation. The strongest obstacles demand a zero included angle between the branches of the dislocation at either side for penetration. This was our second (implicit) assumption: strong obstacles (although not necessarily of the maximum strength). And we have not considered thermal activation over the obstacles so far.

Finally, point-like obstacles are almost never arranged in an ordered pattern, such as a square grid; anyway, in the case of forest dislocations, it is probably impossible to construct a regular grid in each slip plane on the basis of one three-dimensional 'Frank net' of noded dislocations. A random arrangement, at the other extreme, may, in its strict sense, also be a rarity—but it fits the essence much better inasmuch as it embodies local fluctuations in obstacle density: this random arrangement was our third assumption. At low stresses, a regular pattern would exhibit the identical resistance to dislocation motion at every stopping point; upon increasing the stress, there comes a single value, at which penetration begins—and it immediately leads to the condition where no equilibrium position can any longer be found anywhere on the slip plane. In contrast, in a non-regular arrangement, such as a random one, a dislocation finds different resistances at a variety of intermediate equilibrium positions [59]; but, as illustrated above, there is again a stress at which no equilibrium position exists in the slip plane: this is the *mechanical threshold* [25] (when suitably averaged over all parallel slip planes).

Note that the picture of randomly arranged point obstacles (of identical strength) was merely a specific model: the essence is a heterogeneous distribution of glide resistance. It could, for example, also be achieved in a fairly regular arrangement of obstacles with varying strengths. (A small variation in local obstacle strength would lead to an enormous variation of waiting times, because of the very low rate sensitivity observed in most practical cases.)

The 'statistical theory of flow stress and strain hardening' [20] outlined above makes specific predictions about the elasto-plastic transition during loading, and the beginning of (large-scale) 'flow', or of 'plasticity'. The amount of plastic strain at stresses below the flow stress is directly proportional to the total area of all soft regions—'soft' meaning: penetrable at the given stress. More accurately (also to clarify that creep is not considered so far): in the elastic plastic transition below the flow stress, the increase in plastic strain with an increase in stress is given by the increase in the area of soft regions. For a random obstacle distribution, the function of the area swept, a , as a function of the shear stress, τ , has been derived on the basis of a specific model—but it was unfortunately given with a typographic error in Eq. (3.21) of Ref. [59]; the correct form is

$$a = \frac{l^2}{1 - \exp\left\{1 - (\hat{\tau}/\tau)^2\right\}} \quad (34)$$

where l is the average obstacle spacing and $\hat{\tau}$ the mechanical threshold. This is illustrated in Fig. 42.

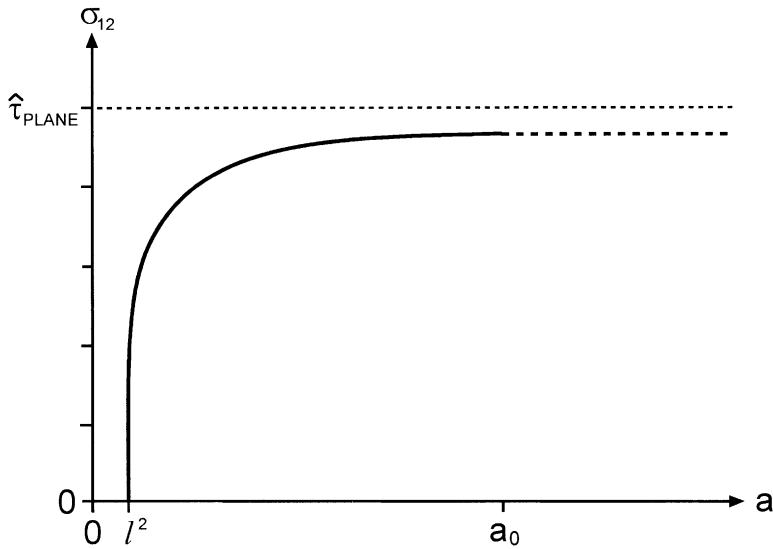


Fig. 42. The area swept by a segment of dislocation (on the abscissa) as a function of the applied stress (on the ordinate); l is the average obstacle spacing; at a_0 (qualitative) the soft regions become continuous [59].

Eventually, the soft regions interconnect: then, the total area of the soft regions is formally infinite. This is the mechanical threshold; it defines ‘long-range’ slip, and thus the flow stress

$$\hat{\tau} = \alpha \mu b / l \quad (35)$$

It is evident, then, that the stress–strain response during the elasto-plastic transition regime is not what is meant by ‘strain hardening’. In the above example, in fact, strain hardening is zero. What is meant by strain hardening is the change in *flow stress* with pre-strain [58] (Section 2.3.3). In the present model, this means a change in the mechanical threshold with a change in dislocation structure (as a consequence of the previous strain step).

Note that the above definition of ‘flow stress’ differs from the ‘elastic limit’—which, in the above example, is zero, or very close to zero. Similarly, the ‘proportional limit’ depends entirely on the size of the deviation from linearity in the stress–strain response that is—explicitly or implicitly—allowed. The beginning of long-range slip depends on different criteria, and it is this beginning that is to be associated with the term ‘flow stress’ (or ‘yield strength’ when one starts with an annealed material).

It is interesting to observe that the critical stress depends only on the situation in a few very special areas, the ‘critical gates’ that—just barely—connect neighboring soft regions, in a sort of ‘saddle point’ between hard regions. Of course, there are many of these critical gates in the specimen, and they are not all the same—but we

treat them in a generic fashion (since in reality fluctuations will be smoothed out by pile-up effects).

Finally, a word about the influence of thermal activation on the flow stress. In the above picture, the flow stress is controlled by the ‘critical gates’: the hardest of the soft-region interconnects. Thus, the macroscopic flow stress depends entirely on the interaction of a mobile dislocation with the ‘trees’ in this specific region—and the flow stress is thermally activated in whatever way this local interaction is thermally activated (in an average sense, of course).

4.1.2. Dislocation percolation

The abrupt change in topology of the slip plane at a critical stress (the ‘mechanical threshold’) may also be termed the ‘percolation limit’ of the soft areas: it is the soft (‘white’) areas that percolate. In this sense, ‘percolation’ would be equivalent to the limit of possible equilibrium states: a static phenomenon. (The term ‘percolation’ was not used in the original proposal [20], but many times since then [39].)

More commonly, the word ‘percolation’ is used in a dynamic sense; for example, a dislocation line that starts at one edge of a (simulated) sample may, under increasing stress, eventually ‘percolate’ through the slip plane. This is a two-dimensional analog of the situation where a liquid is being forced, from one side, through an (open) sponge: the surface tension provides a sharply-defined pressure where this becomes possible.

These descriptions amount to the same physical mechanism. Even in the last Section, we occasionally talked about dislocations that ‘moved’—but in a non-temporal sense: only by way of connecting neighboring states. In dynamic simulations, this is different: the result depends on the initial and boundary conditions, and the actual sequence of motions (and thus the final configuration) may depend on the specific assumptions on the laws of motion. Thus, a dynamic simulation will always give only one specific result; the abstraction into general behavior is a non-negligible subsequent task.

The motion of a dislocation through a random two-dimensional arrangement of point obstacles under stress has been simulated by computer [15,171–174]. The result is that there are two qualitatively different regimes of penetration: ‘unzipping’ when the obstacles are weak, ‘percolation’ when they are strong [173].⁷ In both cases, increasing stress leads to increasing penetration, and at a critical stress, the dislocation can no longer be stopped. Unzipping has been modeled in great detail for the weak-obstacle limit [175] in connection with solute hardening, and the flow stress has been derived with analytical accuracy [176]. The dislocations are quasi-straight, in part because the interaction of one character, say the edge, with the obstacles is much stronger, and therefore rate-controlling, than the other, say the screw.

When the obstacles are stronger, the dislocation has to bow out more; eventually (especially when the resistance is high for both edges and screws), glide becomes ‘areal’ [39], and the spreading (‘convex’) loop becomes more and more unstoppable (because it has more and more potential weak spots on it).⁸ In this case (and only in

⁷ This use of the term percolation is also, more descriptively, called *invasion percolation*; it is somewhat different from the more common description, in physics problems, of a *point* percolating on a lattice [174].

⁸ This was a major flaw of all the first work hardening theories [3].

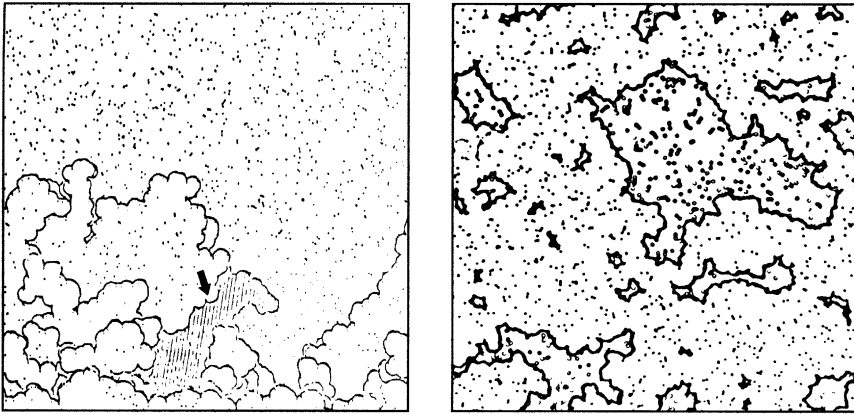


Fig. 43. (a) Percolation of a straight dislocation line from the bottom of the graph, under increasing stress, through randomly placed strong point obstacles; (b) “concave loops” left behind at the percolation threshold [20]. The lines exhibit the fractal nature of dislocation percolation.

this case), it is possible for dislocations to penetrate through some parts of the slip plane (‘soft spots’) but not through others (‘hard spots’). The non-slipped regions have to have dislocation lines surrounding them, i.e. separating them from the slipped regions. (That is as good a definition of a dislocation as any [58].)

The situation just below and just past the percolation threshold is illustrated in Fig. 43 [176]. The hard regions in Fig. 43b are surrounded with ‘concave loops’ (loops that tend to shrink under a positive stress). Note that the size of a ‘hard’ region depends both on the local fluctuation in the obstacle density and on the applied stress. (The oft-made comparison of a concave loop left around a hard spot in a random array of point obstacles with an Orowan loop left around a single second-phase particle is therefore more misleading than helpful.)

The fact that a part of the continually flexing gliding dislocation was left behind (as a kind of ‘debris’), while other parts continued, is the root cause of strain hardening—provided that the debris is not unstable under further loading [20,39,59].

Detailed simulations of the 2-D motion of dislocation lines in the percolation regime [177] have shown two features that we wish to emphasize: (1) the developing dislocation lines exhibit a fractal nature (with a percolation exponent of about $3/4$); and (2) they develop ‘fingers’: regions in which an initially straight dislocation line preferentially advances into a soft region. The first of these observation indicates that there is no length scale along the mobile dislocations; the second, on the other hand, produces a length scale in the width of the finger, which is (how could it be otherwise?) proportional to the average obstacle spacing, l .

As a function of the breaking angle φ_c at the obstacle, Fig. 44 shows the normalized finger width w . The ordinate is actually the inverse of w , raised to $3/4$ (the percolation exponent): this gives an approximate straight line. Around a breaking angle of 90° , the finger width becomes infinite: the percolation model must be replaced by the unzipping model. But in the limit of zero breaking angle, i.e. the strongest obstacles, there is a definite value of the ordinate, which translates to $w = 8.55$.

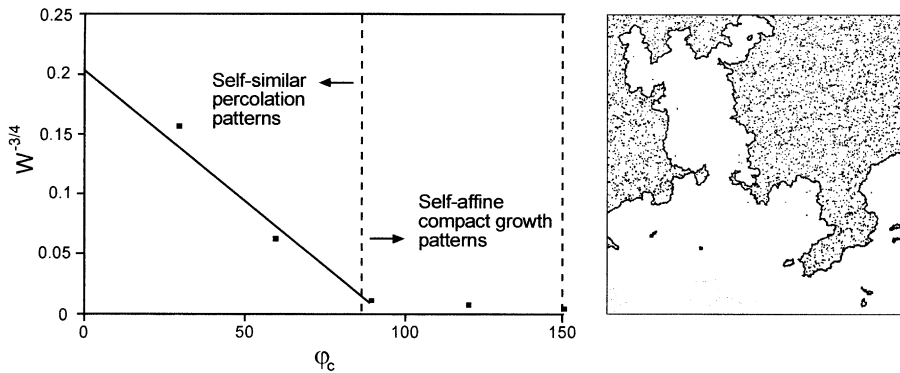


Fig. 44. Left: the mean intercept length, W ; of a dislocation line at the percolation threshold as a function of the obstacle strength i.e. the braking angle ϕ_c [177], plotted is $W^{-3/4}$ vs ϕ_c . Right: configuration of a dislocation line at the percolation threshold for obstacles of medium strength, $\phi_c = 50^\circ$ [173].

There is currently a wide-ranging debate as to whether ‘cell structures are fractal’ and therefore, by implication, have no average size. We suggest that this question, as phrased, is a fallacy: it is the mobile dislocations that are (probably) fractal, and therefore perhaps the last dislocation stored will have no identifiable length scale along its length (so long as there are no upper or lower limits to the obstacle spacing in the random arrangement); but the cells themselves (whether loosely tangled or more sharply defined) do have a size that obeys ‘similitude’ to the average obstacle spacing—as is evidenced by many thousands of observations, since the Swann–Berkeley book at least. And this is not a contradiction.

4.2. Statistical storage of ‘concave loops’

The mean free path, Λ , was identified, in Section 2.3 [20], with the inverse of the length of dislocation stored per area swept. In the current model, this corresponds to the average periphery of the concave loops, L , divided by the square of their average spacing, λ :

$$\frac{d\rho}{d\varepsilon} \equiv \frac{1}{b\Lambda} = \frac{L}{b\lambda^2} \quad (36)$$

Since L is no doubt proportional to the average obstacle spacing, l , the normalized mean free path becomes

$$\frac{l}{\Lambda} \propto \left(\frac{l}{\lambda}\right)^2 \quad (37)$$

If this is inserted into the equation for the strain-hardening rate [Eq. (12)], the athermal-storage contribution becomes

$$\frac{\theta_0}{\mu} \propto \frac{\alpha}{2} \left(\frac{l}{\lambda} \right)^2 \quad (38)$$

It is seen that, in order for θ_0/μ to come out around $1/200$, l/λ must be around 10. The problem of finding an absolute number of order 100 has thus been reduced to finding one of order 10: the square in (38), which is due to the areal nature of glide in the percolation regime, is the essence of this work-hardening model.

We must now introduce two refinements of the above derivation (which will tend to cancel each other). Firstly, the proportionalities used in place of equations were due to the fact that an average of the line length in concave loops was not known. It should be a few times the average spacing of obstacles in the dense region—which itself is less than the average obstacle spacing l . We could set $L = k_L l$, with k_L likely a bit greater than 1.

Secondly, while we have emphasized that the obstacles that determine the flow stress are forest dislocations, we have made no distinction between their density (say, ρ_f) and the density of moving and stored (‘primary’) dislocations ρ . We shall discuss in the next Section how the two may be related; but there is no doubt that the increase in forest density should be subject to a conversion efficiency factor less than 1; let us call it k_p . With these two corrections, the actual athermal strain-hardening rate is then

$$\frac{\theta_0}{\mu} = \frac{\alpha}{2} k_L k_p \left(\frac{l}{\lambda} \right)^2 \quad (39)$$

which leaves the conclusion unchanged that l/λ must be of order $1/10$. (In addition, some investigators [178, 173] include a factor (η) that is supposed to account for dislocation interactions in the cut-off radius of the line energy; it is also of order 1, and always lower.)

Finally, the value of α is subject to debate. From a theoretical point of view, it should be 0.83 for a random array of strong obstacles (breaking angle zero) [20]. But there is no doubt that forest dislocations have a range of effective interaction strengths in the slip plane [179]. The fraction of ‘strong’ ones has been estimated [180] to be about 20%; and the interaction strength of a ‘typical’ attractive junction has been estimated to be $\alpha \approx 0.3$. Values measured from the flow stress vs. dislocation density relation vary between 0.25 and 1. (It is important that the definition of the dislocation density and the shear modulus are given: they differ from author to author.) With our definitions, the range is narrowed to between $1/2$ and 1 [180].

One should therefore not take these derivations too seriously. The magnitudes of the constants in Eq. (39) and those mentioned in the text will never be derived with complete accuracy from first principles; the estimates we have given can perhaps be slightly improved by recourse to experiment or simulation—but not significantly. The product of all the correction factors can, under no circumstances, yield a total factor of order $1/100$: the onus of this large factor remains with the term that is squared. Instead of an exact equation with unknown constants of order 1, one might as well use an estimate, preferably with some aspect of being a bound. We conclude that

$$\theta_0/\mu < 1/100 \quad (40)$$

and typically between this and $1/300$, for fcc metals.

It is perhaps appropriate at this point to remark about the distinction between a theory and a model. It should be clear from the above exposition that an *ab initio* theory of strain hardening, with a quantitative prediction of the numerical constants, is unlikely to ever be derived even for a specific case, and impossible with any generality. A model, however, can postulate, with a generality based on physical insight, that there should be an athermal hardening rate, obeying similitude, of order $\mu/100$, and not greater than that. The actual value can then be taken from experiment, for each specific case.

4.3. *The resulting substructure and internal stress pattern. Plastic relaxation*

The geometrical mechanism of dislocation storage reviewed in the last section has one very attractive feature: it naturally leads to a heterogeneous dislocation substructure—not by the decomposition of an assumed initially homogeneous structure, but by increasing any preexisting fluctuations (which are to be expected). It is just the initially dense areas that become denser through the storage mechanism. Thus there is no need for a separate mechanism of ‘patterning’.

There are, however, a number of problems that were not addressed in the above introduction. The first is that the concave loops stored would be expected to collapse when the next dislocation traverses the slip plane and attempts to add another loop to the same area. Thus, a stabilizing mechanism must be found. The second problem is that the mechanism reviewed in Sections 4.1 and 4.2 is entirely two-dimensional (in each individual slip plane), whereas the structures that actually develop are three-dimensionally interconnected. Finally, and connected with the three-dimensionality, is the problem of converting the primarily stored mobile dislocations into the forest dislocations that are needed to control the flow stress. We shall discuss these problems together in the following.

In polycrystals (and single crystals with symmetrical orientations), most deformation occurs by multiple slip. This may be assumed to facilitate the three-dimensional storage of dislocations. However, as we keep emphasizing, the nature of slip is quite ‘jerky’: the multiple slip systems are rarely going to be operating at precisely the same time in the same region, but rather in a succession of quick, single slip events. Then the problem of stability may persist for the individual stored loops.

An essential aspect of a dislocation loop is that it has a dipole nature: the internal stresses inside the loop are high, those outside decay rapidly [20]. In the case of ‘concave’ loops, the stresses inside the loop are ‘forward’, i.e. in the same direction as the applied stress was during its formation; those outside the loop are ‘back-stresses’. It would thus be natural that some relaxation mechanism would tend to operate in the immediate vicinity of the loop: such as to partially relieve the high local stresses—and, in the process, prevent the generation of substantial non-deforming, and thus incompatible, regions.⁹ It was proposed in [20] that ‘secondary’ dis-

⁹ This problem will be further discussed under the heading of dislocation rearrangements in Section 5.2. Mughrabi [11] also treated polarized internal stresses and demonstrated them experimentally. For a model, however, he assumed constant wall thickness and concurrent deformation in the manner of a two-phase composite.

locations in the neighborhood (i.e. belonging to a slip system not active at this instant) can rearrange locally under the new loop stresses. Some of them would be likely to react by entangling with the primary loop; this would serve as a potent stabilizing mechanism. Another possibility was recently proposed, based on observations in a dislocation simulation code: that bits of the concave loop itself engage in a local cross-slip process [181]. Both processes have in common that non-coplanar slip systems are required—which can explain why the ‘stage II’-type hardening processes occur only in materials that have slip systems on different planes (and not, for example, in Zn, which slips on the basal plane only).

If, then, a stabilizing mechanism exists for an individual concave loop, further loops can be added around it—and further stabilization may take place. To the extent that the total effective loop number becomes greater than one, this arrangement amounts to a ‘dipole pile-up’: a very attractive model for internal stress patterns. This process only ends either when no further stabilization is available (in which case slip could continue without hardening in that plane); or when the impenetrable islands have grown in extent and are now overlapping (in which case slip would stop on that plane). The latter is akin to a mechanism proposed by Fisher, Hart and Pry for dispersion hardened crystals [182]. This solution is preferable because, without stoppage on each individual plane, strain hardening could never occur. This situation is illustrated in Fig. 45 [39]. It has an interesting consequence: in the very process of stopping one slip plane, the dense areas become contiguous and thus the isolated patches are transformed into a loose cell-and-wall structure (within the slip plane)—at a scale much smaller than the glide distance. Note that the situation depicted in Fig. 45a could hardly be observed: whenever a dislocation structure is directly observed (say, by TEM) the actual deformation has stopped and one normally sees the situation in Fig. 45b.

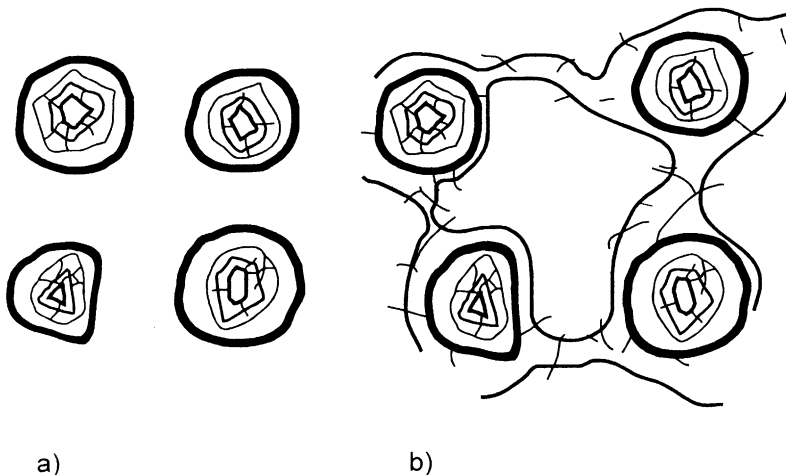


Fig. 45. (a),(b) Tangle and cell formation in a slip plane.

While the closing-up of the tangle structure in one slip plane limits the amount of slip on one plane at one time (and thus determines the slip step height to the order of λ/l [39]), it is important to realize that this is not the mechanism that determines the macroscopic hardening rate—which is determined by the flow stress experienced on the next slip plane, since slip is ‘jerky’ both on the plane and from plane to plane.

This leads us to one of the major problems in strain-hardening models (this one as well as others): how to explain the generation of a three-dimensional dislocation network (as a consequence of two-dimensional glide). The model of concave loops, stabilized by interactions with other, non-coplanar, slip systems (or by cross slip) naturally calls for the formation of a more spherical than circular affected region. This in turn affects neighboring slip planes, which then are more likely to deposit their own concave loops above the previous set. This is somewhat analogous to the concept of glide polygonization, in which individual edge dislocations tend to accumulate above each other; but here it is dipoles that do. (Note that groups of dislocations that are arranged ‘above’ each other, in the manner of an incipient—but not yet complete—subboundary, exert high internal stresses out-of-plane.)

We thus see natural mechanisms that link neighboring storage centers in the slip plane (as slip stops on that plane) and also that propagate storage centers perpendicular to the slip plane. The formation of a complete network of cell walls that enclose cells is a process that might well require some further coordination; perhaps it is helped by the internal stress pattern [183] and certainly by increasing ease of rearrangement through dynamic recovery (Section 5). In any case, if the network forms after the preliminary stages envisaged here, it remains ‘loose’ in the sense that the cell wall strength varies from place to place and dislocations can still penetrate the walls (as is required to satisfy the observation that slip lines are considerably longer than cell dimensions): cells are not boxes.¹⁰

The ‘concave-loop’ concept now needs to be generalized: the fact that dislocations of one sign tend to be on one side of an obstacle, and those of the other sign on the other side, holds for cell walls also. One might say that the concave loops around individual hard spots are now replaced by polarized cell walls [31]. This model automatically leads to a certain distribution of internal stresses: they will be high and ‘forward’ (i.e. in the sense of the stress that was applied when the structure was generated) inside the loops and, as a consequence, inside the (finite-thickness) cell walls. For compensation, there must be a back-stress inside cells which, however, is much smaller, since the volume is so much larger. The forward stress will be the basis for dislocation rearrangements inside the cell walls during the process of (static or dynamic) recovery, as will be elaborated in Sections 5.1 and 5.2. The back-stress, on the other hand, is responsible for the Bauschinger Effect, to be treated in Section 4.5.

An interesting intermediate case occurs just at the ‘critical gates’: here, the internal stress makes no contribution. Perhaps, one might phrase this more accurately by saying: the critical gate will adjust its precise location so that the internal stress contribution vanishes there.

¹⁰ Note that what we term *cell walls* are always loose in structure, ‘tangled walls’; when they are sharp (and without stress), we call them *subboundaries*.

In summary, the two-dimensional structures that form as an immediate consequence of the mobile dislocations having left some parts of themselves behind as concave loops, are associated with an internal stress pattern that tends to propagate these structures into the third dimension. This ‘plastic relaxation’ of the internal stresses is not an event that actually follows the initial storage in a time sequence, but ought to occur simultaneously. (Our description is thus more in the nature of a *gedankenexperiment*). Note that the relaxation processes will always be less than perfectly efficient, so that a polarized internal stress pattern, albeit lowered, remains (and causes subsequent thermally activated rearrangement). They have, however, achieved a solution to the three problems mentioned above: they have stabilized the dislocation structure; they have made it more three-dimensional; and they have (at least in our model) automatically converted the effect of ‘primary’ dislocations into ‘forest’ dislocations.

4.4. *An intermediate assessment of strain-hardening theory*

Plastic straining is produced by dislocations gliding along slip planes. (Twinning and climb are not considered in this section.) It takes many dislocations traveling long distances to achieve macroscopic plasticity; e.g., if the active slip planes are 1 μm apart, it takes six dislocations moving on every slip plane through the whole cross section to produce just 0.2% shear. The generation of such a large number of dislocations is not usually rate controlling (at least in pure fcc materials of a grain size not smaller than about 1 μm); it is the interaction of moving dislocations with obstacles that determines the flow stress. The obstacles must, at the flow stress, be overcome: if all the dislocations that produce the straining were stopped at the obstacles and remained in the crystal, the strain-hardening rate would be of the order of $\mu/2\pi$ [3]. Thus, it is the effective resistance of the obstacles to being overcome by the moving dislocations that controls the flow stress. But if all obstacles were overcome, there would be no dislocation storage.¹¹ The model of dislocations percolating through a slip plane studded with a non-regular array of point obstacles allows for the storage of a fraction of the dislocations that have produced the strain; and any storage mechanism at all would have to store, initially at least, some of those mobile dislocations.

In principle, there could be an alternative mechanism: dislocation storage could occur, not by the interaction of one moving dislocation with a fixed obstacle, but by the interaction between two moving dislocations (not necessarily parallel); i.e. by 2-body collision processes. This possibility may be neglected, even in multiple slip, on the basis of the kinetic observation that slip is ‘jerky’: a slip line is typically formed in 10^{-4} s [63], but the time increment between them is of the order of the waiting time for thermal activation at the critical obstacles, which is of order 1 s (at a strain rate of 10^{-4} s⁻¹). Thus the dislocations that, at any instant, produce the majority of the plastic straining rarely have a chance to meet each other. The interaction between dislocations on parallel planes controls the spacing between active slip planes [39].

¹¹ In the case of linear obstacles, a ‘relay race’ can develop, where each dislocation added to one side leads to one leaving the other (or an opposite dislocation *arriving* at) the other side. In this case, steady-state pile-ups develop [25].

In the absence, then, of significant dynamic storage processes, the percolation model provides a very attractive basis for treating flow and hardening at the same time. In the following, we review some additional points that need to be satisfied by any realistic strain hardening model—and summarize how the percolation model deals with them.

- (A) The elasto-plastic transition is gradual but rapid: from ‘true’, totally reversible elasticity, through ‘anelasticity’ as a consequence of the reversible (though perhaps time-delayed) motion of dislocations, to ‘true’, irreversible plasticity.
 - This sequence is well represented by the generation of convex, spreading dislocation loops through soft areas—until these soft areas join, at the percolation limit—which is approached very rapidly. Here, the remaining dislocations form concave loops and are thus stable.
- (B) Slip occurs in spurts on slip planes: many dislocations must be able to move long distances, and this activity must then be stopped. The last condition also means that there must be local hardening; otherwise, glide would continue on one plane until failure.
 - These conditions are fulfilled by the concept of percolation through a non-regular array, with storage of concave loops around hard spots—until these hard regions touch and prevent further slip on this plane. It is not explained how many dislocations pass through the slip plane before the close-off occurs; i.e. how deep the slip steps are on the surface. The observation that this step height typically corresponds to about 20 dislocations having slipped means that the hard regions must get harder with straining, in a stable way. This problem will be taken up in point F and [Section 5.2](#).
- (C) The dislocation structure becomes more sharply heterogeneous as straining progresses, even when slip is essentially homogeneous.
 - In the percolation model, this follows from the fact that dislocation storage occurs at just those places that had a higher obstacle density at the beginning; in a random array, this would be due to statistical fluctuations. It is not explained why the spacing of the dense areas (the ‘tangle spacing’ or ‘cell size’) is observed to be about 10 times the obstacle spacing (though it is explained why it remains proportional). This number is reasonable in view of simulation results.
- (D) The mean-free path of dislocations (as defined by the inverse of the length of dislocation stored per area swept) must be about two orders of magnitude larger than the dislocation spacing, if it is to explain the low value observed for the maximum rate of strain hardening; in fact, it must be about one order

of magnitude larger even than the cell size; yet it must also be noticeably smaller than the slip line length.

- In the percolation model, the last fact is trivial; in actuality, one would expect the slip line length to be infinite (though limits can be found [39]). The first two comparisons follow immediately from the areal nature of glide: the mean free path is proportional to the square of the tangle spacing.
- (E) It is observed that the maximum strain-hardening rate is an athermal contribution to strain hardening.
- In the percolation model, the original dislocation storage (before dynamic recovery and dislocation rearrangement take place) is a strictly geometric property.
- (F) The developing dislocation structure is observed to be three-dimensional, even though slip is essentially planar. Also, strain hardening can only occur if successive parallel slip planes find an increasing flow stress. And finally, the flow stress is assumed to be related to the forest density of dislocations, whereas the primarily stored dislocations are all primaries.
- In the percolation model, these three related facts follow, in a qualitative sense, from a prediction: that an internal stress pattern develops, through the formation of the concave-loop pile-ups, which must be relieved by some process of plastic relaxation, involving intersecting slip planes. The details of this plastic relaxation are not known and would form an interesting problem for dislocation simulation studies.
- (G) Although plastic deformation is essentially stable upon unloading, a reversal of the loading direction leads (even a bit before zero stress is reached) to a ‘long transient’ (of order 3% strain [184]) before the last-reached flow stress is re-attained. The dislocation structure is also observed to be transiently unstable upon other changes in straining path.
- In the percolation model, the remaining internal stresses are essentially polarized, with most of the material under a slight back stress. The ordered pattern of internal stresses allows, in principle, for the simultaneous existence of stability in the original loading path and some degree of instability under reversals and path changes. A guideline for evaluating the Bauschinger effect will be provided in the next section.

The following will flesh out a few ancillary points. Section 5 will then deal with detailed proposals concerning the formation of the three-dimensional network of tangles and cell walls, and with dislocations rearrangements within them under the action of the forward internal stresses and thermal activation.

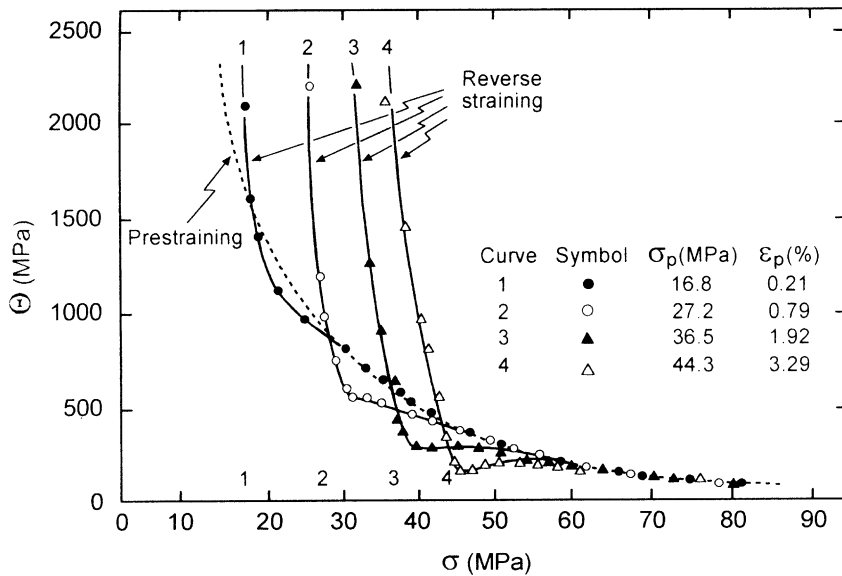


Fig. 46. Strain-hardening rate Θ vs. true stress σ on stress reversals in Al after straining at 293 K and $\dot{\varepsilon} = 3.5 \cdot 10^{-4} \text{ s}^{-1}$ up to the prestresses and the prestrains σ_p and ε_p resp. [41].

4.5. The Bauschinger effect

There are various kinds of Bauschinger effects; all relate to plastic behavior upon stress reversal and are caused by internal stresses. In the presence of a non-deformable phase, at one extreme, substantial back-stresses are generated in the matrix, all of one sign, and they cause an effect that looks like a ‘permanent’ lowering of the flow stress in the reversed direction of straining; only near steady state does the flow stress catch up. An intermediate case is the ‘Masing effect’ in polycrystals, due to the compatibility constraints between different grains. But, at the other extreme, a Bauschinger effect exists even in single crystals. Here, it is a transient effect: the strain hardening behavior is affected only for reloading strains of about 1–3%. That the behavior then returns to normal can best be seen on the Θ – σ diagram in Fig. 46 [41].

This effect can only be due to a polarization of the dislocation structure and consequent patterning of internal stresses. Fig. 47 plots a schematic, in one dimension, of the 3-D internal stress pattern we have described in the last section: there are narrow regions of high forward stresses (not all identical: we show an ‘average’ hard spot and a ‘strong’ one)—balanced by wide regions of much lower back-stresses. Observe the three horizontal lines: the solid one is the zero-line when no external stress is applied. The lower dashed line can be used as a marker for zero total stress when a stress τ_p is applied. If τ_p is the last flow stress that was observed in the ‘forward’ direction, then the total stress, including the developing internal stresses is positive everywhere. (In the center of the hard regions, even this total stress is not

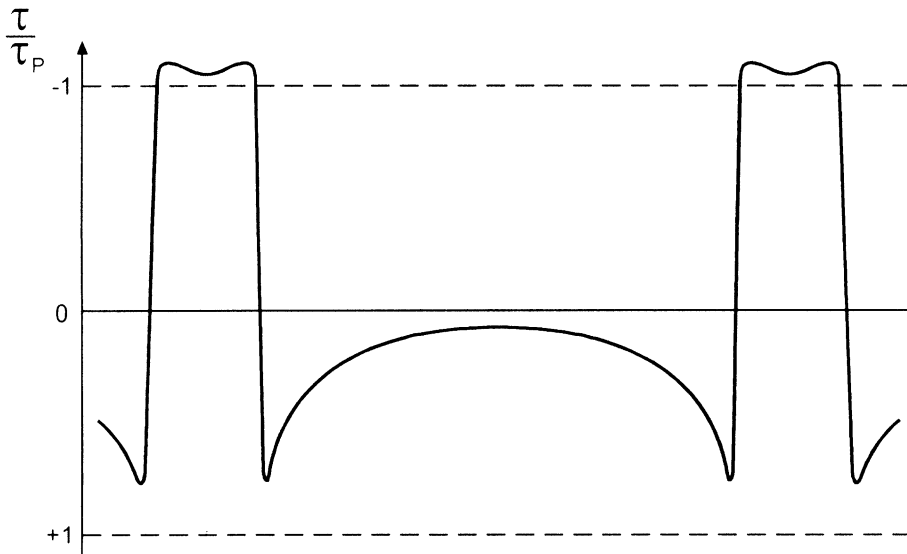


Fig. 47. Concave loops produce a residual internal stress, which is high and forward (i.e. in the direction of the originally applied stress) in hard regions but low and backward in soft regions.

enough to cause flow there except for plastic relaxation by some local flow on secondary systems.)

As the applied stress, τ , is lowered from τ_P , the lower dashed line moves up. Before τ has reached a zero value, substantial regions are under negative total stress (the ‘cell interiors’). As τ moves into negative values, the region in which the total stress is negative becomes larger: its borders climb up the sides of the walls. In the entire region of negative total stresses, all dislocations that are present and free, must move backward, causing reverse flow—long before the ‘flow stress’ has been reached in the reverse direction.

A stunning effect occurs (at least sometimes) when the reverse stress reaches about the magnitude of τ_P : the whole cell structure seems to dissolve, and a substantial ‘free strain’ is observed in the reverse direction; i.e., a ‘stagnation of strain hardening’ [184,185]. Even more surprising: at the end of this episode, a new cell structure forms with the same scale, but with the opposite sign of the internal stresses. This effect has been nicely explained on the basis of Fig. 47 [184]: just at the flow stress, not only do dislocation ‘unpile’, they also ‘untangle’ at the periphery of the cell walls, and in the weaker regions along them, even those many dislocation that were stabilized by the forward stress, can break loose under the reverse stress. Still: the very center of the tangles, at some specific spots along the walls, stable arrangements should remain: they form the nuclei for the formation of the new, inverted wall structure.

In the transition region, between $\tau = \tau_P$ and $\tau = -\tau_P$, one is dealing essentially with dislocation elasticity: the reverse shear strain, γ_B , should be a function of the applied stress, τ —presumably scaled by τ_P ; and it should follow a law of approximately this kind:

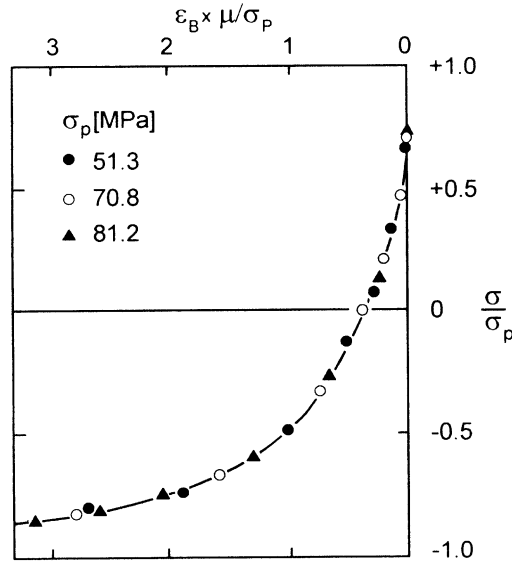


Fig. 48. Plastic back-strain, ε_B , in a Bauschinger test as a function of the applied stress, σ , during continuous stress reduction followed by a stress-reversal after prestraining to different levels of the flow stress σ_p . Normalization by the shear modulus and the flow stress according to Eq. (41) leads to a unique curve [41].

$$\gamma_B = n \frac{b}{d} = \frac{\tau_p}{\alpha \mu} n \left(\frac{\tau}{\tau_p} \right) \quad (41)$$

where d is the slip plane spacing and proportional to $1/\tau_p$. The function $n(\tau/\tau_p)$ is the number of dislocations per potential slip plane that have moved back completely: a number that may be smaller than 1 (and comes out in the order of 1). As Fig. 47 indicates, near the harder spots, there should be more ‘free’ dislocations, or more that can be freed by a small reverse driving force; near weaker hard spots, there will be fewer. We do not know the function $n(\tau/\tau_p)$ —but we can determine it experimentally by plotting data in a normalized form as suggested by Eq. (41); Hasegawa et al. [41,184] showed that the scaling law is indeed obeyed as seen from Fig. 48, where the abscissa is $n/M^2\alpha$ (increasing to the left).

5. The ‘rearrangement model’ for dynamic recovery

The model of dislocation percolation and statistical storage of concave loops, outlined in Section 4, has provided a rather well-established basis for explaining the existence of a sharp flow stress and of a geometric, athermal contribution to strain hardening, of the right order of magnitude; the more new computer simulations

results come in, the more the model seems to get confirmed. One aspect that is still too vague in its details is the process of plastic relaxation and the establishment of three-dimensional tangles and cell walls; much could be learned about this from appropriate simulations, and we will get further insight into this problem in the following (Section 5.2).

The mechanism of dynamic recovery is not quite as well settled; the authors themselves have proposed a number of different formalisms [24,27,30,39]. The basic process, however, has been clear for a long time [20]: the applied stress in combination with the forward internal stresses inside the tangles and cell walls, predicted by the percolation model—even after they have been reduced by plastic relaxation—provide the driving force for a rearrangement of the dislocations into lower-energy configurations (such as dipoles and incipient subboundaries), which are also presumed to lower the local glide resistance. Such rearrangements *can* take place under the action of the *internal* stresses only, plus thermal activation: this is a process of (early) *static recovery*, which will be discussed in Section 5.1. When the stress remains applied, and the internal stresses are continually replenished by the addition of new stored dislocations, a similar process takes place, at a much more rapid rate; this is *dynamic recovery*. While this is a process that is strongly influenced by thermal activation, it can also happen strictly mechanically, even at absolute zero temperature, and we call it ‘dynamic recovery’ even there. As in all processes with a stress-dependent activation energy, it is appropriate to derive the mechanical profile first; thus, we treat the dynamic recovery process first at 0 K (Section 5.2).

None of these recovery processes involve diffusion as a necessary element, and all should lead to similar cell structures—as is indeed observed (Section 2.2).

It is necessary, for the more detailed descriptions to follow, to introduce a term that characterizes the regions of forward internal stress—whether they be inside concave loops, inside loose tangles, or inside a connected network of (thick, irregular) cell walls. We shall merely call these the *recovery sites*. Their behavior responds to a complex interplay between the local *stress* and the local *strength*. In a simple picture, one may think of the local forward stress (τ_F) as the ‘dipole pile-up’ stress inside still-free concave loops, or inside a ‘polarized cell wall’. And one may think of the local strength (τ_M) as the obstacle-dislocation density inside the recovery site. In reality, the situation is complicated by the processes of plastic relaxation, which convert—with the help of secondary dislocations, immediately following the original storage—some of the ‘pile-up’ dislocations into ‘obstacle’ dislocations (and simultaneously lead to some semi-continuous back-and-forth local rotations). This process cannot be 100% efficient; what we consider to be the ‘recovery site’ is the residual situation, *after* all plastic relaxation has taken place. It would be very good to have a better understanding of this structure, including the attendant local rotations, from appropriate simulations.

The processes of dynamic recovery occur under the action of stress *and* thermal activation. The extreme cases are, however, of interest also: *static* recovery will be discussed in Section 5.1, before in Section 5.2 dynamic recovery is treated first for zero absolute temperature and then under the action of thermal activation at elevated temperature.

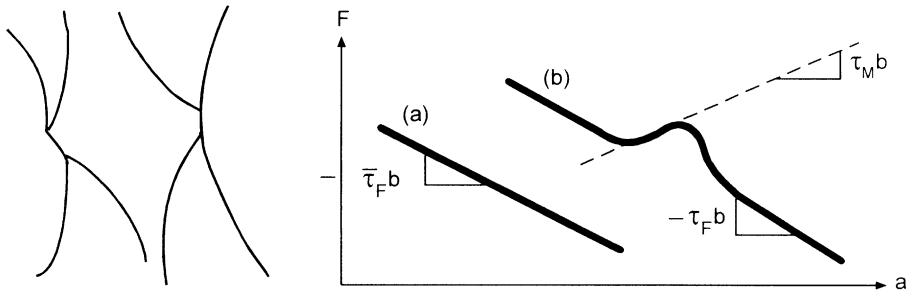


Fig. 49. Left: an example of a potential ‘recovery site’. The nodes are under a local forward force $b\tau_F$, not quite enough to break them; increasing the stress to τ_M would have completed the process. Alternatively, thermal activation under τ_F could have. Right: a diagram of free energy, F , versus area swept, a , of a pair of dislocation segments as in the left figure with a local resistance, τ_M , which, (b), may be overcome by jerky recovery.

5.1. Rearrangement under local stress and thermal activation: static recovery

The high internal stresses in small regions—i.e., at the ‘recovery sites’ defined above—provide a driving force for recovery: this local driving force is a more specific prescription than the overall decrease in stored energy to be achieved. Fig. 49 illustrates, by example, how this could arise. Two junctions of mobile and forest dislocations were formed by mobiles that came from opposite sides. The applied stress was not quite high enough to lead to annihilation (or dipole formation). Thermal activation during subsequent recovery times can complete the process. (This is ‘Type I’ recovery [36,37], which *precedes* the coarsening of the structure.)

If the stress that would have achieved mechanical break-through at a particular tangle is called τ_M (a local ‘strength’ or glide resistance [25]), and the actual local forward stress is called τ_F (Fig. 49b), then the probability of thermally activated collapse of this tangle is proportional to

$$\exp\left\{-\frac{\Delta G(\tau_F/\tau_M)}{kT}\right\} \quad (42)$$

where we have introduced the stress-dependent activation energy as a *Gibbs* free energy ΔG . If all the tangles were the same (or one could represent them by an ‘average tangle’) then, with an attempt frequency ν_r , the number of sites per unit volume at which this can happen N_r , and the length lost per event l_r , the rate of loss of dislocation density would be

$$\frac{d\rho_r}{dt} = -N_r l_r \nu_r \exp\left\{-\frac{\Delta G(\tau_F/\tau_M)}{kT}\right\} \quad [186]^{12} \quad (43)$$

¹² Eq. (43) leads to the observed logarithmic time law [186] whereas smooth—i.e. non-jerky—recovery according to global loss of stored energy typically leads to a power law.

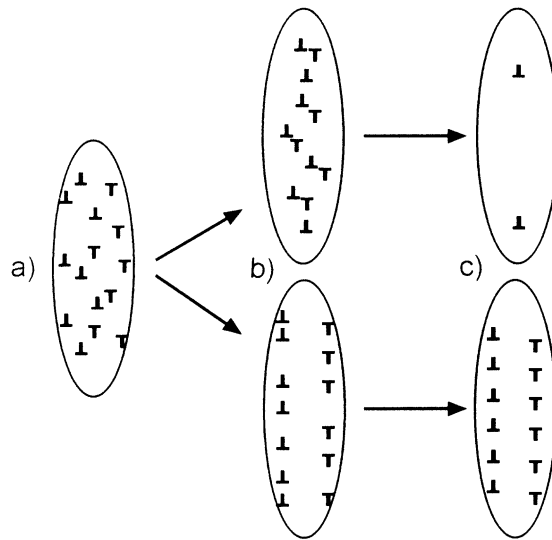


Fig. 50. (a) Schematic of the rearrangement of a cell wall (assumed infinite perpendicular to the paper), in two steps: (b) by glide to form dipoles or imperfect subboundaries (c) completion of the process with eventual aid of diffusive processes.

Note, however, that the dislocation segments that are rearranging themselves will not necessarily annihilate; they could, for example, form dipoles (which do have a lower energy and also affect the flow stress less than monopoles do).

This is illustrated in Fig. 50a for straight edge dislocations in a cell wall. Note that more positive dislocations are on the left side of the wall, more negative on the right: this is the consequence of the way the wall was formed, and the origin of the ‘forward’ stress inside the wall. If the opposite-signed dislocations on either edge of the cell wall were to move closer to each other, the cell walls should tend to get sharper (Fig. 50b)—exactly as observed (Fig. 52). If no annihilation takes place, one will be left with either a series of dipoles or *two* imperfect subboundaries (which are still held apart by some resistance inside the cell wall, shown schematically as a shaded area). In actuality, one would expect *both* of these lower-energy configurations to form.

Diffusion would be required only for the completion of these processes (Fig. 50b): annihilation of the dipoles (upper part of the figure); or (lower part) sharp, regular subboundaries (one or two of them), with no dipole content (Fig. 50c). This does occur at the highest recovery temperatures and times; only when it has been completed, can ‘Type II’ recovery commence: the *coarsening* of the subgrain structure [36].

An interesting side-light observed in Fig. 50c is that there are two distinct *misorientations*: the one from one cell to the other, which is due to the small number of *excess* dislocations of one sign, which does not change during this recovery process; and a *misorientation* in the *inside* of a ‘cell wall dipole’ which can be quite large [36,187]. The observed substructures will be addressed in Section 5.5.

The concept that Type I recovery occurs only in the regions of forward internal stress led to an interesting prediction: that a net macroscopic strain (however small) in the forward direction should accompany static recovery. This was verified to take place (after a small initial reverse strain associated with backward flow in the cell interiors), as shown in Fig. 51 [188]. Both effects cease at high temperature, when the cell walls are thin.

The picture of the recovery process as consisting of cell walls shrinking in thickness was a bit of an oversimplification. We have emphasized that cell walls are not uniform in strength or thickness; in fact, at the location of former critical gates, they would be very thin. As a consequence, we think that some more significant untangling could occur at such places, which may even trigger long-range motion of a few of the previously stored dislocations as an additional recovery event.

The microstructure at the *end* of the kind of recovery processes discussed here would consist of *sharp* cell walls, i.e. *subboundaries*—on the same scale as the previous cell walls (Fig. 52c). Only after completion of this process of subboundary formation can the subboundaries move and the subgrains grow (Fig. 52d); this is a *secondary* recovery mechanism (diffusion aided [36]). Note the similarity of the progression of structures with *time* in static recovery (Fig. 52) and that with *strain* in dynamic recovery (Section 2.2.2; Fig. 8).

In all probability, the activation energy in Eq. (43) does not stay the same throughout an annealing cycle; rather, it will increase continuously as recovery progresses. This is so since the soft recovery sites, where, presumably, the internal stress is rather high, disappear first so that the forward stress decreases continuously in consequence of the loss of dislocations. Therefore the recovery rate will slow down quickly with time: due to the decrease of the exponential term in Eq. (43), much faster than the number of potential recovery sites (in the preexponential factor) decreases.

If we assume that dynamic recovery is basically the same physical process as type I recovery and that it follows the same general rules, then Eq. (43) can be applied after two important modifications have been introduced. First the applied flow stress has to be added to the internal forward stress so that the activation energy becomes a function of τ and τ_F

$$\Delta G = \Delta G([\tau + \tau_F]/\tau_M) \quad (44)$$

At this point it becomes necessary to be more specific about the precise meaning of the introduced variables τ_F and τ_M . They are not characteristic averages; rather, together with the activation energy ΔG , they must be considered as local quantities, which fluctuate from site to site at a given level of the flow stress τ and which vary continuously with the increase of τ in the course of straining.

The second modification concerns the superposition of simultaneous storage of dislocations and the thermally assisted reduction by thermally activated recovery. Before a detailed formalism will be set up in the next section we like to point out here one aspect that is important for the understanding of dynamic recovery. Different from static recovery, in the dynamic case, the activation energy does not

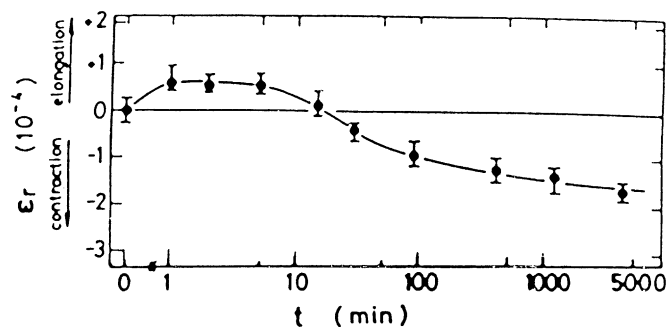


Fig. 51. Microstructural changes in (111) Al crystals deformed to 42 MPa at room temperature (a) and recovered at 453 K for (b) 3 min, (c) 2 h, (d) 30 h [36].

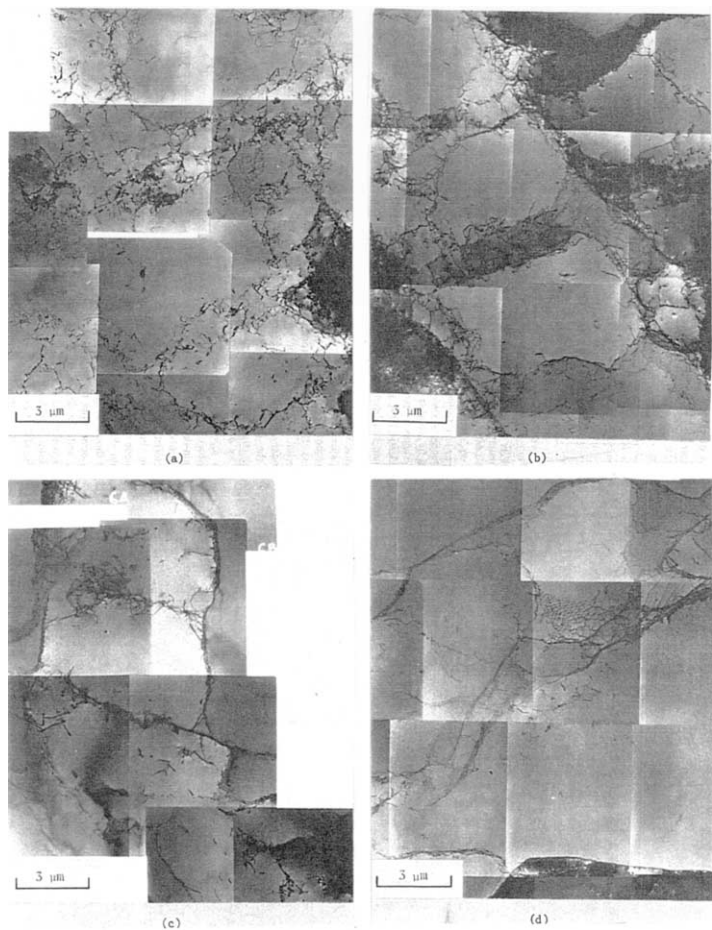


Fig. 52. Changes in the length of a (111) aluminum specimen with recovery at 453 K after compression to 59 MPa at room temperature (ϵ_r is the recovery strain, t the recovery time) [188].

necessarily change with progressing recovery. Due to the continuous replenishment of annihilated dislocations by new ones, and due to additional storage, more dislocations are stored than annihilated. As the dislocation density increases with strain, also the quality of the activated sites may gradually change, in such a fashion that the activation energy adjusts itself at a certain level. In this context a key role falls to the experimentally found scaling rule in connection with the Cottrell–Stokes analogy. As already discussed in Section 3.3 [see also Eq. (23)], it implies that, within the limits of experimental accuracy, ΔG stays constant along a stress–strain curve, for a given temperature and strain rate

$$\Delta G = \text{const.}(\dot{\gamma}, T) \quad (45)$$

5.2. Storage and rearrangement inside tangles and cell walls under continued straining: dynamic recovery at 0 K

During straining, dislocations will be accumulated in hard spots, of which some are barely stable, others may sustain a considerable increase of the stress in the course of continuing work hardening. The resistance of a particular spot depends on the local density of the trapped glide dislocations and of additional dislocations produced during plastic relaxation by local slip on secondary systems. Nonetheless, every hard spot will collapse mechanically or rearrange under some value of the locally acting stress, call this value τ_M . Thereby the τ_M value of each individual site is not interpreted as a fixed value, since the characteristics of the site may change in the course of further straining, rather τ_M is meant to be the current value at the instant of collapsing or rearrangement processes at that site which are accompanied by a reduction of the local dislocation density.

In our model, to be formulated in the following, the fundamental assumption is that these collapse and rearrangement events are the key process which cause dynamic recovery and we thus name the potential sites in the dislocations structure for such events to occur, the recovery sites. Since they represent fluctuations in local dislocation density, also the τ_M values vary from site to site, and can be characterized by a distribution function $f(\tau_M)$. At what level of the attained flow stress a recovery site will be activated does not only depend on its individual strength, but also on the locally acting forward stress τ_F , which in addition is influenced by local occurrences in the vicinity of that site. Therefore, also the different values of τ_F can be characterized by a distribution function $F(\tau_F)$.

The consumption and supply of recovery sites is described by the evolution of the distribution functions $f(\tau_M)$ and $F(\tau_F)$. These functions are likely to be related through the process of local storage and relaxation, but neither of them is likely to be ever derived on a first-principles basis. Yet the very existence of a distribution (which we could afford to gloss over in our brief exposition of static recovery) is essential for understanding the phenomena of strain hardening.

As explained in the last section, the experimentally observed constancy of the activation energy along stage III of a stress–strain curve, suggests that the variables τ_M and τ_F also maintain a fixed relationship with the flow stress τ in the course of straining. This will allow us to drop one of the variables by folding $f(\tau_M)$ and $F(\tau_F)$

into a new distribution function of solely τ_F (or alternatively solely τ_M), by making suitable assumption about one of them such as $\tau_M = \text{const.}$, or alternatively $\tau/\tau_F = \text{const.}$ In this way the modeling is reduced to considering the variation with T and $\dot{\gamma}$ of either $\tau + \tau_F$ or alternatively τ/τ_M . We have discussed resulting formalisms for these two possibilities in the literature [24,30,189] and we will follow up the latter one here, since the interpretation appears to be simpler.

The distribution function to be considered is then $\phi(\tau_M)$, and the underlying physical picture is, that the dislocation density as a whole can be considered as consisting of individual sites, each of which is characterized by its individual collapse or rearrangement stress τ_M . The total contribution $\delta\rho$ of all sites belonging to the strength interval $\delta\tau_M$ is then $\delta\rho = \phi(\tau_M)\delta\tau_M$ and the total dislocation density would be the integral over all increments. This integral changes with dislocation density, i.e. with the flow stress, so that ϕ must be looked at as a function of not only τ_M but also of τ . The total dislocation density is then given, as

$$\rho b^2 = \int_0^\infty \phi(\tau, \tau_M) d\tau_M / \mu \quad (46)$$

using the dimensionless variables ρb^2 and τ_M/μ .

For the purpose of adjusting unknown quantities in this formalism to experimental results it is convenient to consider the derivative with strain of the above distribution function

$$\Phi(\tau, \tau_M) \equiv \frac{d\phi(\tau, \tau_M)}{d\gamma} \quad (47)$$

Φ is a measure of the incremental increase of the number of recovery sites with the individual collapse strength τ_M caused by a strain increment.

The crucial fact that will allow a rudimentary model to be formulated of the simultaneous supply (hardening) and loss (softening) of recovery sites, is that the upper and the lower limit of the activation stress τ_M of the recovery sites are quite well defined. The lower limit is the flow stress τ since it is postulated that no sites can exist that would be unstable under the current applied stress. The upper limit $\hat{\tau}_M$ in the extreme would be the theoretical shear strength, but it can be much lower (not least because the forward stress most probably happens to be quite high at extremely strong sites) and, most important, $\hat{\tau}_M$ can be derived from comparison with experiments.

In this scheme the increase of dislocation density with strain at any particular value of the flow stress τ becomes

$$b^2 \frac{d\rho}{d\gamma} = \int_{\tau/\mu}^{\hat{\tau}_M/\mu} \Phi(\tau, \tau_M) d\tau_M / \mu \quad (48)$$

Not much is known about the function Φ , and thus we start out in a heuristic way with the simplest possible assumption, namely, that Φ does not depend on τ_M within the interval of existence of recovery sites, i.e. it is assumed that Φ stays constant in

the τ_M -interval between the (applied) flow stress τ and $\hat{\tau}_M$, and that it vanishes outside this range. In order to establish a connection with the model on statistical storage of dislocations outlined in Section 4, the actual value of Φ is supposed to increase during straining in proportion to the average dislocation spacing

$$\Phi = \Phi_0 b \sqrt{\rho} \quad (49)$$

Based on these two assumption, as the first step, a solution of the above integral will be presented for zero temperature, where the recovery sites are activated purely mechanically by the applied flow stress $\hat{\tau}/\mu = \alpha_0 b \sqrt{\rho}$ [see Eqs. (3), (4) and (7)]. In a second step (Section 5.3), the formalism will be extended to include thermal activation at $T > 0$. To illustrate, the assumed dependence of Φ on τ_M is sketched in Fig. 53 for two different levels of the flow stress at zero temperature, $\hat{\tau}_1$ and $\hat{\tau}_2 > \hat{\tau}_1$.

Under these conditions, execution of the above integral leads to

$$b^2 \frac{d\rho}{d\gamma} = \frac{\Phi_0 \hat{\tau}}{\alpha_0 \mu_0} \left(\frac{\hat{\tau}_M}{\mu_0} - \frac{\hat{\tau}}{\mu_0} \right) \quad (50)$$

This relation is of the Voce-type; it allows to determine the involved parameters by comparison with experimental and theoretical quantities of the foregoing sections as follows: the upper limit of the strength of the recovery sites, $\hat{\tau}_M$ appears to be identical with the Voce stress at $T=0$,

$$\frac{\hat{\tau}_M}{\mu_0} = \frac{\tau_{v0}}{\mu_0} \quad (51)$$

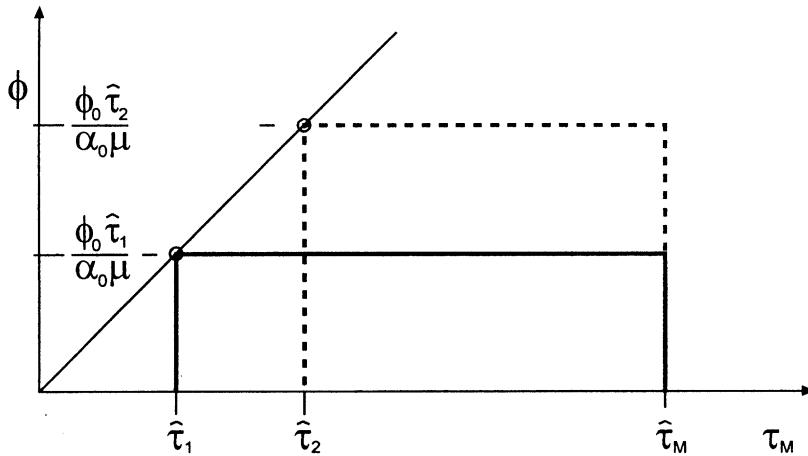


Fig. 53. Sketch of the assumed distribution function for the incremental change of the number of recovery sites with the individual collapse strength τ_M for the two flow stress levels $\hat{\tau}_1$ and $\hat{\tau}_2 > \hat{\tau}_1$ at zero temperature. Φ increases in proportion to $\hat{\tau}$ so that the normalized quantity $\Phi/(\hat{\tau}/\mu)$ is a constant.

and replacing the left hand side by $2\hat{\tau}\hat{\theta}/\alpha_0^2\mu_0^2$ leads with $\hat{\theta} \equiv d\hat{\tau}/d\gamma$ and $\hat{\theta}_0 \equiv \Phi_0\alpha_0\tau_{v0}/2$ to the Voce law at $T=0$ K

$$\hat{\theta} = \hat{\theta}_0 \left(1 - \frac{\hat{\tau}}{\tau_{v0}} \right) \quad (52)$$

In this model the highest possible value for the strength of all recovery sites in the dislocation network, $\hat{\tau}_M = \tau_{v0}$, is a material constant. It can be determined experimentally by extrapolation of the Voce stress to τ_{v0} at $T=0$ (see Fig. 28).

Finally, a short comment regarding the possible existence of a true stage II in the original sense of an initially straight-line portion of the stress–strain curve: it would be included in the above formalism by the simple assumption that, besides the upper limit, $\hat{\tau}_M$, of the distribution function, also a natural value of the lower limit exists, let's say at $\hat{\tau}_{III}$. Then recovery processes would not come into play until the flow stress has reached this level. The slope of the stress–strain curve up to that stress level would be

$$\hat{\theta}_{II} = \hat{\theta}_0 \left(1 - \frac{\hat{\tau}_{III}}{\tau_{v0}} \right) \quad (53)$$

Such an approach would be equivalent to the traditional τ_{III} -concept, where dynamic recovery has been ascribed to the activation of cross slip processes from τ_{III} upwards [55,56,92]. It is mentioned here for the sake of completeness, but it will not be followed up in detail.

A constant slope in stage II, typically, occurs in deformation by single slip at low temperature (see Fig. 31). It seems to be correlated with the existence of stage I of the work hardening curve. Possibly, it can be ascribed to the specific relaxation and stabilization processes of stage II which, to a large part, are driven by internal stresses in single slip orientations and not primarily by the applied stress, as is the case in polyslip orientations. In any case, we believe that effects of this nature account for the generally observed differences in hardening behavior of differently oriented single crystals and also differently textured polycrystals. Their influence is particularly pronounced at low strains where athermal hardening prevails, due to the sensitiveness of athermal hardening (θ_{II} and θ_0) about the actually set-up internal stress pattern and the developing obstacle structure. stage III behavior seems to be less affected, possibly, since the upper limit of the strength of the recovery sites, $\hat{\tau}_M$, reflects an inherent material property, not dependent on crystal orientation.

5.3. Dynamic recovery at finite temperature, thermal activation

So far the derived equations apply only for deformation at zero temperature, where recovery is a purely mechanical process and where, out of the spectrum of recovery sites, the weakest ones are eliminated for which the flow stress reaches the corresponding τ_M value. At higher temperature, however, thermally assisted recovery comes into play, the more the higher the temperature and the lower the strain rate. Then, the weak recovery sites with τ_M -values in a certain range immediately above the (applied) flow stress τ will already become unstable.

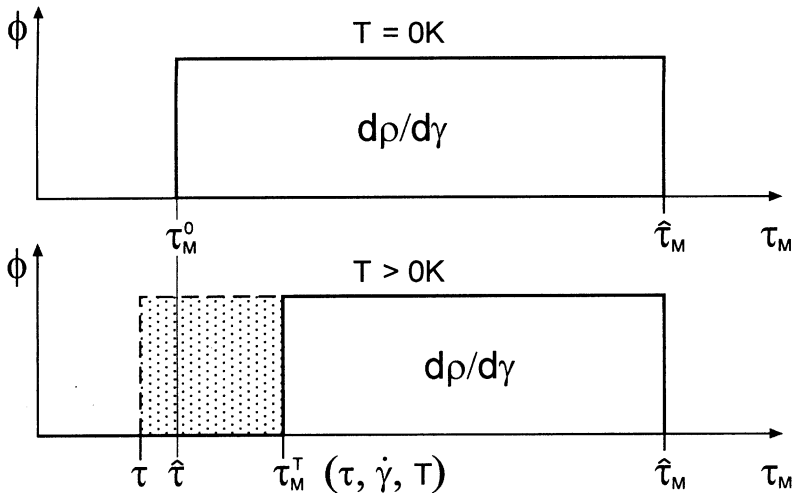


Fig. 54. Upper part: the area of the box represents the net storage of dislocations, $d\rho/d\gamma$; at 0 K where the distribution function Φ is truncated at the flow stress $\hat{\tau} = \tau_M^0$. The lower figure shows that, for the same value of $\hat{\tau} = \tau/s(\dot{\gamma}, T)$, at a higher temperature the truncation is at $\tau_M^T = \tau/r(\dot{\gamma}, T)$, since in the dashed region recovery sites are unstable under the action of the applied stress τ and thermal activation. (Note: for simplicity the normalization of all stresses by the shear modulus is left out here.)

As illustrated in Fig. 54 the effect of thermal activation can be easily incorporated in our scheme just by inserting as the lower limit of the integral in Eq. (48) a rate and temperature dependent τ_M -level, $\tau_M^T(\tau, \dot{\epsilon}, T)$, below which recovery sites are unstable under the action of the applied stress τ and thermal activation. The dependence of τ_M^T on τ at given $\dot{\gamma}$ and T is controlled by the stress dependence of the activation energy for triggering the recovery sites. Fortunately, as explained at the beginning of this section, in Section 5.1, this activation energy remains constant during a continuous stress–strain test, meaning that also the ratio τ/τ_M^T stays constant along a stress–strain curve:

$$\tau_M^T = \tau/r(\dot{\gamma}, T) \quad (54)$$

If this value is inserted as the lower limit of the integral in Eq. (48) the solution will be basically the same as Eq. (50); merely in the last term on the r.h.s. $\hat{\tau}/\mu_0$ is replaced by τ_M^T/μ . If also τ_M^T is substituted with the help of Eq. (54) then, as the equivalent to Eq. (52), we obtain

$$\theta = \theta_0 \left(1 - \frac{\tau/\mu}{r(\dot{\gamma}, T) \cdot \tau_{v0}/\mu_0} \right) \quad (55)$$

where $\theta = \hat{\theta} \cdot s(\dot{\gamma}, T)$ and $\theta_0 = \hat{\theta}_0 \cdot s(\dot{\gamma}, T)$. In this way also for the last remaining open quantity of the model, r , the empirical determination is at hand from a comparison with Eq. (29),

$$r = \left\{ 1 - \left(\frac{1}{g_0} \frac{kT}{\mu b^3} \ln \frac{\dot{\gamma}_0}{\dot{\gamma}} \right)^{1/2} \right\}^2 \quad (56)$$

A convenient alternative is the power relationship

$$r \cong \left(\frac{\dot{\gamma}_0}{\dot{\gamma}} \right)^n \quad (57)$$

with the temperature dependent exponent $n = kT/A$. As discussed in Section 3.3, this equation reflects a logarithmic stress-dependence of the activation energy, which is a good approximation in a limited stress interval. Inserted into Eq. (55) it leads to a relation of the same type as Eq. (19), which had been introduced in Section 3.1 empirically, in order to demonstrate the difference between time recovery and observed behavior.

It is seen that the complex interplay between work hardening and softening is described surprisingly well by the rather simple approach, assuming a box-like shape of the distribution function Φ for the accumulation of recovery sites of different strengths. An essential element of the model is the new variable τ_M^T as the characteristic of the width of this distribution. It can be interpreted as some measure of how narrowly the link length is distributed within the dislocation net work. In this way the main features of the dislocation structure are classified by two macroscopic variables.

The primary variable is $\hat{\tau}$, it quantifies the total dislocation length per unit volume. The secondary variable τ_M^T characterizes the uniformity of the dislocation network. It is a measure of how far dynamic recovery has advanced in converting the virtually random pattern, formed at low strain where $\tau_M^T \approx 0$, to a perfectly regular pattern at larger strain, where, in the end, only a single collapse strength survives when τ_M^T has advanced to its upper limit $\tau_M^T = \hat{\tau}_M$. As seen from the regrouping of Eq. (55) in connection with Eq. (54),

$$\frac{\tau_M^T}{\mu} = \frac{\tau_{v0}}{\mu_0} \left(1 - \frac{\theta}{\theta_0} \right) \quad (58)$$

τ_M^T is uniquely linked to the work hardening coefficient, at any deformation temperature or strain rate in the same way. This is an important consequence of the model; some implications will be discussed in the following sections.

Two small additions: The current scheme would lead to a steady state limit only if the collapse of dislocations inside tangles were accomplished by complete annihilation. If instead of annihilation there is a constant rate of debris formation, this should lead to a continuous increase of the collapse strength $\hat{\tau}_M$, except presumably of the highest of all probable ones; the theoretical shear strength. This mechanism is independent of whether one assumes [190] that there is a critical distance below which both screw and edge dislocations annihilate mechanically. Rather it corresponds to a secondary storage mechanism, namely, debris accumulation inside collapsing tangles—rather than tangle formation due to stage II processes—as

it has been proposed for a stage IV mechanism. It can be quantitatively described by [149]

$$\frac{d\tau_v}{d\dot{\gamma}} = \theta_{IV}; \tau_s(\dot{\gamma} = 0) \equiv \tau_v = (\tau_{v0}\mu/\mu_0)r(\dot{\gamma}, T) \quad (59)$$

Replacing in Eq. (55) τ_v by τ_s with the help of the above equation would then give the complete strain hardening behavior, assuming that all the hardening processes of the stages II, III and IV, are simultaneously active (though with different weightings in the different regimes).

The second additional remark is about the interdependence between the inverse stress exponents $1/n$ and $1/m$ in Eqs. (23) and (24). In many investigations work hardening is treated as the evolution of only a single state variable, namely, the total dislocation density, represented by $\hat{\tau}$. As a consequence the inverse stress exponent $1/n$ in Eq. (23) would then be the sum of the constant structure rate sensitivity, $1/m = d \ln s / d \ln \dot{\gamma}$, and the one that has to do with the rate sensitivity of the evolution of $\hat{\tau}$. The difference $1/n - 1/m = (\partial \ln \hat{\tau} / \partial \ln \dot{\gamma})_{\hat{\theta}/\hat{\theta}_0}$ is then assumed to be the relevant quantity that is solely related to dynamic recovery processes [29,191]. Follansbee [192] has actually performed as a function of temperature the painstaking experiment that allowed him to extrapolate to zero temperature, to obtain as a state parameter the mechanical threshold.

$\hat{\tau}$ is the ruling quantity, indeed. In the above model, however, we have identified τ_M^T as a second state variable, that is needed for a more accurate description of the observed phenomena. It is related with the flow stress τ and the work hardening coefficient θ by Eqs. (54) and (58), leading to a direct connection of the stress exponent n with the recovery function r

$$\frac{1}{n} = \left(\frac{\partial \ln r}{\partial \ln \dot{\gamma}} \right)_T \quad (60)$$

without any modification by the constant-structure rate-sensitivity $1/m$. We believe that, beyond our modeling, this finding has general validity and is a reflection of the fact that, at a given temperature and strain rate, dynamic recovery is driven by the applied stress (which is always the flow stress τ and not $\hat{\tau}$). In the present context, a fortunate consequence of this result is: τ and θ (and their counterparts σ and Θ), used throughout this paper, are the physically adequate quantities for the analysis of work hardening behavior; the use of the (experimentally) rather inconvenient quantity $\hat{\tau}$ even turns out to be superfluous in this context.

5.4. Transients, state parameters and structure characteristics

The effective lower limit of the distribution of the strength of the hard spots was labeled τ_M^T . If one were to abruptly change the temperature or the strain rate, the equilibrium value $\tau_M^T(T, \dot{\gamma})$ would change immediately—but the dislocation structure corresponding to the original value cannot change instantaneously: it can only

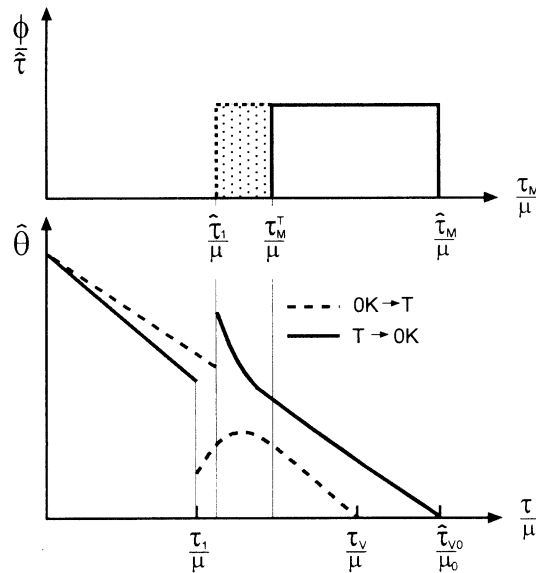


Fig. 55. Change of the position of the effective τ_M^T value and corresponding transients in strain hardening behavior upon an abrupt change of the external conditions. Sketched is the specific case of temperature changes between zero temperature and $T > 0$ at the flow stress level τ_1 : dashed curves for an increase of temperature (initial deformation at 0 K up to $\hat{\tau}_1$ and then the temperature is changed to $T > 0$ and the test continued from $\tau_1/\mu = s \cdot \hat{\tau}_1/\mu_0$ on), solid curves for a decrease of temperature to 0 K (after initial deformation at $T > 0$ the distribution function for the strength of recovery sites is truncated at τ_M^T , which is higher than after initial deformation at 0 K where the truncation is at $\tau_M^0 = \hat{\tau}_1$).

adjust itself in the process of straining, and this leads to transients in the stress–strain behavior.

Let us first consider the case that the temperature was lowered or the strain rate raised. Then, due to the drop in thermal activation as shown in Fig. 55, τ_M^T assumes a lower value than the one just before the change in conditions and the interval between the two, for which the distribution function is empty (shaded area), needs to be filled up with dislocations. This will occur at a high hardening rate—initially presumably θ_0 , then decreasing to meet the new conditions.

To estimate the length of the transient in terms of strain for a change of strain rate by $\Delta \ln \dot{\epsilon}$, one can set $\Delta \gamma = \Delta \tau_M^T / \theta_0$, and then $\Delta \tau_M^T = \tau_M^T / n \Delta \ln \dot{\gamma}$, where $1/n$ is the rate sensitivity of τ_M^T . If the change in strain rate was made near steady state, one can set $\tau_M^T \cong \tau_v$ and obtains

$$\Delta \gamma = \frac{\tau_v}{\theta_0} \frac{\Delta \ln \dot{\gamma}}{n} \quad (61)$$

Note that the first ratio is the Voce strain: the length of the ‘transient’ that represents the whole stress–strain curve on its way to steady state. If the rate was changed by a factor of 10, and the ‘stress exponent’ is, say, 20 (as typical for Cu at room

temperature), then the length of the rate-change transient is about 1/8 of the Voce strain. It is in fact typically of order 10% (3% in terms of ε)—and 100% is not an unreasonable value for the Voce strain either.

Now consider the converse effect: raise the temperature. Now, as sketched in Fig. 55, the τ_M^T value mandated by the new conditions is *higher* than the previous one (at $\tau_M^0 = \hat{\tau}_1$), which is recorded in the current state of the dislocation structure, and all the recovery sites of the interval between the two values (shaded area in Fig. 55) will become unstable and will be removed quickly. Presumably this will happen preferentially in regions where new slip activity takes place since the established forward stresses are expected to be rather high in these regions. To some extent, this is an unstable process, and indeed strong slip traces are observed under these conditions; but the mechanism that stops slip on an active plane (Fig. 45b) is still operative—and it must be so, because otherwise the specimen would slip off on the first such newly active plane. Thus, the process of ‘work softening’ is complete when the whole specimen has experienced new deformation. If, say, 10 dislocations pass on every active slip plane and these are 0.1 μm apart, it would take 5% shear strain to screen the whole sample: a reasonable estimate.

Every transient is evidence for a *state parameter* that evolves toward *its* steady state under the given applied conditions. Thus, as mentioned before, the whole stress–strain curve may be viewed as a (very long) transient during which the state parameter τ evolves toward its steady-state value (or toward a moving target, see Section 5.2). Similarly, the existence of a transient upon a change in externally prescribed conditions part-way up the stress–strain curve calls for an additional state parameter. We have identified this as τ_M^T , the lower cut-off of the distribution of ‘hard-spot’ strengths.

To summarize: We have identified two state parameters which evolve with deformation in a coordinated way. The first one is the flow stress, τ . It is a measure of the total dislocation density and depends only weakly on strain rate and temperature, which is measured as the instantaneous response to sudden changes of temperature and strain rate. The second one is the lower limit of the strength distribution of the individual recovery sites of the dislocation structure τ_M^T . The difference between the material constant $\hat{\tau}_M = \tau_{v0}$ and the state variable τ_M^T is a measure of the distribution width of the strength of the recovery sites, which can be interpreted as a measure of how regularly the dislocations are arranged in the network. It depends much more strongly on temperature and strain rate than the flow stress does. Also it correlates with the work hardening coefficient: any level of the hardening rate, θ , in continuous stress–strain tests corresponds to a certain value of τ_M^T independent of the temperature or strain rate; at $\theta = 0$ is always $\tau_M^T = \hat{\tau}_M$.

Other transients we have mentioned before are the Bauschinger transient upon stress reversal and the post-recovery transient in strain-hardening rate; both are related to the change in dislocation structure as discussed here—so far a quantitative assessment has (to our knowledge) not been made to ascertain that they all depend on the *same* second state parameter. We believe, however that the developed formalism for the description of hardening behavior in stress–strain tests is also applicable for transient creep and forming behavior under changing conditions.

Even if our model does not refer to the spatial distribution of the dislocations the question to be discussed is whether it agrees with microscopic observations of the evolving dislocation patterns. Our physical picture is that during straining dislocations are accumulated by storage processes in randomly distributed tangles. However, these are not assumed to represent stable configurations, but rather to consist of hard spots, the recovery sites, each characterized by an individual stress value, τ_M , where it collapses or rearranges into a new configuration. The upper bound of the strength of recovery sites has the fixed value $\hat{\tau}_M$ but the lower bound, τ_M^T , increases continuously in the course of strain hardening until they both coincide and only a single site-strength survives, namely, $\hat{\tau}_M$. From the point of view of structure evolution the above scenario offers the following interpretation: The loose patterns formed in the beginning of deformation possess a wide distribution of segment lengths. As straining goes on, it will be transformed gradually into a regular arrangement, whereby the distribution of segment lengths narrows down continuously until only a single length survives, the one corresponding to $\hat{\tau}_M$. From there on the characteristics of the structure do not change anymore since the storage and recovery processes have established a dynamic equilibrium by which every collapsing hard spot will be replaced by a new one with the same properties, though not necessarily at the same location. In this process the temperature and strain rate control the overall density of the hard spots, but have no effect on their quality.

5.5. The cell structure. Misorientations

This paragraph deals with the questions, whether and which characteristic parameters of the dislocation substructure can be linked to the model parameters, and whether the physical concept, for the interpretation of macroscopic quantities and their evolution with strain, is consistent with microscopic observations.

The cell structure observed in heavily deformed pure metals gives the impression of being a two-parameter arrangement. But that is misleading: the cell diameter (for reasonably equiaxed cells) is known not to be an independent parameter, but be inversely proportional to the square root of the average dislocation density and thus to the ‘first’ state parameter, the flow stress.

Then the question arises whether the second parameter, that is demanded for a fuller description of macroscopic behavior, is at all identifiable in TEM pictures of the dislocation substructure (which, of course, contain many more than two parameters). The ratio of the dislocation densities in the cell interior and the cell walls becomes so low at large strain that it seems to be no meaningful second parameter. A quantity that offers itself from the basic model outlined above and in Section 5.2 is the *thickness* of the cell walls, which decreases with increasing total amount of (dynamic or static) recovery. It could be connected with the width of the distribution of the strength of hard spots, which also shrinks with progressing recovery in proportion with the decreasing difference between $\hat{\tau}_M$ and τ_M^T , the second state parameter.

A third structural feature is the misorientation between adjacent cells. They are sometimes prominently discussed as an important aspect of the substructure. This is

quite plausible when, for example, subsequent recrystallization behavior is discussed. On the other hand, there is no *prima facie* reason why they should be relevant to the flow stress or hardening behavior.

There are two types of misorientations. One relates to the difference in orientation between cells, i.e. from one side of a cell wall to the other. In dislocation language, this is proportional to the *excess* dislocations of one sign; it relates to small differences in deformation in two adjacent cells, and is always small compared to the *balanced* distribution. The misorientation of this type typically saturates at about $2\text{--}3^\circ$ [77]. This angle appears to be quite insensitive to changes of the external conditions such as temperature and strain rate or the mode of deformation, tempting us to speculate, that it is possibly interconnected with the upper bound for the strength of the recovery sites, $\hat{\tau}_M$, which plays a key role in our model, even if it is only a material constant and not a state parameter. Furthermore, this type of cell walls (subboundaries in the limit) seems to behave in a similar manner as the (unspecified) recovery sites of our model. The fact that the cells remain equiaxed up to rather large strain is a reflection of a dynamic equilibrium between dissolution of existing and creation of new cell walls. Thereby the average misorientation angle remains always the same and changes of the deformation conditions do affect the volume fraction but not the quality of the considered cell walls. This corresponds very well with behavior of the recovery sites in our model, which always approach the same quality, characterized by the unique strength $\hat{\tau}_M$.

There can, however, be a different type of misorientation that is much larger; namely, between the *interior* of a (thick) cell wall and either side. This would be expected on the basis of the fact that the dislocations that come from one side (and are therefore stored on that side) have the opposite sign of those that come from the other side, can build up a substantial misorientation [187]. This is especially evident when the two sides have formed separate cell walls. Since only a small fraction of cell walls is of this second type, we assume that they are of minor importance for the macroscopic properties but that they can play a key role in recrystallization as nucleation sites for new grain orientations.

Although cell walls certainly are dominant in the substructure at large strains, also other wall-like structures are set up in the course of straining including large-angle boundaries [193]. These are not recoverable by slip processes and are assumed to be of no particular importance for work hardening in the regime of dynamic recovery, though they might have some influence in stage IV as in the model of Beaudoin and coworkers [130] discussed in a preceding paragraph.

What is the general outcome of this discussion? We have demonstrated, not only that the model adopts the macroscopic phenomena very well, but also that its general physical frame is in excellent agreement with observations on the evolution of the substructure. Furthermore, it goes very well together with the concept of similitude discussed in Section 2.3.4, by correlating the second state variable with the thickness of the cell walls.

At this point [see also 85] a short comment seems to be appropriate regarding the relation between flow stress and substructure: It has become the fashion in literature

[11–13,171] to describe the flow stress of a material containing a cell structure by the rule of mixture

$$\tau = f_i \alpha_i \mu b \sqrt{\rho_i} + f_w \alpha_w \mu b \sqrt{\rho_w} \quad (62)$$

where f_i and f_w denote the volume fraction of the cell interiors and the cell walls, ρ_i and ρ_w are the corresponding dislocation densities and α_i , α_w are numerical constants, which depend on the respective geometrical arrangements of the dislocations. For the sake of simplicity $\alpha_w = \alpha_i = \alpha_r$ is assumed here i.e. the dislocations are considered to be randomly distributed in each subvolume as marked by the subscript r of the constant α_r . Application of Eq. (62) implies that the cell interiors and the cell walls can be treated like two separate phases. This, in our opinion, can be misleading in the present situation: Different from a two-phase mixture, in a cell structure the volume fractions change continuously with straining and there are no volume areas where cell walls preferentially develop, a spot belonging to a cell wall may be part of a cell interior at a later stage of deformation. Even more important is the fact, cell walls are not separated from the cell interior by interfaces, rather dislocations can be easily exchanged between the constituents. Therefore (and by additional reasons given in the last section of this chapter) it is doubtful whether the above equation—if at all applicable—is more appropriate than the simple relation used throughout this paper

$$\tau = \alpha \mu b \sqrt{\rho} \quad (3)$$

where ρ is the average dislocation density. As will be seen below, however, even if this convenient equation were physically less accurate than Eq. (62), application of the latter would still lead to a useless complication of the mathematics. Elimination of the flow stress by combining Eqs. (62) and (3) together with the trivial relationship

$$\rho = f_i \rho_i + f_w \rho_w \quad (62a)$$

leads us to

$$\alpha = \alpha_r \left(f_i^{1/2} + [\zeta f_w]^{1/2} \right) (1 + \zeta)^{-1/2} \quad (63)$$

where $\zeta = f_w \rho_w / f_i \rho_i$ is the ratio of the total dislocation length in the cell walls and in the interior (generally $\zeta \approx 1$ seems to be a good estimate). It turns out that, at least for the cell patterns presented in this paper, α would be rather insensitive to the details of the dislocation arrangement but would stay always close to α_r ; even for the rather extreme case that f_w is gradually reduced to 10% of f_i (i.e. the cell wall thickness decreases to merely 3% of the cell diameter) and that ρ_i approaches a level one order of magnitude smaller than ρ_w the value for α would stay in the interval $\alpha_r \geq \alpha \geq 0.9 \alpha_r$. This difference is much smaller than the uncertainty in any theoretical or experimental determination of these constants.

5.6. Summary: state variables, master curves and dislocation patterns

We have abstracted from the experimental analysis (Section 3), and derived from the general features of a model (Sections 5.2 and 5.3), that there should be two state variables: The flow stress τ and the width of the distribution of the collapse strengths of the recovery sites, characterized by τ_M^T . We will summarize in this section the basic equations for the evolution of the two state variables as a function of temperature and strain rate and in relation to structure evolution.

To start with the flow stress: the variable that evolves is really the dislocation density of the material, as it may be characterized by the mechanical threshold $\hat{\tau}$, according to

$$\hat{\tau}/\mu_0 = \frac{\tau/\mu}{s(\dot{\gamma}, T)} = \alpha_0 b \sqrt{\rho} \quad (64)$$

obtained by combining Eqs. (3), (4) and (7). For comparing the various mechanical states that have been established in the course of deformation, the proper variable to be used is then $\hat{\tau}$, the flow stress at $T=0$ K where $s=1$.

In many investigations $\hat{\tau}$ has been considered to be the only relevant state parameter. It is the ruling quantity, indeed, by modeling dislocation storage, however, we have identified a second state variable, τ_M^T . It is needed, in particular, for the description of the long transients in work hardening rate upon sudden changes of temperature or strain rate.

In a continuous test the two state variables evolve in an equilibrated way, whereby the actually established equilibrium is dictated by the chosen testing method. It will be different in a stress–strain test, with $\dot{\gamma}, T = \text{const}$, as compared to a creep test, with $\tau, T = \text{const}$, or as compared to a relaxation test, with $\gamma + \gamma_{\text{el}}, T = \text{const}$ (where γ_{el} includes the total elastic strain of the sample and the testing frame). The present investigation, exclusively, deals with stress–strain test, and it has been found that in this testing mode the two state variables evolve in proportion to each other

$$\tau_M^T/\mu = \frac{\tau/\mu}{r(\dot{\gamma}, T)} \quad (65)$$

τ_M^T is a measure of how far dynamic recovery has advanced, at a given temperature and strain rate, under the action of the applied stress τ with the help of thermal activation. The dynamic-recovery function r is equivalent to the inverse of g in Eq. (26). A specific form is given in Eq. (56) and as an alternative the popular power law in Eq. (57).

$\hat{\tau}$, τ and τ_M^T attain identical values at 0 K, where $s=r=1$. However, they will be separated at higher temperature due to thermal activation, whereby the impact on τ_M^T is much stronger than on τ (i.e. $s>r$). Then the actual level of τ_M^T moves ahead of τ by the factor $1/r$, which becomes the larger the higher the temperature and the lower the strain rate.

The flow stress follows an evolution function that describes the rate of strain hardening in a scaled form

$$\frac{\theta}{\theta_0} = \mathbf{E}\left(\frac{\tau}{\tau_v}\right) \quad (66)$$

Here θ_0 is proportional to $\mu(T)$ and of the order $\theta_0 \approx \mu/200$. As shown in Sections 5.2 and 5.3, strictly speaking, $\theta_0/\mu = \theta_0/\mu s(\dot{\gamma}, T)$ is a constant in our model, rather than θ_0/μ . We believe, however, not much would be gained by such sophistication, because the use of plots of $\hat{\theta}$ instead of θ versus τ will lead to virtually the same results. τ_v in the above equation is a stress scaling parameter that can be chosen to be the extrapolation to zero hardening rate of the essentially linear decay of θ with τ in the middle range. Its purpose is to unify strain-hardening curves for a large range of temperatures and strain rates, according to another general function that describes dynamic recovery:

$$\tau_v = \mathbf{R}(\dot{\gamma}, T) \quad (67)$$

Eqs. (66) and (67) represent two ‘master curves’ that should be derived from experiment. The primary purpose of these two equations is to provide combinations of variables that are based on the underlying physical mechanisms.

We have given specific forms of these equations (still with free parameters in them), which we have found to match both experimental and theoretical observations—but which we feel are not *ab initio* mandates. For the evolution function $\mathbf{E}$, the linear decay over a substantial range [like in Eqs. (52) and (55)] is a key feature, but never true over the entire range. Two variations that we have employed are: one based on a physical mechanism for ‘stage IV’ behavior in the form of Eq. (55) modified with the help of Eq. (59); and another strictly heuristic one obtained by replacing σ and Θ by τ and θ resp. in Eq. (22), which was first proposed by Bronkhorst et al. [91] and which also decays linearly initially and extrapolates to $\tau_v(T, \dot{\gamma})$.

The dynamic recovery function $r = \mathbf{R} \cdot \mu_0 / \tau_{v0} \mu$ is the inverse of g in Eq. (26); the best fit we know is based on Eq. (29) and is explicitly given in Eq. (56) which contains the two material parameters τ_{v0} and g_0 (of order $10^{-2} \mu$ and 1, respectively). However, the customary plot of the *logarithm* of τ_v/μ versus the normalized activation energy (the argument of the function in Eq. (56) is also useful; it is equivalent to a ‘power law’ stress–strain-rate relation with the stress exponent $n = A/kT$ (72) where A is proportional to μb^3 and is moreover, when the $\log(\tau)$ vs ΔG plot is not straight, a function of stress.

We like to continue this summary with a few remarks about the general behavior and the interplay between the two state variables. Eqs. (64) and (65) both contain the flow stress explicitly, reflecting the fact that the dislocation density is the dominant physical quantity. It may change during deformation by several orders of magnitude, depending on external conditions. At low temperature the flow stress reaches values up to about 0.01μ corresponding to $\rho = 10^{11}/\text{cm}^2$ while at higher tem-

perature (at $g = g_0$) where the stress level is as low as, let us say, 0.0003μ , the dislocation density does not exceed $10^8/\text{cm}^2$. The second state variable τ_M^T , in a certain sense, operates in a catalytic way by controlling the rate of dislocation storage, which decreases continuously towards zero as τ_M^T advances on its way towards the limiting value $\hat{\tau}_M = \tau_{v0}$. Independent of temperature and strain rate, τ_M^T covers always the same range and is uniquely linked to the work-hardening rate by Eq. (58). It is a measure of the progression of dynamic recovery e.g. by converting the random dislocation tangles, forming at the beginning of deformation, into a regular cell structure at later stages of straining. Like the flow stress τ , also τ_M^T enters both Eqs. (64) and (65), though Eq. (64) only indirectly, where it is hidden in the geometrical constant α_0 ; the variable τ_M^T can be interpreted as a measure of the cellularity and, as such, it is expected to come along with a gradual, though minor decrease of α_0 in a 10% range.

In summary, the emerging picture of the interplay between dislocation glide, hardening, softening and structure evolution is as follows: On an atomistic level plastic flow is always very heterogeneous. It is restricted to the weak channels in the dislocation network where the shear rate is concentrated in slip lines. An individual slip line will be clogged by work hardening, after it has produced locally a rather large amount of shear in an incrementally small volume so that it adds only an increment of strain on a macroscopic scale. It is then replaced by slip activity at a different location at another weak spot with an incrementally higher strength level. Thus, probably assisted by dynamic pile ups, the squeezing of dislocations through the critical gates in weak channels is assumed to control the actual level of the flow-stress. In this concept there is no room for the rule of mixture which, in a strict sense, is a continuum approach. It implies, that everywhere in the sample the locally acting stress is equal to the local glide-resistance or, equivalently, that the plastic strain rate is the same everywhere. Only under these conditions would the macroscopic flow stress be the volume average of the flow stresses in soft and hard regions.

Of course, the applied stress causes some slip activity everywhere in the dislocation structure also at a stress level below the local flow stress. This, however, is short range slip, contributing only a negligibly small fraction to the total macroscopic strain. It is connected with the local collapse of metastable dislocation configurations and it is generally accompanied with a loss of stored dislocations. These are the genuine processes of dynamic recovery in our model. What is then the role of the second state variable τ_M^T and how is it recorded in the dislocation structure? No site exists in the dislocation net work with a collapse strength lower than τ_M^T and as such it characterizes (in combination with $\hat{\tau}_M$) the uniformity of the dislocation structure.

The presented model is a strictly phenomenological one, and therefore the developed formalism is generally applicable. Its limitations lie in the fact that it operates with global averages but does not specify the precise nature and the spatial distribution of individual recovery events. It also leaves open to what extent the recovery events occur consecutively or in parallel with the formation of slip lines i.e. whether we deal with stress controlled work softening (triggered by the globally applied stress in combination with short range slip) or with strain softening (triggered by the local stress fields in the vicinity of active slip lines). Most probably, these two possibilities occur simultaneously in reality. Thereby, the stress controlled

recovery possibly dominates in the early part of stage III where, due to the rather high work hardening rate, the stress increases rapidly with deformation, while strain softening takes over at large strains where the work hardening rate is low. Treating these questions in more detail would require a refined description of the distribution functions for the locally acting forward stresses $F(\tau_F)$ and for the local collapse stresses $f(\tau_M)$ and how these functions evolve in the course of straining [24,30,189].

The following remark concerns the possible presence of other contributions to the flow stress. The only evolution we have considered is that of the *dislocation contribution* to the flow stress. When other contributions are known to exist, one must find the superposition law and adjust for them [78]. For example, when there is a small amount of substitutional solute hardening, as is frequently the case, one may consider this additive; the quantity τ used above is the actually observed flow stress minus the yield strength, and similarly the quantity τ_v is the actually observed ‘Voce stress’ minus the yield strength.

Finally, the entire discussion in Sections 4 and 5 has dealt with the glide resistance to dislocation motion, i.e. the shear flow stress τ in a *crystallographic* slip system. What is typically observed is the behavior of polycrystals, and one must have a viable model to relate the two. The situation gets complicated by the fact that the texture evolves during deformation and that thus there is no constant ‘orientation factor’. It has been outlined in detail in Refs. [47,116,117,122], and an example is given in Section 3.6, how one can proceed in analyzing polycrystal experiments: by *assuming* a unique hardening law in the crystals and feeding it into a computer simulation of polycrystal plasticity, including texture changes—then modify the input until the output matches a subset of experiments.

6. Conclusions

6.1. Crucial phenomena

The fundamental process of model development starts with an assessment of which physical phenomena are considered crucial (and which ancillary). Some of the facts we consider crucial are.

The strain-hardening behavior for different fcc metals does not follow a sequence based on melting temperature or diffusion coefficient, but of (normalized) stacking-fault energy (SFE), $\chi/\mu b$.

It does not change in a qualitative way from near zero absolute temperature to well above half the melting point. In this context, it is an important realization that it is possible for diffusion to occur without exerting any significant influence on plasticity.

The microstructures that develop become more heterogeneous the larger the strain, the higher the temperature, the lower the strain rate. They are similar in appearance to those observed during static recovery with increasing time.

Comparing pure fcc metals with other materials, one can conclude that the hardening behavior described above occurs only when slip is areal rather than lineal, i.e.

when screw and edge dislocations move with comparable ease; and only when slip systems on intersecting slip planes are available.

Thermal activation, with a stress-dependent activation energy, plays a central role, as evidenced by the fact that the temperature and strain-rate dependencies can be unified under this umbrella.

The highest shear-strain-hardening rate observed in single-phase single crystals and—in the presence of a sufficient number of slip systems—polycrystals, is of the order of, but less than $\mu/100$, and athermal.

The decrease of the hardening rate with strain is strongly dependent on temperature and weakly on strain rate—indicating a need for a stress-dependent activation energy.

The entire strain-hardening regime (athermal and ‘dynamic recovery’, and perhaps even ‘stage IV’ processes) can be treated as a single-parameter set, demanding ‘similitude’ of the relevant structural elements.

One aspect of the structure that changes with strain in a non-self-similar manner is the thickness of the cell walls; one aspect of strain-hardening behavior that cannot be described by a single state parameter is the appearance of transients upon a change in conditions.

6.2. *Main features of the model*

A main aspect of model building is the format of the model, and its aims.

A crucial requirement for models in materials science is that they must be based on the structure/properties connection. In phenomenological terms, this means that they can only involve state parameters; history variable such as strain and time can only appear differentially. Thus, strain-hardening behavior must be expressed by *differential* equations; the integral over a specific path gives the stress–strain curve.

Another desirable feature of models is that they should be expressed in terms of combinations of variables that follow from the physical basis and are dimensionless; in plasticity, typical examples are θ/μ , τ/μ , $kT/\mu b^3$ (so long as diffusion is not relevant), and $\chi/\mu b$. The actual relation between these variables is often not obtainable in closed form, but may well have to be specified by a graph or table [45]; only over a small range, or in an approximate overall form, should one be able to verify the *kind* of behavior and the right orders of magnitude. In this article we have identified a new graph that holds over a very wide range

Finally, an obvious criterion for predictive value is that the number of free parameters should be as small as possible—typically, the number of macroscopic variables minus one.

The model, summarized in this article, fulfills all the requirements emphasized above. It has the following main features.

The flow stress is described by the percolation limit of dislocations in areal glide through the weakest links on the hardest lines in an array of point-like obstacles (probably representing primarily the breaking of attractive junctions with forest dislocations), with significant variations of glide resistance from place to place.

This leads to a geometrical storage of dislocations (in the form of concave loops around ‘hard spots’) which leads to an athermal contribution to strain hardening of the right order of magnitude—while keeping the slip line length large compared to the hard-spot spacing.

An interaction of these concave loops with surrounding dislocations of intersecting slip systems (‘plastic relaxation’) leads to a stabilization of the local arrangement and its spreading into the third dimension in the form of ‘tangles’. (Note that our plastic relaxation achieves the opposite of that used by Hirsch et al., viz. a *decrease* in the calculated hardening rate.)

Linking-up of hard spots after the passage of a certain number of dislocations on a slip plane causes this slip plane to stop and deformation to spread through the bulk. In the process, the tangles get connected into loose cell walls.

A prediction of the model is the build-up of an ordered pattern of internal stresses: high forward stresses inside the tangles or cell walls; low back-stresses in the cell interior; and no internal-stress contribution to the flow stress at the ‘critical gates’. (Note that these internal stresses are connected with pile-ups—but they have a certain *dipole pile-up* character inasmuch as they consist of concave, not convex, loops.)

The forward internal stresses inside the tangles and cell walls provide the driving force first for the stabilization of tangles by plastic relaxation involving secondary slip systems, and then, in a second step, for static and dynamic recovery. Thermal activation helps complete the process of junction breaking in the hard spots.

Dynamic recovery gradually converts the dislocation structure from a more-or-less random into a cellular pattern. The impact of thermal activation on dynamic recovery, is much stronger than on the glide resistance.

While the flow stress is controlled by the total dislocation density, dynamic recovery is uniquely linked to the distribution of the strengths of junctions in the hard spots, the recovery sites. The interplay between these two quantities can be described by introducing, in addition to the flow stress, a second state variable, the lower limit of the strength of recovery sites. It is a measure of how far dynamic recovery has advanced in establishing a unique cell structure. It allows to model transients, occurring in connection with abrupt changes of the deformation temperature and strain rate.

The rearrangements within the dense areas rarely lead to an annihilation of dislocations, but more typically to the formation of a dipole debris—which could be responsible for an increase in the hard-spot strength and ultimately for a contribution to ‘stage IV’ processes.

The mechanisms of athermal storage, dynamic recovery, and the storage of dipole debris all occur simultaneously, not in ‘stages’ of work hardening.

6.3. Data analysis

The prescriptions for an analysis of experiments which follow from the model (and were in turn the factual basis for the model) are the following.

A set of experiments at a number of different temperatures and strain rates should be plotted as Θ/μ versus σ . It should be possible to unify this set of curves by pick-

ing a single scaling factor Θ_0/μ (using $\mu(T)$) and a set of scaling factors σ_v , one for each condition. (If this does not seem to work well enough, there may be other mechanisms influencing flow stress and strain hardening.) The resulting first master curve should consist of a linear decay at intermediate flow stresses, normally with deviations at higher stresses such that it is concave up.

The resulting scaling stresses should be plotted as $\ln(\sigma_v/\mu)$ versus the normalized activation energy, $\Delta G_{\text{norm}} = (kT/\mu b^3) \ln(\dot{\epsilon}_0/\dot{\epsilon})$, with the free parameter $\dot{\epsilon}_0$ to be picked such as to unify the temperature and strain-rate dependencies (and within a couple of orders of magnitude of 10^7 s^{-1}). If an analytical description is desired, try plotting $\text{sqrt}(\sigma_v/\mu)$ versus $\text{sqrt}(\Delta G_{\text{norm}})$, and determine the intercept $\text{sqrt}(g_0)$.

Find g_0 on the plot (Fig. 30) of g_0 versus $\chi/\mu b$, and ascertain that the latter is well ordered with respect to other materials. Then Fig. 30 may serve as a guideline where the master curve of a material should be located in a $\ln \sigma_v/\mu$ – g -plot.

For a more detailed and more accurate determination of the physical hardening behavior, one must measure the initial texture, use the above results as a guideline for a postulated grain-level hardening law [47], and feed this into a polycrystal plasticity simulation that predicts reasonable texture changes with deformation. The resulting stress–strain curve, after suitable variations in the input law, must match a set of observed stress–strain curves—and can then serve to predict the behavior under other test conditions (so long as the deformation mechanisms do not change).

6.4. Range of application

The presented mechanistic interpretation and the developed model is restricted to monotonous deformation at low enough temperature where dynamic recovery is governed by conservative glide of dislocations, rather than by time recovery in conjunction with self diffusion and dislocation climb.

It is still an open question where exactly the control of diffusion begins. We have discussed the transition to the diffusion regime several times in literature [26,27,40]. Based on the analysis of data on single-crystal [40] the transition was assumed to occur at a temperature where the stress exponent n in Eq. (23) approaches $n \cong 4$ (a value which has been established as the ultimate value of the stress exponent for steady state creep of pure fcc metals in [194]). Differentiation of the empirical Eq. (29) [or the alternative Eq. (68)], leads to

$$n = \left(\sqrt{\frac{\mu b^3 g_0}{kT \cdot \ln(\dot{\epsilon}_0/\dot{\epsilon})}} - 1 \right) \ln(\dot{\epsilon}_0/\dot{\epsilon}) \quad (68)$$

according to which n does continuously decrease with temperature and depends also somewhat on strain rate. However, this rate dependence appears to be relatively small as long as the temperature is not too large.

In the interval of strain rates $10^{-8} \text{ s}^{-1} \leq \dot{\epsilon} \leq 1 \text{ s}^{-1}$ the value $n=4$ is attained at the reduced temperature $0.023g_0 < kT/\mu b^3 < 0.04g_0$. The corresponding ranges of the

temperature are $0.6T_m < T < 0.8T_m$ for Al and $0.65T_m < T < 0.83T_m$ for Ag, the two extremes. This means that dislocation glide remains the governing process up to a temperature of about $0.6T_m$ for the low strain rates usually attained in creep tests, while it controls practically up to the melting temperature at high deformation rates beyond 1 s^{-1} , such as in technical forming operations. We also believe, that in many cases the so called “five power law creep” regime [195] is significantly affected by glide rather than climb of the dislocations.

Our last point concerns path changes. Apart from the discussion of the Bauschinger effect due to strain reversals, our analysis does not include work-hardening behavior in combination with changes of the strain path. It is known since a long time [196,197] that changes of the direction of straining (not only reversals) cause significant transients of the hardening coefficient, sometimes even initially negative values i.e. work softening.

Although effects of this kind have particular technical relevance, e.g. for the anisotropy of material flow (earring) in forming operations, microstructure based investigations such as in Ref. [198], are rarely found in the literature. We feel, however, (and propose) that research on work hardening behavior under changing conditions will be intensified in a direction that includes path changes. The tools are now available and we hope that this article will be of some benefit for advancing research on crystal plasticity in new directions.

Acknowledgements

This work has been made possible by the continuous financial support of the Deutsche Forschungsgemeinschaft (SFB 371) and the Center for Material Science of Los Alamos National Laboratory funded by the US Dept. of Energy. It has profited very much from interaction with many people. We are particularly indebted to: A. Beaudoin and D. Embury for many helpful discussions, corrections, suggestions and comments, C. Tomé for reading and commenting on the manuscript, for generous support and technical assistance of the members of the materials-physics group at TUHH, in particular R. Bormann A. Bartels and Ch. Hartig. T. Janke deserves specific thanks for his willing, professional help in drafting the figures. Last but not least, we acknowledge with great gratitude the sustained support and never ending patience of our spouses, Mathilde and Marianne.

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